Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai)

[Knowledge is Nectar]

Liberal, Pi-Model of Engineering Education @ SPIT

(Department of Computer Science and Engineering)

Department of Computer Science and Engineering

A common scheme for "Computer Science and Engineering" and "Computer Engineering" till Semester V.

Effective from Academic Year 2023-24 [2023-2027 BATCH]

Revised -Board of Studies Approval: 18th July 2025,
Revised - Academic Council Approval: 30th August 2025

D. R. Kalbande	**Dr. Y. S. Rao**	**Dr. B. N. Chaudhari**
H.O.D., CSE	**Dean (Academics & Research)**	**Principal**

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(Department of Computer Science and Engineering)

CURRICULUM STRUCTURE FOR UNDERGRADUATE ACADEMIC PROGRAMS IN COMPUTER SCIENCE AND ENGINEERING AT SPIT

W.E.F. A.Y. 2023-24 [2023-2027 BATCH]

A common scheme for "Computer Science and Engineering" and "Computer Engineering" till Semester V.

Preamble: The government of Maharashtra has directed Autonomous Colleges to revise their curriculum and step into the implementation of the National Education Policy (NEP) 2020. We commit ourselves to the effective and fruitful implementation of NEP 2020 in its spirit. The holistic development of learners has always been the priority and center of focus for "BharatiyaVidyaBhavan". S.P.I.T. started implementing the philosophy of NEP in the year 2019 itself. We have in fact graduated the first batch of our holistic curriculum in 2023. Now based on our learnings from the implementation and recent recommendations of the Government, we are pleased to offer a 2nd iteration of our holistic curriculum for 2023-27, a Liberal Pi Model of Engineering Education.

This curriculum aims at the development of an **all-rounded** personality. It follows a **holistic** approach to education, ensures a strong science, and mathematics foundation and program core, develops expertise in domain vertical through the sequel of electives, ensures significant exposure to additional discipline through "Multidisciplinary Minor" courses, imparts state-of-the-art practical knowledge through a semester-long industry/research internship, collaborates outside world for the imparting relevant skill courses, challenges good learners through "Honors" evaluation, and systematically develops soft skills, and social, physical, mental, spiritual personality through carefully articulated **Liberal Learning** and **Humanities** sequels. Thus, it offers a unique, liberal **"Pi-Model"** of Engineering Education.

Table 1: Nomenclature of the courses in the curriculum

Groups	Abbreviation	Course Category
Basic Sciences and Engineering Sciences Courses (BSES)	BSESC	Basic Science & Engineering Science Courses
	BSESEC	Basic Science & Engineering Science Elective Courses
Skill Based Courses (SBC)	SEC	Skill Enhancement Course
	CC	Co-curricular Courses
Humanities, Social Science and Management (HSSM) Courses	HSSMC	Humanities, Social Science and Management Courses
	CP	Community Project
Ability Enhancement Courses (AEC)	IKS	Indian Knowledge System
	UHV	Universal Human Values
Program Related Courses (PRC)	PCC	Program Core Courses
	PEC	Program Elective Courses
	ELC	Experiential Leaning Courses
Multi-Cross-Trans disciplinary courses (MCTD)	OEC	Open Elective Courses
	MDM	Multidisciplinary Minor

Indicative List of BSE SE Courses:

- Engineering Physics
- Engineering Chemistry
- Biology for Engineers
- Engineering Mechanics
- Engineering Graphics
- Material Science
- Environmental Science
- Thermal & Fluid Engineering

Table 2: Comparison of S.P.I.T. credit structure with the G.R. recommendations

S.P.I.T.											
Sem	**BSES**	**SEC**	**AEC**	**HSSM**	**CC (LLC)**	**PCC**	**PEC**	**OE**	**EXP LEARNING**	**MDM**	**Total**
I	11	5	2		1						**19**
II	11	5	2		1				2		**21**
III	6	2		2	1	12					**23**
IV	3	2		2	1	12				3	**23**
V						19			2	4	**25**
VI		2				7	6		1	3	**19**
VII							6	3	3	4	**16**
VIII								3	11		**14**
Total	31	16	4	4	4	50	12	6	19	14	**160**
%	**19.38**	**10**	**2.5**	**2.5**	**2.5**	**31.25**	**7.5**	**3.75**	**11.87**	**8.75**	**100**
G.R. (NEP-2020) Recommended											
Tota	**30**	**10**	**8**	**4**	**4**	**44**	**20**	**8**	**22**	**14**	**164**
%	**18.3**	**6.1**	**4.88**	**2.44**	**2.44**	**27**	**12.2**	**4.88**	**13.42**	**8.54**	**100**

Figure 1:Comparison of S.P.I.T. credit structure with the G.R. recommendations

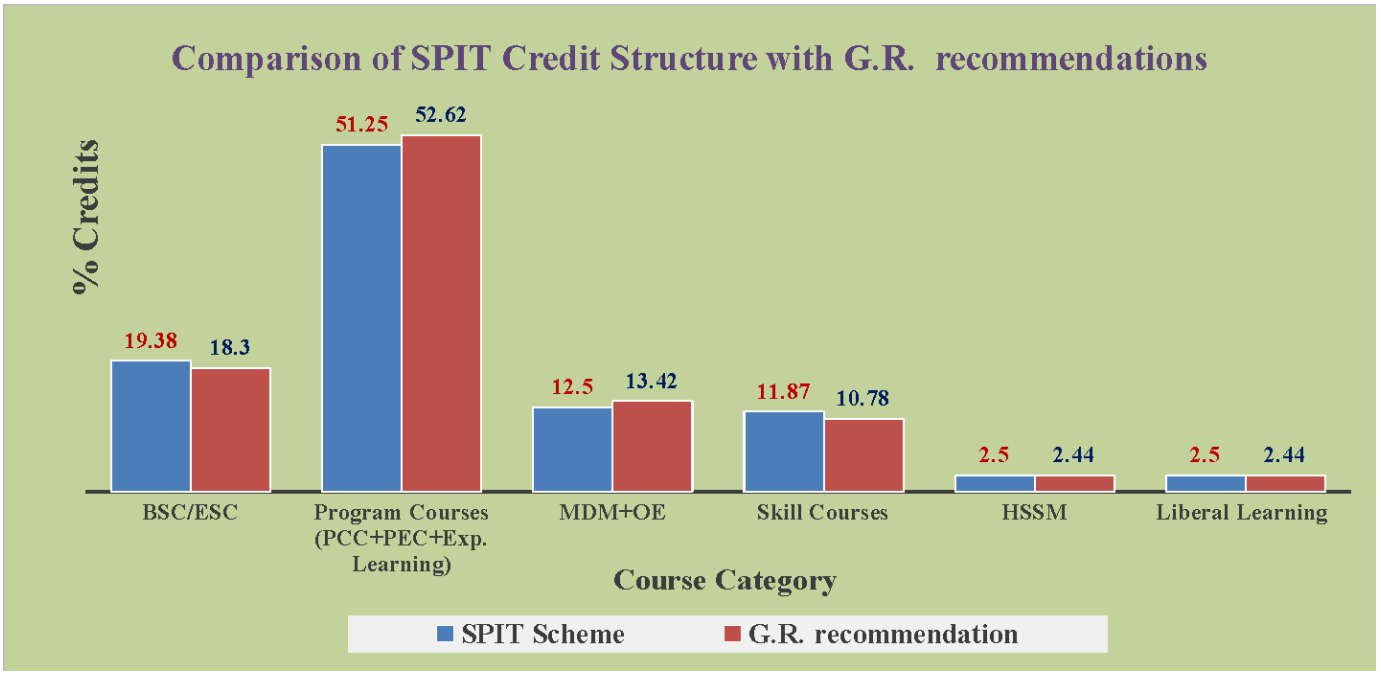


Figure 2: Pie-chart of vertical-wise allocation of credits

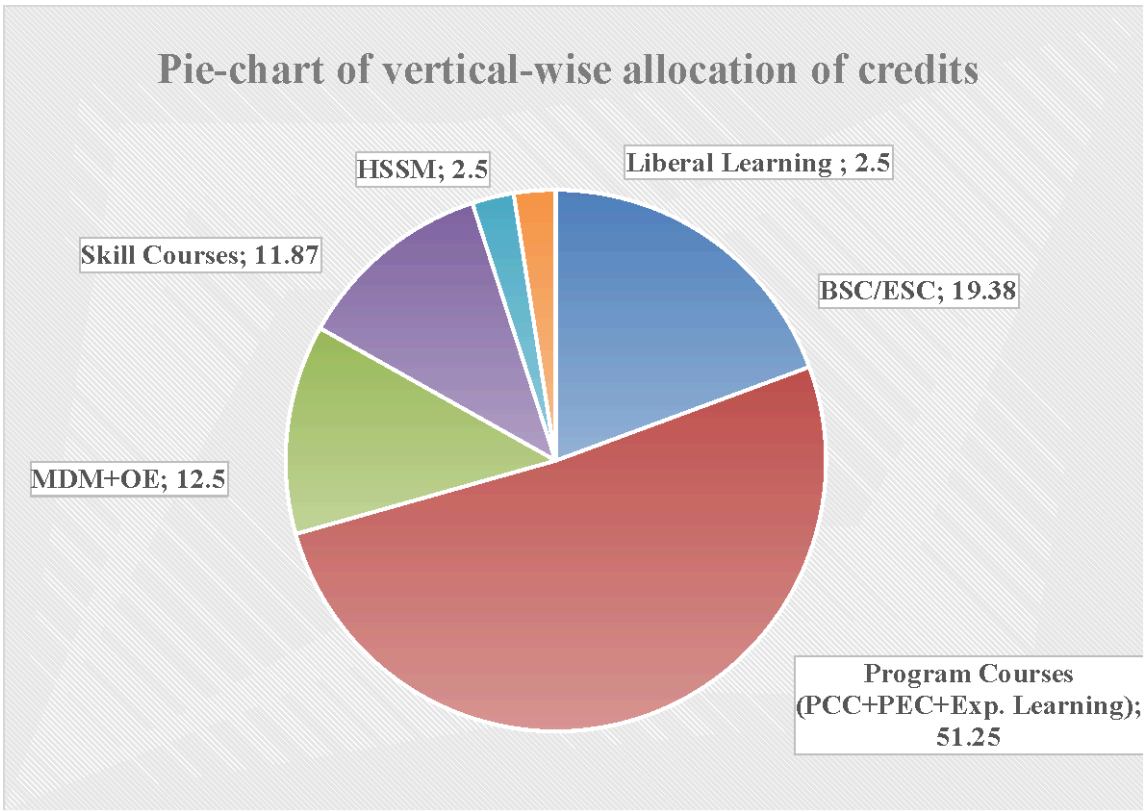


Table 3: Semester-wise allocation of credits to different verticals

SEM I										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic & Engg. Sciences	BSES	MA101	Mathematics I (ECL)	3	1	0	8	12	4
2	Skill Enhancement Course	SEC	CE101	Problem Solving using Imperative Programming Lab	0	1	2+2	4	9	3
3	Basic & Engg. Sciences Elective	BSESE		**Course I**						3
			AS101	Engineering Physics	2	0	2	4	8	
			AS102	Engineering Chemistry	2	0	2	3	7	
			AS103	Biology for Engineers	3	0	0	3	6	
			AS104	Engineering Mechanics	2	0	2	4	8	
			AS105	Engineering Graphics	1	0	2+2	2	7	
			AS108	Material Science	2	0	2	4	8	
			AS109	Environmental Science	3	0	0	3	6	
			AS110	Energy Science	2	0	2	3	7	
			AS111	Thermal & Fluid Engineering	3	0	0	3	6	
4	Skill Enhancement course	SEC	AS106	Tech Shop	1	0	2	2	5	2
			AS107	Soft Skill I						
5	Basic & Engg. Sciences	BSES	EC102	Basic Electrical Engineering	3	0	2	5	10	4
			EC101	Digital Systems	3	0	2	6	11	
6	Ability Enhancement -*Course	AEC	AS108	IKS	2	0	0	1	3	2
			AS109	UHV						
7	Cocurricular Courses	CC (LLC)	LLCXX	LLC--I	1	0	0	2	3	1
				Total	**12**	**2**	**10**	**25**	**49**	**19**

SEM-II										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic & Engg. Sciences	BSES	MA102	Mathematics II (DECA)	3	1	0	8	12	4
2	Skill Enhancement Course	SEC	CE102	Problem Solving using Object Oriented Programming Lab	0	1	2+2	4	9	3
3	Basic & Engg. Sciences Elective	BSESE		**Course I**						3
			AS101	Engineering Physics	2	0	2	4	8	
			AS102	Engineering Chemistry	2	0	2	3	7	
			AS103	Biology for Engineers	3	0	0	3	7	
			AS104	Engineering Mechanics	2	0	2	4	8	
			AS105	Engineering Graphics	1	0	2+2	2	7	
			AS108	Material Science	2	0	2	4	8	
			AS109	Environmental Science	3	0	0	3	6	
			AS110	Energy Science	2	0	2	3	7	
			AS111	Thermal & Fluid Engineering	3	0	0	3	6	
4	Skill Enhancement course	SEC	AS106	Tech Shop	1	0	2	2	5	2
			AS107	Soft Skill I						
5	Basic & Engg. Sciences	BSES	EC102	Basic Electrical Engineering	3	0	2	6	11	4
			EC101	Digital Systems	3	0	2	5	10	
6	Ability Enhancement -*Course	AEC	AS108	IKS	2	0	0	1	3	2
			AS109	UHV						
7	Cocurricular Courses	CC (LLC)	LLCXX	LLC--I	1	0	0	2	3	1
				Total	**12**	**2**	**10**	**25**	**49**	**19**

Summer Term										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Experiential Learning	CP (in Summer)	PR101	Community Project	0	0	4	4	8	2
2	HSSE	COI	AS112	Constitution of India (2Hrs/Week)	1	**0**	**0**	**1**	**2**	**NC**

SEM III										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic & Engg. Sciences	BSES	CS201	Discrete Structures and Graph Theory	3	0	0	5	8	3
2	Basic & Engg. Sciences *	FOM-I	MA202	Foundation of Mathematics-I*	2	1	0	0	3	3
3	Skill Enhancement Course	SEC	AS201	Soft Skill II-Professional Communication Skills	0	1	2	4	7	2
4	Basic & Engg. Sciences Elective	BSESE		**Course I**						3
			AS101	Engineering Physics	2	0	2	4	8	
			AS102	Engineering Chemistry	2	0	2	3	7	
			AS103	Biology for Engineers	3	0	0	3	7	
			AS104	Engineering Mechanics	2	0	2	4	8	
			AS105	Engineering Graphics	1	0	2+2	2	7	
			AS108	Material Science	2	0	2	4	8	
			AS109	Environmental Science	3	0	0	3	6	
			AS110	Energy Science	2	0	2	3	7	
			AS111	Thermal & Fluid Engineering	3	0	0	3	6	
5	Humanities	HSSM-I	HS2XX	**Course I**	2	0	0	3	5	2
7	Program Core Courses (12 Credits)	PCC	CS202	Data Structures	3	0	2	4	9	4
8		PCC	CS203	Computer Organization and Architecture	3	0	2	4	9	4
9		PCC	CS204	Database Management Systems	3	0	2	4	9	4
10	Cocurricular Courses	CC (LLC)	LLCXX	LLC--III	1	0	0	1	2	1
				Total	**17**	**1**	**10**	**28**	**56**	**23**

SEM IV										
Sr. No.	**Course Category**	**Abbrevia-tion**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic & Engg. Sciences	BSES	CS205	Statistical Methods in Computer Science	3	0	0	6	9	3
2	Basic & Engg. Sciences *	FOM-II	MA204	Foundation of Mathematics-II*	2	1	0	0	3	3
3	Skill enhancement course	SEC	AS202	Python Programming for Data science	0	1	2	4	7	2
4	Humanities	HSSM-II	HS2XX	**Course II**	2	0	0	3	5	2
5	Program Core Courses (12 credits)	PCC	CS206	Operating Systems	3	0	2	4	9	4
6		PCC	CS207	Design and Analysis of Algorithms	3	0	2	4	9	4
7		PCC	CS208	Computer Communications and Networks	3	0	2	4	9	4
8	Cocurricular Courses	CC (LLC)	LLCXX	LLC--IV	1	0	0	1	2	1
9	Multidisciplinary Minor	MDM	MDEC1X	MDM-I	To be defined by others					3
				Total	**15**	**1**	**8**	**26**	**50**	**23**

**Only for Lateral Entry Students*

Summer term (For Lateral Entry Students)										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic & Engg. Sciences	BSES	CS201	Discrete Structures and Graph Theory	3	0	0	5	8	3
2			CS205	Statistical methods in Computer Science	3	0	0	6	9	3

- Students are expected to start working for the Mini Project I during the summer.
- Summer Research Phase I of minimum 2 months and Course on Research Methodology for the "Honors by Research" for 7(4+3) credits- HR21
- For Enrollment to Honors by research, Minimum CGPA must be 8.25

SEM V										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Experiential Learning	ELC	PR1	Mini Project I	0	0	4	4	8	2
2	Program Core Courses (19 Credits)	PCC	CS301	Distributed Computing	3	0	2	6	10	4
3		PCC	CS302	Software Engineering	3	0	2	6	10	4
4		PCC	CS303	Artificial Intelligence and Soft Computing	3	0	2	6	11	4
5		PCC	CS304	Theory of Computation	3	0	0	5	8	3
6		PCC	CS305	Cryptography and Network Security	3	0	2	5	10	4
7	Multidisciplinary Minor	MDM	MDEC 2X	MDM-II	To be defined by others					4
		Total			**13**	**0**	**12**	**32**	**57**	**25**

SEM VI										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Program Core Courses (7 credits)	PCC	CS306	Human Machine Interaction	3	0	2	4	9	4
2		PCC	CS307	Machine Learning	2	0	2	5	9	3
3	Multidisciplinary Minor	MDM	MDEC3X	MDM-III	To be defined by others					3
4	Experiential Learning	ELC	PR3-I	Main Project Stage I	0	0	2	4	6	1
5	Program Elective Courses	PEC	CS3X1	PE-I	2	0	1	4	7	3
6	Program Elective Courses	PEC	CS3X2	PE-II	2	0	1	4	7	3
7	Skill Enhancement Course	SEC	CS308	DevOps Lab	0	1	2	2	5	2
				Total	**9**	**1**	**10**	**23**	**43**	**19**

- Summer Research Phase II of minimum 2 month and A course on Optimization for the "Honors by Research" for 7(4+3) credits HR31
- For Enrollment to Honors by research, Minimum CGPA must be 8.25

SEM VII									
Course Category	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
Multidisciplinary Minor	MDM	MDEC4X	MDM-IV	To be defined others					4
Program Elective Courses	PEC	CS4X3	PE-III	2	0	1	4	7	3
Program Elective Courses	PEC	CS4X4	PE-IV	2	0	1	4	7	3
Open Elective	OE	OE1	OE-I	2	0	1	4	7	3
Experiential Learning	ELC	PR3-II	Main Project Stage II	0	0	6	4	10	3
			Total	**6**	**0**	**9**	**16**	**31**	**16**

SEM VIII										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Open Elective	OE	OE2	OE-II**	2	0	1	4	7	3
2	Experiential Learning	ELC	INTR INTI PR4	Research/ Industry Internship/Major Project Stage III***	0	0	24	12	36	11
				Total	**2**	**0**	**25**	**16**	**43**	**14**

- **Extended Research Internship of minimum 2 month for the "Honors by Research" for 4 credits HR41**
- **For Enrollment to Honors by research, Minimum CGPA must be 8.25**

***** To be completed from MOOCs***

******Students neither taking research or industry internship nor willing to extend their project work can earn additional 11 credits from Swayam Platform or NPTEL or registering courses from any peer institution of higher learning., besides open elective program elective courses offered by the institute.***

Indicative List of Humanities courses (HSSM-I):

Course Code	Course Title	Course Code	Course Title
HS211	Law for Engineers-I	HS212	Law for Engineers-II
HS221	Psychology -I	HS222	Psychology -II
HS231	Finance for Engineers-I	HS232	Finance for Engineers-II
HS241	Economics-I	HS242	Economics-II
HS251	French-I	HS252	French-II
HS261	German-I	HS262	German-II
HS271	Japanese-I	HS272	Japanese-II
HSNP	NPTEL (HSS/Management)	HSNP	NPTEL (HSS/Management)

Indicative List of Co-curricular Courses (LLC)

Course Code	Course Title
LLC01	Dance (Kathak)
LLC02	Dance (Bharatnatyam)
LLC02	Fundamentals of Photography
LLC03	Art of Short Film Making / Cinematography
LLC04	Film Appreciation
LLC05	Basics of Music Composition
LLC06	Basics of Keyboard playing
LLC07	Physical Fitness
LLC08	Self Defense for Women
LLC09	Pran-Vidya (Combo of Yoga and Pranayam)
LLC10	JeevanVidya (Work Life Balance)
LLC11	Integrated Personality Development-I
LLC12	Indian Knowledge System-I
LLC13	Design Thinking
LLC14	Innovation and Creativity
LLC15	Principle Centered Leadership
LLC16	Social Psychology
LLC17	Mentoring of School Children at SPIT (Abhudaya)
LLC18	Basics of Fire Safety
LLC19	Study of one of the Identified Books
LLC20	Teaching Assistantship
LLC21	Trekking
LLC22	Kannada Language
LLC23	Telugu Language
LLC24	Tamil Language
LLCXX	Any other Course approved by Dean Academics and Research

PROGRAM ELECTIVE COURSES

4 Electives are sufficient to specialize in a particular domain.

Track	PE-I CS3X1	PE-II CS3X2	PE-III CS4X3	PE-IV CS4X4
Emerging Networking Technologies	CS311: Digital Forensic	CS312: Cloud Computing	CS413: Blockchain Technology	CS414: IT Infrastructure Monitoring and Management
Emerging AI	CS321: Natural Language Processing	CS322: Deep Learning	CS423: Generative AI	CS424: Explainable AI
Data Analytics	CS331: Big data Analytics	CS332: Business analytics with Python	CS433: Data Warehouse and Mining	CS434: AI for Healthcare Analytics
Fundamentals of Visualization	CS341: Digital Signal & Image Processing	CS332: Augmented Reality &Virtual Reality [AR-VR]	CS433: Computer Vision	CS434: Visual Intelligence

Indicative list of Multidisciplinary Minors

MDM Sequels for EXTC

- Computer Engineering
- AIML
- Data Science
- Interface and Experience Design
- IT Infrastructure

Course Category of Multidisciplinary Minor	MDM-I (Semester IV)	MDM-II (Semester V)	MDM-III (Semester VI)	MDM-IV (Semester VII)
Computer Engineering	**M051:** Database Management Systems	**M052:** Data Structures and Algorithms	**M053:** Cloud Computing	**M054:** Internet and Web Technology
Artificial Intelligence and Machine Learning	**M061:** Fundamentals of NNFL (NN, Fuzzy)	**M062:** Machine Learning	**M063:** Deep Learning	**M064:** Generative AI and application
Data Science	**M071:** Fundamentals of Data Science	**M072:** Data Analytics and Visualization	**M073:** Decision Making and Business Intelligence	**M074:** Social Media Analytics
Interface and Experience Design	**M081:** UI/UX Fundamentals	**M082:** Fundamentals of Experience Design	**M083:** Human Machine Interaction	**M084:** Prototyping and Interaction Design [Capstone project]

MDM Sequels for CE/CSE

- Industrial IoT
- Digital Signal Processing
- Electronics Communication

- VLSI
- Mathematics and Statistics
- Finance
- Economics

Course Category of Multidisciplinary Minor	MDM-I (Semester IV)	MDM-II (Semester V)	MDM-III (Semester VI)	MDM-IV (Semester VII)
Industrial IoT	**M011:** Introduction to Internet of Things	**M012:** Embedded "C" and Micro Python for IoT	**M013:** Communication Protocols for IoT	**M014:** IoT Applications and Analytics
Signal Processing and Communication	**M021:** Digital Signal Processing	**M022**: Digital Communication	**M023:** Digital Image & Video Processing	**M024:** Wireless Communication
VLSI	**M031:** Hardware Description Language programming	**M032:** Digital CMOS VLSI Design	**M033:** VLSI Physical Design	**M044:** ASIC Verification

Notes:

1. Learners who earn a minimum of total 160 credits will be awarded "B. Tech in Engg. /Tech. with Multidisciplinary Minor" degree.
2. Learners will have the following options to earn B. Tech. in Engg. /Tech. degree with MDM andHonors Certification. If they earn top grade in any 8 Program core courses and earn 80 percentiles in Gate exam.
3. Learners who earn 18 additional credits through 6-month (2+1+2+1) Research Internships during summer and winter breaks, as mentioned in the scheme, are eligible for the degree: "B. Tech in Engg. /Tech. with Multidisciplinary Minor" and Honors by Research, subject to earning CGPA of 8.25 throughout all semesters.
4. Learner can earn the certificates based on his/her exit from the program as follows:
   a. After a one-year (40 credits to be earned) and 8-week summer workshop: Certificate in Engineering.
   b. After two-years (80 credits to be earned) and 8-week summer workshop: Diploma in Engineering.
   c. After three-years (120 credits to be earned) and 8-week summer workshop: B. Sc. Engineering

D. R. Kalbande	**Dr. Y. S. Rao**	**Dr. B. N. Chaudhari**
H.O.D., CSE	**Dean (Academics & Research)**	**Principal**

SYLLABUS

SEMESTER I

SEM I										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic & Engg. Sciences	BSES	MA101	Mathematics I (ECL)	3	1	0	8	12	4
2	Skill Enhancement Course	SEC	CE101	Problem Solving using Imperative Programming Lab	0	1	2+2	4	9	3
3	Basic & Engg. Sciences Elective	BSESE		**Course I**						3
			AS101	Engineering Physics	2	0	2	4	8	
			AS102	Engineering Chemistry	2	0	2	3	7	
			AS103	Biology for Engineers	3	0	0	3	6	
			AS104	Engineering Mechanics	2	0	2	4	8	
			AS105	Engineering Graphics	1	0	2+2	2	7	
			AS108	Material Science	2	0	2	4	8	
			AS109	Environmental Science	3	0	0	3	6	
			AS110	Energy Science	2	0	2	3	7	
			AS111	Thermal & Fluid Engineering	3	0	0	3	6	
4	Skill Enhancement course	SEC	AS106	Tech Shop	1	0	2	2	5	2
			AS107	Soft Skill I						
5	Basic & Engg. Sciences	BSES	EC102	Basic Electrical Engineering	3	0	2	5	10	4
			EC101	Digital Systems	3	0	2	6	11	
6	Ability Enhancement -*Course	AEC	AS108	IKS	2	0	0	1	3	2
			AS109	UHV						
7	Cocurricular Courses	CC (LLC)	LLCXX	LLC--I	1	0	0	2	3	1
				Total	**12**	**2**	**10**	**25**	**49**	**19**

SEMESTER II

SEM-II										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**

1	Basic & Engg. Sciences	BSES	MA102	Mathematics II (DECA)	3	1	0	8	12	4
2	Skill Enhancement Course	SEC	CE102	Problem Solving using Object Oriented Programming Lab	0	1	2+2	4	9	3
3	Basic & Engg. Sciences Elective	BSESE		**Course I**						3
			AS101	Engineering Physics	2	0	2	4	8	
			AS102	Engineering Chemistry	2	0	2	3	7	
			AS103	Biology for Engineers	3	0	0	3	7	
			AS104	Engineering Mechanics	2	0	2	4	8	
			AS105	Engineering Graphics	1	0	2+2	2	7	
			AS108	Material Science	2	0	2	4	8	
			AS109	Environmental Science	3	0	0	3	6	
			AS110	Energy Science	2	0	2	3	7	
			AS111	Thermal & Fluid Engineering	3	0	0	3	6	
4	Skill Enhancement course	SEC	AS106	Tech Shop	1	0	2	2	5	2
			AS107	Soft Skill I						
5	Basic & Engg. Sciences	BSES	EC102	Basic Electrical Engineering	3	0	2	6	11	4
			EC101	Digital Systems	3	0	2	5	10	
6	Ability Enhancement -*Course	AEC	AS108	IKS	2	0	0	1	3	2
			AS109	UHV						
7	Cocurricular Courses	CC (LLC)	LLCXX	LLC--I	1	0	0	2	3	1
				Total	**12**	**2**	**10**	**25**	**49**	**19**

Summer Term										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Experiential Learning	CP (in Summer)	PR101	Community Project	0	0	4	4	8	2
2	HSSE	COI	AS112	Constitution of India (2Hrs/Week)	1	**0**	**0**	**1**	**2**	**NC**

First Year Evaluation Scheme (Percentage)

Sr. No.	**Course Code**	**Course Name**	**Credits**	**ISE (%)**	**MSE (%)**	**ESE (%)**	**Sub Total (%)**	**Total**
1	MA101	Engineering Calculus-Theory	4	20	20	60	100	100
2	EC101	Digital Systems-Theory	3	15	15	45	75	100

		Digital Systems-Lab	1	15	0	10	25	
3	EC102	Basic Electrical Engineering-Theory	3	15	15	45	75	100
		Basic Electrical Engineering -Lab	1	15	0	10	25	
4	AS101	Engineering Physics-Theory	2	13.5	13.5	40	67	100
		Engineering Physics-Lab	1	26	0	07	33	
5	AS102	Engineering Chemistry-Theory	2	13.5	13.5	40	67	100
		Engineering Chemistry-Lab	1	26	0	07	33	
6	AS103	Biology for Engineers-Theory	3	20	20	60	100	100
7	AS104	Engineering Graphics -Theory	1	13	0	20	33	100
		Engineering Graphics-Lab	2	40	0	27	67	
8	AS105	Engineering Mechanics-Theory	2	13.5	13.5	40	67	100
		Engineering Mechanics-Lab	1	26	0	07	33	
9	AS108	Material Science and Engineering-Theory	3	20	20	60	100	100
10	AS109	Environmental Science -Theory	3	20	20	60	100	100
11	AS110	Energy Science -Theory	3	20	20	60	100	100
12	AS111	Thermal & Fluid Engineering -Theory	3	20	20	60	100	100
13	AS106	Tech Shop- Lab	2	80	0	20	100	100
14	AS107	Soft Skill I	2	80	20	0	100	100
15	CE101	Programming I -Lab	3	80	0	2	100	100
16	CE102	Programming II -Lab	3	80	0	20	100	100
17	AEC01	IKS	2	80	0	20	100	100
18	AEC02	UHV	2	80	0	20	100	100
19	LLC01	LLCXX	1	100	0	0	0	100
20	LLC02	LLCXX	1	100	0	0	0	100
21	PRJ01	Project-01	2	80	0	20	100	100

Note: Course coordinators desirous of changing the percentage allocation to ISE: MSE: ESE for Theory / Lab of any course, must seek approval of Dean Academics and the Principal,

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(BSES)	Mathematics I (Engineering Calculus)	3	1	0	8	12	3	1	0	4
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
MA101		Theory	20%		20%		60%		100%	
		Laboratory	--		--		--		--	

Pre-requisite Course Codes, if any.	Basics of derivatives and integration done earlier.
Course Objective: To develop mathematical skills for solving engineering problems.	
Learning Outcomes (LO): ***At the End of the course students will be able to:-***	
MA101.1	Differentiate a function partially.
MA101.2	Find extreme values of a given function.
MA101.3	Find the nth order derivative of a given function.
MA101.4	Expand a given function as a power series.
MA101.5	Calculate the value of integrals in one variable using different techniques.
MA101.6	Solve multiple integral in various coordinate systems and use to calculate Area.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
MA101.1	3											
MA101.2	3											
MA101.3	3											
MA101.4	3											
MA101.5	3											
MA101.6	3											

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2
MA101.1						
MA101.2						
MA101.3						
MA101.4						
MA101.5						
MA101.6						

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref	Hrs.
1	Title	Partial Differentiation	1,2,3	10

	1.1	Partial derivatives of first and higher order. Partial derivatives of composite functions.		4
	1.2	Euler's theorem for homogeneous functions with two and three independent variables, deductions from Euler's theorem.		3
	1.3	Application of partial derivatives: i) Local Maxima and Minima of functions of two variables. ii) Lagrange's Method of undetermined multipliers.		3
2	**Title**	**Successive Differentiation and Series**	1,2,3	**10**
	2.1	Successive Differentiation: Proofs of nth derivatives of standard functions. Use of De Moivre's theorem and partial fractions to calculate nth derivatives of given functions.		3
	2.2	Leibnitz's Theorem on nth derivative of product of two functions		2
	2.3	Infinite series: 1) McLaurin's series (without proof) and derivation of series of some standard functions using the Maclaurin series. Expansion of functions in powers of x by using i) Standard series method ii) Method of differentiation and integration. 2) Taylor's series and applications.		5
3	**Title**	**Integral Calculus (one variable)**	1,2,3	**8**
	3.1	Gamma functions: properties of gamma functions and integrals reducible to gamma functions.		2
	3.2	Beta functions: properties, relation between Beta and Gamma functions, integrals reducible to Beta functions, Duplication formula.		4
	3.3	Differentiation under Integral sign: differentiating integrals with constant limits of integration for one parameter.		2
4	**Title**	**Integral Calculus (multi variable)**	1,2,3	**14**
	4.1	Tracing of curves. Sketching standard solids (Spheres, Ellipsoids, Cylinders, Cones, Tetrahedrons, planes).		2
	4.2	Double Integration: definition and evaluation. Evaluate by changing the order of integration and by changing to polar form.		7
	4.3	Application of double integral to finding area of given regions.		2
	4.4	Triple integration: definition and evaluation (Cartesian and cylindrical coordinates).		3
5	**Self Study**	1.1 Partial differentiation of implicit functions. 2.3 Series by method of Substitution. 3.2 Proof of Duplication Formula. 3.3 Differentiation under Integral sign using two parameters. 4.1 Finding lengths of curves in Cartesian and polar form.	1,2,3	**08**
Total				**42***

***Total of 42 hours does not include self-study hours.**

Textbooks

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Higher Engineering Mathematics	Forty Fourth	Dr. B. S. Grewal	Khanna Publications	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Advanced Engineering Mathematics	Twenty Eighth	H.K Das	S. Chand	2014
2	Advanced Engineering Mathematics	Tenth	Erwin Kreysizg	John Wiley & Sons	2011
3	Advanced Engineering Mathematics	Fourth	Jain and Iyengar	Narosa Publications	2014
4	Higher Engineering Mathematics	Eleventh	B.V. Ramana	Tata McGraw-Hill	2010
5	Advanced Engineering Mathematics	Sixth	Dennis G. Zill	Jones and Bartlett	2016

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Differential Equations and Complex Analysis: Sem-II

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(BSES)	Mathematics II (Differential Equations and Complex Analysis)	3	1	0	8	12	3	1	0	4
		Examination Scheme								
		Component		ISE		MSE		ESE		Total
MA102		Theory		20%		20%		60%		100%
		Laboratory		--		--		--		--

Pre-requisite Course Codes, if any.	Topics from Engineering Calculus- Semester I
Course Objective: To develop mathematical skills for solving engineering problems.	
Learning Outcomes (LO): ***At the End of the course students will be able to:-***	
MA102.1	Solve differential equations of first order.
MA102.2	Solve differential equations of higher order using operators.
MA102.3	Solve differential equations in electrical engineering problems.
MA102.4	Find powers, roots & logarithm of a complex number and to separate the function of a complex number into real and imaginary.
MA102.5	Check whether a given function is analytic and construct analytic functions.
MA102.6	Compute integrals of complex valued functions.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
MA102.1	3											
MA102.2	3											
MA102.3	3											
MA102.4	3											
MA102.5	3											
MA102.6	3											

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
MA102.1							
MA102.2							
MA102.3							
MA102.4							
MA102.5							
MA102.6							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref	Hrs.
1	**Title**	**Linear Differential Equations of first order**	1,2,3	11
	1.1	Exact Differential Equations, Integrating Factors, and equations reducible to exact form.		3
	1.2	Linear differential equations (Definition), equations reducible to linear form, Bernoulli's equation.		2
	1.3	Simple application of differential equation of first and second order to electrical engineering problems.		2
	1.4	Numerical solution of ordinary differential equations of first order and first degree using (a)Taylor's series method (b)Euler's method (c) Modified Euler method (d) Runge-Kutta fourth order formula		4
2	**Title**	**Linear Differential Equations of higher order**	1,2,3	**11**
	2.1	Linear Differential Equation with constant coefficient- complementary function, particular integrals of differential equation of the type f(D)y = X where X is e^{ax}, sin(ax+b), cos(ax+b), x^m, e^{ax}V, xV, where V is a function of x.		7
	2.2	Cauchy's homogeneous linear differential equation and Method of variation of parameters for second order.		2
	2.3	System of Differential Equations.		2
3	**Title**	**Complex Numbers**	1,2,3	**8**
	3.1	De Moivre's Theorem and its application to determine powers of complex numbers. Roots of complex numbers by De Moivre's Theorem.		3
	3.2	Expansion of sin$n\theta$and cos$n\theta$in terms of powers of sinθ and $cos\theta$. Expansion of $sin^n\theta$ and $cos^n\theta$ in terms of sines and cosines of multiples of θ.		1

	3.3	Hyperbolic Functions: relation between circular and hyperbolic functions, Inverse hyperbolic functions. Separation into real and imaginary parts of complex functions.		3
	3.4	Logarithm of a complex number.		1
4	**Title**	**Analytic functions and Complex Integrals**	1,2,3	**12**
	4.1	Analytic functions, Cauchy Riemann equations in Cartesian and polar form, construction of analytic functions using Milne-Thompson's method, Harmonic functions, poles of f(z).		6
	4.2	Line Integral of complex valued function, Cauchy's Integral theorem for simply connected regions, Cauchy's Integral formula (for poles lying inside or outside the curve).		6
5	**Self Study**	1.3 To form D.E. for given L-C-E-R circuit. 2.1 Method of undetermined coefficients to solve differential equations. 2.2 Legendre's differential equation, Method of variation of parameters for third order differential equations. 3.2 Complex examples using De Moivre's Theorem. 4.1 Construction of analytic function f(z) =u+iv when u+v or u-v is given.		**08**
			Total	**42***

***Total of 42 hours does not include self-study hours.**

Textbooks

Sr. No.	Title	Edition	Authors	Publisher	Year
1.	Higher Engineering Mathematics	Forty Fourth	Dr.B.S. Grewal	Khanna Publications	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1.	Advanced Engineering Mathematics	Twenty Eighth	H.K Das	S.Chand	2014
2.	Advanced Engineering Mathematics	Tenth	Erwin Kreysizg	John Wiley & Sons	2011
3.	Advanced Engineering Mathematic	Fourth	Jain and Iyengar	Narosa Publications	2014
4.	A First Course in Differential Equations with Modelling Applications	Ninth	Dennis G. Zill	Cengage India	2009

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Engineering Physics

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
BSESE	Engineering Physics	2	0	2	4	8	2	0	1	3
		Examination Scheme								
		Component			ISE %		MSE %		ESE %	Total %
AS101		Theory			13.5		13.5		40	67
		Laboratory			26		–		7	33

Pre-requisite Course Codes, if any.	HSC level physics
Course Objective: To provide the knowledge and methodology necessary for solving problems in the field of engineering	
Learning Outcomes (LO): At *the End of the course students will be able to*	
AS101.1	Demonstrate the conceptual knowledge of quantum mechanics, semiconductors, lasers and wave optics.
AS101.2	Solve the problems by applying the basics concepts of physics.
AS101.3	Explain the working principle of various LASERs and its practical applications.
AS101.4	Develop experimental skills and practical abilities.
AS101.5	Use scientific apparatus and comprehend the importance of precision, accuracy of the experimental data.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AS101.1	1											
AS101.2		1										
AS101.3	1											
AS101.4				1								
AS101.5				1								

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
AS101.1							
AS101.2							
AS101.3							
AS101.4							
AS101.5							
AS101.6							
AS101.7							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Quantum Mechanics**		**09**
	1.1	de-Broglie hypothesis; experimental verification of de Broglie hypothesis; wave packet, group velocity and phase velocity; Wave function, Physical interpretation of wave function; Heisenberg's uncertainty principle; Applications of uncertainty principle	1,2,3	
	1.2	Schrodinger's time dependent wave equation, time independent wave equation; Application of time-independent Schrodinger equation - Particle trapped in one dimensional box and Potential barrier (Tunneling)	1,2,3	
2	**Title**	**Physics of Semiconductors and Semiconductor devices**		**13**
	2.1	Conduction in metals and semiconductors; Fermi-Dirac distribution function and Fermi level in a conductor, insulator and semiconductor	5	
	2.2	Intrinsic and extrinsic semiconductors; intrinsic conductivity and extrinsic conductivity; Law of mass action, charge neutrality condition; intrinsic carrier concentration, electron and hole concentration; Extrinsic carrier concentration as a function of temperature; Effect of impurity concentration and temperature on the Fermi Level; Hall Effect and its applications. Drift and Diffusion current density	5	
	2.3	Formation of a P-N junction, depletion region and barrier potential; concept of carrier current densities in p-n junction in equilibrium, forward bias and reverse bias	5,6	
	2.4	P-N junction devices: LED, Zener diode, photoconductors, solar cells	5,6	
3	**Title**	**LASERS**		**06**
	3.1	Processes - Absorption of light, spontaneous emission, stimulated emission; Einstein's equations, Population inversion; metastable states; pumping and pumping schemes; optical resonance cavity	3,4	
	3.2	Ruby and Helium-Neon laser, semiconductor laser; Applications of lasers in industry, medicine and holography (construction & reconstruction of holograms)	3,4	
4	**Self Study**	Interference of light in thin films having uniform thickness, Newton's rings, Applications of interference in anti-reflecting and highly reflecting thin films. Diffraction of light, Diffraction due to single slit, double slit and diffraction grating.		**05***
		Total (*Not included)		**28**

Laboratory Component

Sr. No.	Title of the Experiment*
1	Determination of energy band gap of a semiconductor using four probe method
2	Study of I-V characteristics of a Zener diode
3	Determination of the type of semiconductor sample, concentration of charge carriers and its mobility using Hall Effect
4	Determination of Planck's constant using photo vacuum tube
5	Measurement of ultrasonic velocity in liquid medium using ultrasonic interferometer
6	Determination of radius of curvature using Newton's Rings
7	Determination of thickness of a thin foil or wire using the interference pattern of a wedge- shaped film
8	Determination of wavelengths of a mercury source and resolving power of a plane diffraction grating
9	Study of single slit diffraction
10	Determination of grating element of a diffraction grating using a laser source
11	Determination of the numerical aperture of an optical fibre
12	Uses of a Cathode-Ray Oscilloscope

*Students will perform any 10 of the above experiments

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	A Textbook of Engineering Physics	Eleventh	Dr. M. N. Avadhanulu & Dr. P. G. Kshirsagar	S. Chand	2018
2	Engineering Physics	First	D. K. Bhattacharya & Poonam Tandon	Oxford University Press	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Concepts of Modern Physics	Sixth	Arthur Beiser	McGraw Hill Education	2009
2	Modern Physics	Third	Serway, Moses and Moyer	Thomson Learning	2005
3	Fundamentals of Physics	Tenth	Halliday and Resnick	Wiley	2013
4	Solid State Physics	Eighth	S. O. Pillai	New Age International Publishers	2018
5	Solid State Electronic Devices	Seventh	Ben G. Streetman and Sanjay Kumar Banerjee	Pearson Education	2016
6	Lasers: Fundamentals and Applications	Second	Ghatak and Thyagarajan	Springer	2011
7	Engineering Physics	First	B. K. Pandey and S. Chaturvedi	CENGAGE	2023

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Engineering Chemistry

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
BSESE	Engineering Chemistry	L	T	P	O	E	L	T	P	Total
		2	0	2	3	7	2	0	1	3
AS102		Examination Scheme								
		Component	ISE %		MSE %		ESE %		Total %	
		Theory	13.5		13.5		40		67	
		Laboratory	26				7		33	

Course Education Objective:
To provide necessary background of applied chemistry suited for relevant areas of engineering.

Pre-requisite Course Codes		HSC Level Chemistry
After successful completion of the course, learner will be able to		
Learning Outcomes	AS102.1	Relate thermodynamic principles and laws to crucial applications like heat engines (Understanding)
	AS102.2	Summarize properties and applications of different materials like polymers, ceramics, alloys , nanomaterials, (Understanding)
	AS102.3	Compare different sources of energy like conventional fossil fuels, alternative fuels, batteries, fuel cells with respect to availability, working principles, constitution, efficiency of performance and environmental impact (Understanding)
	AS102.4	Apply knowledge of electrochemistry and green chemistry in the interest of public health and environment (Application)
	AS102.5	Make use of analytical techniques (potentiometric and iodometric titrations) and instruments (pHmeter, conductometer and Orsats's Apparatus) for various purposes like redox reactions, neutralization reactions etc.
	AS102.6	Estimate key properties of lubricants like flash point, Viscosity and acid value

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AS102.1	1											
AS102.2	1											
AS102.3	2											
AS102.4	2						1					
AS102.5							2					
AS102.6				1								

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
AS101.1							
AS101.2							
AS101.3							
AS101.4							
AS101.5							
AS101.6							
AS101.7							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No	Module Name	Unit No.	Topics	Ref.	Hrs
1	Thermody-namics	1.1	Introduction, Terminology, Concepts of Internal Energy and Thermodynamic equilibrium, Zeroth and First Law of Thermodynamics, Implications and Limitations of First law	1, 2	1
		1.2	Joule Thomson Effect and significance of inversion temperature, Carnot's Cycle, Carnot's theorem and related numericals, Second Law of thermodynamics	1, 2	1
		1.3	Overview of applications of thermodynamics	1,2	1
2	Polymers	2.1	Introduction to polymers, Effect of heat on polymers : Glass transition temperature and melting with significance;	2,3,4	2
		2.2	Latest Applications: Conducting polymers, Liquid crystal polymers, Engineering Polymers,	2,3,4, 5	2
3	Fuels and combustion	3.1	Definition and classification of fuels, Calorific value : Definition, Gross or Higher calorific value & Net or lower calorific value, Dulong‟s formula & numericals for calculations of Gross and Net calorific values.	2,3	2
		3.2	Knocking, Octane number, Antiknock agents, unleaded petrol, Cetane number	2,3	1
		3.3	Calculations for requirement of only oxygen and air (by weight and by volume only) for given solid & gaseous fuels.	2,3	2
		3.4	Disadvantages of fossil fuels, Alternative (Green) Fuels : Power alcohol , Biomass, Biogas, Biodiesel, Natural Gas and CNG (Description, Utility, advantages and disadvantages)	2,3	1
4	Electro-chemistry	4.1	Introduction, concept of electrode potential, electrochemical cells, Nernst Equation with applications and numericals,	2,3,4	2
		4.2	Reference Electrodes (Hydrogen electrode, calomel electrode), Glass Electrode for pH measurements, Electrochemical sensors : Working principle, construction and applications	2,3,4	2
		4.3	Faraday's Laws of electrolysis and numericals,	2,3,4	1

5	**Batteries and Battery Technology**	**5.1**	Introduction, Important terms, Lead Acid storage Cell, Nickel-Hydrogen(metal hydride), Rechargeable Lithium ion batteries, Sodium ion batteries	2,3, 6	2
		5.2	Reserve Batteries, Fuel cells, characteristics, description, construction and working of Hydrogen-oxygen fuel cells	2,3, 6	2
6	**Green Chemistry**	**6.1**	12 principles of green chemistry with examples, Numericals on Atom Economy, Green Solvents (Water, Supercritical Fluids),	2,3	3
7	**Engineering materials**	**7.1**	Eutectic mixtures and soft solders Advanced Ceramic materials and cermets and applications Carbon nanomaterials : Fullerenes and Carbon nanotubes, Structure, Properties and applications	2,3,4	1
		7.2	Semiconductors, Stoichiometric and defect semiconductors.	2,3,4	1
		7.3	Superconductors, perovskite structure and 1:2:3 compound $YBa_2Cu_3O_{7-y,}$ properties and applications	2,3,4	1
			Total		**28**
	Self Learning		Challenges pertaining to use of Li ion batteries, Different types of fuel cells, Calculation of hardness of water and COD of wastewater		

Laboratory Component

Sr. NO	Title Of the Experiment
1	Determination of total hardness of water sample
2	Removal of hardness using ion exchange column
3	Determination of Viscosity of oil by Redwood Viscometer
4	To determine flash point of a lubricating oil
5	Analysis of flue gas for its composition (by Orsat's Apparatus)
6	Estimation of acid value of lubricant
7	Determination of amount of strong acid present in a solution using a conductometer
8	Determination of strength of acid using a pH meter
9	Estimation of Copper in brass by Iodometric Titration
10	Potentiometric titration (acid-base, redox)
11	Estimation of Iron in plain Carbon steel
12	Determination of COD of wastewater sample

References:

[1] Peter Atkins, Physical Chemistry, XIth ed, Oxford, United Kingdom, Oxford University Press, 2017

[2] P. C. Jain & M. Jain, *Engineering Chemistry*, XVIth ed , New Delhi, India:Dhanpat Rai Publishing Co. (P) Ltd., 2014

[3] S. S. Dara & S. S. Umare,*A Textbook of Engineering Chemistry*, XIIth ed., New Delhi, India: S. Chand & Co. Ltd., 2013

[4] S. Chawla, *A Textbook of Engineering Chemistry*, IIIrd ed.,Delhi, India:Dhanpat Rai & Co. (Pvt.) Ltd., 2015

[5] S. Agarwal, *Engineering Chemistry Fundamentals and Applications*, Ist ed, Delhi, India: Cambridge Univ. Press., 2015

[6] Dell, Ronald M Rand, David A J, 'Understanding Batteries', Royal Society of Chemistry, (2001).

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Biology for Engineers

Course (Category) Code	Course Name	Teaching Scheme (Hrs./week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(BSESE)	Biology for Engineers	3	0	0	4	7	3	0	0	3
		Examination Scheme (%)								
		Component		ISE		MSE		ESE		Total
AS103		Theory		20		20		60		100
		Laboratory		0		--		0		0

Pre-requisite Course Codes, if any.	12th
Course Objective: To motivate the students to develop a system with a multidisciplinary approach using basic biological concepts and Artificial Intelligence. Learners will gain a comprehensive understanding of how AI is revolutionizing healthcare and biomedical research.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
AS103.1	Understand the basic biomolecules and their engineering applications.
AS103.2	Grasp how biological systems can be re-designed as substitute products for natural systems.
AS103.3	Acquire the interdisciplinary vision of biological engineering
AS103.4	Explore the use of AI in medical imaging for diagnosis and treatment planning
AS103.5	Understand AI-driven approaches for disease diagnosis, prediction, and personalized medicine with ethical, legal and social implications.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO 8	PO9	PO10	PO11	PO12
AS103.1	2											
AS103.2	3											
AS103.3	3											
AS103.4	2											
AS103.5	2				2							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember √	Understand √	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1	**Biomolecules and their applications**	1,2	8
	1.1	Carbohydrates (cellulose-based water filters, PHA and PLA as bioplastics), Nucleic acids (DNA Vaccine for Rabies and RNA vaccines for Covid19, Forensics – DNA fingerprinting),		
	1.2	Proteins (Proteins as food – whey protein and meat analogs, Plant based proteins), lipids (biodiesel, cleaning agents/detergents), Enzymes (glucose-oxidase in biosensors, alginolytic enzyme in bio-bleaching).		
2	2	**Cardiovascular and Nervous System**	1,2	10
	2.1	Anatomy of the heart; Heart valves, systemic and pulmonary circulation; Conduction system of the heart; Cardiac action potential, electrocardiogram (ECG); Cardiac cycle; Cardiac output; Blood pressure.		
	2.2	Divisions of the nervous system (central and peripheral nervous system); Structure and functions of the brain and spinal cord; Reflex actions and reflex arc; Functions of sympathetic and parasympathetic nervous system; Nerve action potential and nerve conduction.		
3	3	**Human Organ Systems and Bio-Designs**	1,2	10
	3.1	Brain as a CPU system, signal transmission, EEG, Robotic arms for prosthetics. Engineering solutions for Parkinson's disease).		
	3.2	Eye as a Camera system (architecture of rod and cone cells, optical corrections, cataract, lens materials, bionic eye).		
	3.3	Heart as a pump system (architecture, electrical signaling - ECG monitoring and heart related issues, reasons for blockages of blood vessels, design of stents, pacemakers, defibrillators).		
4	4	**AI for Biomedical Applications**	1	14
	4.1	Introduction to AI in Biomedical Applications: Overview of artificial intelligence and machine learning, Challenges and opportunities in applying AI to biomedicine, Ethical considerations in AI-driven healthcare		
	4.2	Medical Imaging and AI: Basics of medical imaging modalities, Image preprocessing and feature extraction, AI applications in medical image analysis and interpretation		
	4.3	Personalized Medicine and Precision Health: Concept of personalized medicine, AI-driven approaches to patient stratification, Precision health initiatives and challenges	2	
	4.4	Genomics and Bioinformatics: Basics of genomics and next-generation sequencing, AI-driven analysis of genomic data, Precision genomics and its applications in healthcare		

	4.5	Regulatory and Ethical Considerations: Regulatory landscape for AI in healthcare, Ethical and privacy considerations in AI-driven healthcare, Bias and fairness in AI algorithms		
5	**Self Study**	Trends in Bioengineering: Bioprinting techniques and materials, 3D printing of ear, bone, and skin. 3D printed foods. Electrical tongue and electrical nose in food science, DNA origami and Biocomputing, Bioimaging and Artificial Intelligence for disease diagnosis. Self-healing Bio concrete (based on bacillus spores, calcium lactate nutrients and biomineralization processes) and Bioremediation and Biomining via microbial surface adsorption (removal of heavy metals like Lead, Cadmium, Mercury, Arsenic).	1,2,3	6*
			Total (6*Not included)	42

ISE Component: Presentations on the following topics

Sr. No	Biology for Engineers
1	Signal Conditioning for ECG
2	Signal Conditioning for EEG
3	Study of Robotic arm for prosthetics
4	Study of Parkinson's disease
5	Utilize Python libraries such as OpenCV and TensorFlow to preprocess medical images and develop a deep learning model for automated diagnosis or classification of medical images (e.g., X-rays, MRI scans).
6	Collect and preprocess relevant clinical data (e.g., patient demographics, lab results, symptoms and implement predictive analytics algorithms to forecast disease progression or patient outcomes.
7	Design personalized treatment plans based on genetic profiles using decision support systems
8	Apply machine learning techniques (e.g., random forests, deep learning) to predict gene function or classify genomic variants [Access genomic datasets from repositories such as the Genomic Data Commons (GDC) or The Cancer Genome Atlas (TCGA).]
9	Develop predictive models for hospital readmission or patient risk stratification using clinical data.
10	A case studies involving ethical dilemmas in AI-driven healthcare (e.g., patient privacy, algorithmic bias). To study and analyze regulatory frameworks (e.g., HIPAA, GDPR) governing the use of AI in healthcare and Propose strategies for addressing ethical concerns and ensuring fairness and transparency in AI algorithms.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Human Physiology	16th	Stuart Fox, Krista Rompolski	McGraw-Hill	2022
2	Biology for Engineers	3rd	Thyagarajan S., Selvamurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jaganthan M.K.	Tata McGraw Hill	2012
3	Artificial Intelligence in Medicine		Adam Bohr and Gopalakrishnan Venkitaraman	Academic Press; 1st edition (21 June 2020)	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Biology for Engineers	--	Arthur T. Johnson	CRC Press, Taylor and Francis	2011
2	Biology for Engineers		Danial Schodek	MIT Press	

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Engineering Graphics

Course (Category) Code	Course Name	Teaching Scheme (Hrs./week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
BSESEC	Engineering Graphics	0	1	4	2	7	0	1	2	3
		Examination Scheme								
		Component		ISE %		MSE %		ESE %		Total %
AS105		Theory		13		0		20		33
		Laboratory		40		–		27		67

Pre-requisite Course Codes, if any.	10+2 Basics
Course Objective: To develop technical drawing and visualization skills using instrumental drawing and soft tools, required for design and modeling, in Engineering Applications and Solutions.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
AS105.1	Construct basic engineering curves.
AS105.2	Draw projection of points and lines.
AS105.3	Draw projection of regular solids inclined to both the reference planes.
AS105.4	Read the 3-dimensional view and draw the orthographic and sectional orthographic projections.
AS105.5	Read the orthographic projection and draw an isometric view.
AS105.6	Draw the development of lateral surfaces of solids.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AS105.1	3	3										3
AS105.2	3	3										3
AS105.3	3	3										3
AS105.4	3	3										3
AS105.5	3	3										3
AS105.6	3	3										3

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Unit1**	**Introduction to Engineering Drawing and Construction of engineering curves**	1,2	**4**
	1.1	Types of Lines, Dimensioning Systems as per IS conventions. First angle method of projection only		
	1.2	Basic construction of Cycloid and Involutes.		
2	**Unit2**	**Projection of Points and Lines: -**	1,2	**5**
	2.1	Projection of points in all four quadrants		
	2.2	Projection of lines parallel to one principal reference plane.		
	2.3	Lines inclined to both the Reference Planes (Excluding Traces of lines).		
3	**Unit3**	**Projection of solid** (Regular solids like Prism, Pyramid, Cylinder, Tetrahedron, Hexahedron and Cone only)	1,2	**5**
	3.1	Projection of solid resting on plane (Single step projection)		
	3.2	Projection of solid such that base inclined to one reference plane (Two step projection)		
	3.3	Projection of solid such that base inclined to both reference planes (Three step projection/problem) (Exclude Spheres, Composite, Hollow solids and frustum of solids)		
4	**Unit4**	**Orthographic Projection**	1	**Lab session (4)**
	4.1	Orthographic views of a simple machine part as per the first angle method of projection recommended by I.S.		
	4.2	Full Sectional views of the Simple Machine parts.		
5	**Unit5**	**Isometric visualization**	2	**Lab session (2)**
	5.1	Isometric view (Natural scale only)		
6	**Unit6**	**Development of lateral surface of solids**		**Lab session (2)**
	6.1	Development of lateral surface of solids (Exclude DLS of a solid with a hole in it and Reverse Development)		
7	**Self Study**	1.2 Construction of Engineering curves like ellipse, parabola, hyperbola, helix, other types of cycloid etc. by using different method of construction. 2.1 Solve more practice examples of orthographic views. Draw different views of a machine part/any object using third angle method of projection. (Axonometric view, oblique view, perspective etc.) 2.2 Half sectional orthographic views. 3.1 Projection of lines with traces, application-based problems on Projection of lines 4.1 Projection of section of solid with cutting planes. 5.1 Development of lateral surface of solids with openings, reverse development of solid.	1,2,3	**6***
			Total (*Not included)	**14**

Laboratory Component (Minimum 10 Laboratory experiments are expected)

Sr. No.	Engineering AutoCAD Laboratory
1	Introduction to Auto-CAD: -Basic Drawing and Editing Commands. Knowledge of setting up layers, Dimensioning, Hatching, plotting and Printing. **Auto-Cad Practice sheet-1** (Two Examples)
2	**Auto-Cad Practice sheet-2** (Two Examples)
3	**Introduction to Orthographic projections sheet-1** (Two Problems)
4	**Orthographic projections sheet-2** (Two Problems)
5	**Introduction to Sectional Orthographic projections sheet-1** (Two Problems)
6	**Sectional Orthographic projections sheet-2** (Two Problems)
7	**Introduction to Isometric Projection/View:** - Isometric View/Drawing of blocks of plain and cylindrical surfaces using plain/natural scale only. (Exclude Spherical surfaces). **Isometric Projection/View sheet-1** (Two Problems)
8	**Isometric Projection/View sheet-2** (Two Problems)
9	**Draw development of lateral surface of solids with simple sections** **DLS sheet-1** (Prism and Cylinder)
10	**DLS sheet-2** (Pyramid and Cone)

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Engineering Drawing	53rd	N D Bhatt	Charotar	2016
2	Engineering Drawing	3rd	Dhananjay A Jolhe	Tata McGraw Hill	2011

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	AutoCAD 2017	--	Sham Tickoo	DreamTech Press, Delhi	2017
2	Engineering Drawing and Graphics	Fifth	K Venugopal	New Age International	2011

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Engineering Mechanics

Course (Category) Code	Course Name	Teaching Scheme (Hrs./week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
BSESE	Engineering Mechanics	2	0	2	4	8	2	0	1	3
		Examination Scheme								
		Component			ISE %		MSE %		ESE %	Total %
AS104		Theory			13.5		13.5		40	67
		Laboratory			26		--		7	33

Pre-requisite Course Codes, if any.	10+2 Basics
Course Objective: To provide knowledge of force analysis methods required in engineering applications and solutions. Also, to develop analytical and computational ability.	
Learning Outcomes (LO):*At the End of the course students will be able to*	
AS104.1	Draw free body diagram and determine reactive forces using conditions of equilibrium and Lami's theorem
AS104.2	Determine coefficient of friction for various contact surfaces
AS104.3	Analyze the three-dimensional system of space forces.
AS104.4	Analyze the kinematics of particles and obtain the various parameters of motion.
AS104.5	Determine Instantaneous center of rotation (ICR).
AS104.6	Design and conduct an experiment to demonstrate principles of statics and dynamics

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AS104.1	3	3							3			
AS104.2	3	3							3			
AS104.3	3	3							3			
AS104.4	3	3							3			
AS104.5	3	3							3			
AS104.6	3	3							3			

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2
AS104.1						
AS104.2						
AS104.3						
AS104.4						
AS104.5						
AS104.6						

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Unit1	**Equilibrium of forces**		8
	1.1	Equilibrant force, conditions of equilibrium for concurrent forces, parallel forces and general force system, equilibrium of connected bodies, Lami's theorem.	1,3	
	1.2	Types of supports, types of loads, Beams, Determination of reactions at supports for various types of loads on beams	3	
2	Unit2	**Friction**		4
	2.1	Introduction to Laws of friction, Cone of friction, Equilibrium of bodies on inclined plane, Application to problems involving wedges, ladders.	1,2	
3	Unit3	**Forces in space**		5
	3.1	Rectangular Components of Forces in Space, Resultant of Space forces, Moment of a Force about a point, axis and line. Equilibrium of a particle in space.	1	
4	Unit4	**Kinematics of Particle**		8
	4.1	Motion along straight and curved path, Rectangular component of velocity and acceleration, Tangential & Normal component of acceleration, Motion curves(a-t, v-t, s-t curves), Projectile motion,	2,3	
5	Unit5	**Kinematics of Rigid Bodies**		3
	5.1	Instantaneous center of rotation for the velocity of bodies in plane motion, (up to 2 linkage mechanism)	3	
6	Self Study	1. Applications of resultant of forces, concept of couple and moments, 2. Centroid and center of gravity, analysis of trusses. 3. Kinetics of rigid body, work energy principle. 4. Principle of Law of Conservation of momentum, Impact and collision.	1,2, 3	6*
			Total (*Not included)	28

Laboratory Component

Sr. No.	Title of the Experiment
1	Draw the force polygon and determine the equilibrant force for a concurrent coplanar force system.
2	Use the conditions of equilibrium for the parallel force system and determine the support reactions.
3	Apply the principle of moment for equilibrium of levers.
4	Determine the coefficient of friction for glass slab and a metal plate on an inclined plane.
5	Determine the axial forces using Lami's theorem for Jib crane apparatus.
6	Use the conditions of equilibrium for a non-concurrent non-parallel force system and draw the force polygon.
7	Measure the acceleration due to gravity with the help of simple pendulum apparatus.
8	Determine the range of projectile and the time of flight for the projectile motion.
9	Verify the law of conservation of momentum and determine the coefficient of restitution for collision
10	(Plot the motion of the projectile using an air-cushion table apparatus.) A small project based on the Engineering Mechanics concept.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Vector Mechanics for Engineers statics and dynamics	Nineth	Beer and Johnston	McGraw Hill	2010
2	Engineering Mechanics	Fifth	Bhavikatti S and Rajsekharappa	New Age International	2009
3	Engineering Mechanics Statics and Dynamics	Fourteenth	A K Tayal	Umesh Publication, Delhi	2012

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Engineering Mechanic: Statics and Dynamics,	Fifth	E.W. Nelson, C.L. Best, W.G. McLean	McGraw Hill	1998
2	Singer's Engineering Mechanics Statics and Dynamics	Third	Vijaya Kumar Reddy. K. and Suresh Kumar. J	BS Publication	2012

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Material Science and Engineering

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
BSESE	Material Science and Engineering	3	0	0	4	07	3	0	0	3
		Examination Scheme								
		Component		ISE (%)		MSE (%)		ESE (%)		Total
AS108		Theory		20		20		60		100
		Laboratory		-		--		-		-

Pre-requisite Course Codes, if any.	12th Physics and Chemistry
Course Objective: To make learners understand the fundamental physical origins of material behavior to optimize the properties for various engineering materials.	
Learning Outcomes (LO): ***At the end of the course students will be able to***	
AS108.1	Demonstrate the knowledge of the basics of crystallography.
AS108.2	Visualize Planes and directions using Miller Indices.
AS108.3	Classify the various magnetic materials and identify the magnetic materials for various engineering applications.
AS108.4	Comprehend the basics of physics and chemistry on the nanometer scale and applications of nano materials.
AS108.5	Comprehend the basic physical properties of smart materials with various engineering applications.
AS108.6	Comprehend the use of various analytical techniques to characterize various materials.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AS108.1	1											
AS108.2	1											
AS108.3	1											
AS108.4	1											
AS108.5	1											
AS108.6	1											

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2	PSO3
AS108.1						
AS108.2						
AS108.3						
AS108.4						
AS108.5						
AS108.6						

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember✔	Understand✔	Apply✔	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Crystallography**	1&2	10
	1.1	Introduction, types of solids, space lattice, basis, unit cell and its characterization, lattice parameters, Bravais space lattices, seven crystal systems		
	1.2	Miller Indices of planes and directions, Interplanar spacing in cubic system		
	1.3	X-rays, Bragg's law, Bragg's X-ray spectrometer and determination of crystal structure		
	1.4	Crystal imperfections: Point and Line defects		
2	Title	**Magnetic properties of materials**	2,3 &4	8
	2.1	Introduction, magnetic moment of an atom, useful magnetic parameters		
	2.2	Classification of magnetic materials		
	2.3	Hysteresis, soft and hard magnetic materials and their applications		
3	Title	**Nanomaterials and their applications**	4	6
	3.1	Introduction, types of nanoparticles (carbon, metals, organic, and composites)		
	3.2	Molecular self assemblies, and Applications		
4	Title	**Modern, Smart and Intelligent Engineering Materials**	2,4	10
	4.1	Introduction, distinguish features of smart materials, Types of smart materials and their various engineering applications		
	4.2	Dielectric materials: polarization, temperature and frequency effects, electric breakdown, ferroelectric materials and its various applications		
	4.3	Metallic glasses and their various engineering applications		
	4.4	Wide-bandgap (WBG) semiconductors: Silicon Carbide, Gallium nitride. Applications of WBG materials		

5	**Title**	**Materials' Characterization Techniques**	4	**8**
	5.1	UV and visible spectroscopy Fluorescence spectroscopy		
	5.2	X-ray photo luminescence spectroscopy, Transmission Electron Microscopy, Scanning Electron Microscopy		
	5.3	Raman effect and Raman spectroscopy		
6	**Self Study**	Superconductivity, Type I and Type II superconductors, Meissener effect, High temperature superconductors, various applications of superconductors.	1,2 & 4	**06**
			Total	**42**

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Electrical Engineering Materials	tenth	A.J. Dekker	Prentice- Hall India	2008
2	Nanotechnology: Principles and practices	third	Sulbha K. Kulkarni	Springer	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Solid State Physics	tenth	S.O.Pillai	New Age International	2022
2	Materials Science and Engineering	sixth	V. Raghavan	PHI	2004
3.	Fundamentals of Materials science and engineering; an integrated approach	sixth	William Callister and Jr. David Rethwisch	John Wiley & Sons.	2021
4.	Engineering Physics	second	B.K. Pandey and S. Chatur	Cengage	2022

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Problem Solving using Imperative Programming SEM-I

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
SEC	Problem Solving using Imperative Programming Lab	0	1	4	4	10	0	1	2	3
		Examination Scheme								
		Component		ISE %		MSE %		ESE %		Total %
CE101		Theory		-		-		-		-
		Laboratory		80		-		20		100

Pre-requisite Course Codes, if any.	
Course Objective: To develop problem solving skills using imperative programming.	
Learning Outcomes (LO): At *the End of the course students will be able to*	
CE101.1	Explain the problem-solving aspects using various programming paradigms.
CE101.2	Solve real world problems using imperative programming approach.
CE101.3	Solve problems using control structures for real world problems.
CE101.4	Solve problems using Arrays and Text processing.
CE101.5	Develop modular code for a given problem.
CE101.6	Solve real world problems using Structures and Unions

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CE101.1	1	-										
CE101.2	2	2										
CE101.3	2	2										
CE101.4	2	2										
CE101.5	2	2										
CE101.6	2	2										

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CE101.1							
CE101.2							
CE101.3							
CE101.4							
CE101.5							
CE101.6							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Problem Solving and Programming Paradigms**		
	1.1	What is a Problem, Characteristics of good Program, Overview of ProgrammingParadigms – Declarative and Imperative, Problem- solving using Algorithm and Flowcharts	3,4	2
2	Title	**Basic Elements of Computer Programming and Control flow**		
	2.1	Variables, keywords, Data types, Operators**:** Arithmetic, Relational and Logical, Assignment, Unary, Conditional, Bitwise, Expression, Statements.	1,2	1
	2.2	Branching Structures: if statement, if-else statement, multi-way decision, switch statement, continue statement, break statement Iterative Structures: while, do-while, for, nested loops	1,2	3
3	Title	**Problem Solving using Array Techniques**		
	3.1	Introduction to Arrays: Declaration, Definition, accessing array elements, one-dimensional array, two-dimensional array, array of characters, Solving using Arrays like Array Order Reversal, Array Counting or Histogramming.	1,2	2
	3.2	Strings and Operations on Strings.	1,2, 4	1
4	Title	**Problem Solving using Modular Approach**		
	4.1	Defining a Function, accessing a Function, Function Prototype, Passing Arguments to a Function, call by value, pointers and call by reference.	1,2	2
	4.2	Recursion	1,2, 4	1
5	Title	**Structures and Unions**		
	5.1	Structures and Union: Declaration, Initialization, structure within structure, Array of Structure, Operation on structures, Concept of Union, Difference between structure and union,	1,2	2
6	Self-Study	File handling**:** Types of Files, File operation-opening, Closing, Creating, Reading, Processing File, Command line arguments, Dynamic Memory Allocation	1,2, 4	**4***
Total (* not counted in total hours)				**14**

Laboratory Component (Minimum 10 Laboratory experiments are expected) Note: All problems should be implemented using C language.

Sr. No.	Title of the Experiment
1	Use the formatted input/output statements, operators and expressions of C language
2	Apply various control structures to solve given problems.
3	Apply the concept of functions to incorporate modularity.
4	Demonstrate the use of one-dimensional arrays to solve a given problem.
5	Demonstrate the use of two-dimensional arrays to solve a given problem.
6	Apply the concept of recursion to solve a given problem.
7	Implement various text processing problems.
8	Apply the concepts of structures/union to solve a given problem.
9	Demonstrate the use of pointers to solve a given problem.
10	Implement various operations on files to solve a given problem.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Programming with C	Fourth	Byron Gottfried	McGraw Hill (Schaum's outline series)	2018
2	The C programming Language	Second	Kernigha, Ritchie	Pearson	2015
3	Foundations of Programming Languages	Second	Kent D. Lee	Springer	2017
4	How to Solve it by Computer	First	R. G. Dromey	Prentice Hall India	1998

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Let Us C	Sixteenth	Yashavant Kanetkar	BPB	2017
2	Programming Language Concepts	Third	Carlo Ghezi, Mehdi Jazayeri	John Wiley & Sons	2008
3	Computer Programming in C	Second	V. Rajaraman & Neeharika Adabala	PHI Learning, Eastern Economy Edition,	2014

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Problem Solving using Object Oriented Programming SEM-II

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
SEC	Problem Solving using Object Oriented Programming	0	1	4	4	8	0	1	2	3
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
CE102		Theory	--		--		--		--	
		Laboratory	80		--		20		100	

Pre-requisite Course Codes, if any.	Problem Solving using Imperative Programming
Course Objective: To learn problem solving using Object-Oriented programming paradigm	
Learning Outcomes (LO):*At the End of the course students will be able to*	
CE102.1	Apply concepts of object oriented programming using classes and objects
CE102.2	Apply Inheritance for a given scenario
CE102.3	Apply polymorphism for solving a given problem
CE102.4	Apply abstraction and exception handling to create efficient programs.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CE102.1	3	2			3							2
CE102.2	3	2			3							2
CE102.3	3	2			3							2
CE102.4	3	2			3							2

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CE102.1							
CE102.2							
CE102.3							
CE102.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory (This course content delivery will be in Java. Course Contents to be taken care accordingly)

Module No.	Unit No.	Topics	Ref.	Hrs.
1		**Introduction and Encapsulation**		**4**
	1.1	Introduction to Object Oriented Programming, Procedural versus Object Oriented Programming, Principles, Benefits and applications of Object-Oriented Programming.	1,2	
	1.2	Encapsulation: Problem solving with Objects and Classes		
	1.3	static data members and methods, constructors and their types. Strings, Arrays		
2		**Inheritance**		**3**
	2.1	Concept of Inheritance, parent class, derived class, this and super keyword	1,2	
	2.2	Types of inheritance: single, multiple, multilevel, hierarchical, hybrid		
3		**Polymorphism**		**3**
	3.1	Static Polymorphism: Method overloading and Constructor overloading	1,2	
	3.2	Dynamic Polymorphism: Method overriding		
	3.3	Data conversion		
4		**Abstraction**		**2**
	4.1	Abstraction: abstract class, interface	1,2	
5		**Exception Handling**		**2**
	5.1	try, throw, and catch exceptions	1,2	
	5.2	Function exception declaration		
6	**Self Study**	File Handling, Multithreading, Packages, Collection Framework, Swings, Aggregation and Composition	1,2	**5***
			Total	**14+ 5***

Laboratory Component

Sr. No.	Title of the Experiment
1	Program on Encapsulation: Write a program to demonstrate classes and objects
2	Program on Encapsulation: Write a program to demonstrate constructor
3	Program on Polymorphism: Implement a Program to demonstrate method overloading,
4	Program on Polymorphism: Implement a Program to demonstrate constructor overloading
5	Program on Polymorphism: Implement a Program to demonstrate method overriding
6	Program on Inheritance: Implement a Program to demonstrate single level Inheritance
7	Program on Inheritance: Implement a Program to demonstrate multilevel Inheritance
8	Program on Abstraction: Implement a Program to demonstrate multiple Inheritance/ interface
9	Program on Abstraction: Implement a Program to demonstrate Abstraction using abstract class
10	Program on abstraction: Implement a Program to demonstrate multithreading

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Java -The Complete Reference	10th	Herbert Schildt	Tata McGraw-Hill	2017
2	Java Programming From the Ground Up	1st	Ralph Bravaco,Shai Simoson	Tata McGraw-Hill	2009

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	An introduction to Programming and Object Oriented Design using Java	3rd	Jaime Nino, Frederick A. Hosch	Wiley Student Edition	2010
2	Head First Java	3rd	Kathy Sierra & Bert Bates	O'Reilly	2022

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Digital Systems

Course(Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(BSES)	Digital Systems	3	0	2	5	10	3	0	1	4
		Examination Scheme								
		Component		ISE %		MSE %		ESE %		Total %
EC101		Theory		15		15		45		75
		Laboratory		15		--		10		25

Pre-requisite Course Codes, if any.	
Course Objective:	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
EC101.1	Explain various logic gates, SOP, POS forms and their minimization with k- map for given combinational circuits.
EC101.2	Construct combinational circuits using given MSI devices.
EC101.3	Apply the knowledge of flip-flops and MSI to design sequential circuits
EC101.4	Compare the logic families based on their characteristics
EC101.5	Comprehend the architectural features of 8085 with basic assembly language programming

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
EC101.1	3	2	1									
EC101.2	3	2	1									
EC101.3	3	2	1									
EC101.4	3	2	1									
EC101.5	3	2	1									

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
EC101.1							
EC101.2							
EC101.3							
EC101.4							
EC101.5							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Combinational Circuits**		11
	1.1	**Logic Gates:** Basic gates, Universal gates, Sum of products and products of sum, minimization with Karnaugh Map (up to four variables), Quine Mc'Clusky method and realization.	1,4	
	1.2	**Combinational Circuits using basic gates as well as MSI devices:** Half adder, Full adder, Half Subtractor, Full Subtractor, Multiplexer, Demultiplexer, Decoder, Comparator	1,4	
2	Title	**Sequential Circuits**		11
	2.1	**Sequential Logic:** Latches and Flip-Flops. Conversions of Flip-Flops.	1,4	
	2.2	**Counters:** Asynchronous Counters, Synchronous Counters, Up Down Counters, Mod Counters, Ring and Twisted Ring Counters , Shift Registers, Universal Shift Register	1,4	
	2.3	MSI counters (IC 7490, IC 74160, IC 74163, IC 74169), MSI Shift registers (IC 74194) and their applications	2,5	
3	Title	**Clocked Synchronous Machines**		05
	3.1	Mealy and Moore Machines, Clocked synchronous state machine analysis, State reduction techniques.	2,5	
4	Title	**Logic Families**		05
	4.1	Types of logic families (TTL and CMOS), characteristic parameters (propagation delays, power dissipation, Noise Margin, Fan-out and Fan-in), transfer characteristics of TTL NAND.	1,4	

5	**Title**	Introduction to Microprocessors		**10**
	5.1	Evolution of computers and Microprocessors	3	
	5.2	Essential components of a conventional Central Processing Unit (CPU)	3	
	5.3	Architecture of 8-bit microprocessor 8085	3	
	5.4	Basic instruction set with its addressing modes and concepts of Instruction cycle, Machine cycle and T states. Elementary programming in assembly language.	3	
	5.5	Elements of I/O data transfer with the concept of interrupts	3	
6	**Self Study**	Concepts of PROM, PAL and PLA. Timing Considerations and Metastability in Flip-Flops. Clocked synchronous state machine design. Interfacing CMOS to TTL and TTL to CMOS. Concepts of peripherals and memory with its interfacing with 8085 microprocessor		
			Total	**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	To implement the combinational logic for given function using basic gates/MSI ICs. a. To study and verify the truth table of logic gates b. To study the universal NAND and NOR gate c. To study the working of half adder, full adder, half subtractor, Full subtractor along with truth table
2	To implement TTL and CMOS logic family a. To study TTL NAND gate (BJT implementation). b. To study CMOS NAND gate c. To study interfacing of the TTL /CMOS
3	To implement 4-bit, 5-bit and 8 bit comparator using given MSI
4	To design implement gate level multiplexers and MSI multiplexers
5	To design and implement gate level and MSI circuits of flip-flops
6	To design counters a. To design a MOD4 synchronous up/down counter b. To study IC 7490 – Asynchronous Decade Counter
7	To synchronous counters, synchronous counters and shift register using given MSI. a. To study IC 74160 as Synchronous Decade Counter and Mod 6 counter b. To study IC 74163 as Synchronous MOD 16 Counter and Mod 10 counter c. To verify the truth table of IC 74194 as Universal Shift Register and implement Ring and Twisted Ring Counter.
8	To perform basic arithmetic operations through assembly language program in 8085
9	To simulate COPY and PASTE operation through 8085 assembly program
10	To write a program to add N elements stored in an array of 8 bit numbers (8085)

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Modern Digital Electronics	Fourth Edition	R. P. Jain	Tata McGraw Hill	**2009**
2	Digital Design Principles And Practices	Third Edition	John F. Wakerly	Pearson Education	**2001**
3	Microprocessor Architecture, Programming, and Applications with the 8085	Sixth Edition	Ramesh S. Gaonkar	Penram International	**2013**

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Digital Design	Fourth Edition	Morris Mano	Pearson Education	**2008**
2	Fundamentals of digital logic design with VHDL	Second Edition	Stephen Brown and Zvonko Vranesic	McGraw Hill	**2006**

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Basic Electrical Engineering

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(BSES)	Basic Electrical Engineering	3	-	2	6	11	3	0	1	4
		Examination Scheme								
		Component		ISE %		MSE %		ESE %		Total %
EC102		Theory		15		15		45		75
		Laboratory		15		--		10		25

Pre-requisite Course Codes, if any.	Basic concepts of electric charge, current, voltage and Power
Course Objective: To impart a basic knowledge of electrical quantities, Circuits and components.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
EC101.1	Compute various electrical quantities of a given dc circuit using circuit simplification techniques and various network theorems.
EC101.2	To expose the students to different terms and concepts in AC Circuits at fundamental frequency and to expose them to basics of effects of harmonics in the waveforms
EC101.3	To study the working principles of electrical machines and their applications
EC101.4	To expose the students the fundamental concepts in Controllable Switch and Modulation based Power Conversion
EC101.5	To study Electrical Parameters of the Batteries and their selection and design criteria for a specific application

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
EC101.1	3	3										
EC101.2	3	3										
EC101.3	3	3										
EC101.4	3	3										
EC101.5	3	1										

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
EC101.1							
EC101.2							
EC101.3							
EC101.4							
EC101.5							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs. (Theory+Tut)
1	**Title**	**DC Circuits**	1,2	**12**
	1.1	Electrical circuit elements (R, L and C), Voltage and current sources, Equivalent resistance of circuits, Simplification using delta-star and star-delta transformation.		
	1.2	Kirchoff's current and voltage laws, Analysis of simple circuits with dc excitation. Mesh analysis, Superposition, Thevenin, Norton and Maximum Power Transfer Theorems		
	1.3	Time-domain analysis of first-order DC Transients in RL and RC circuits.		
2	**Title**	**AC Circuits**	1,2	**12**
	2.1	Representation of sinusoidal waveforms, peak and rms values, phasor representation, real power, reactive power, apparent power, power factor. Harmonics, Total Harmonic Distortion, Power supplied by Harmonic voltages and currents, Power factor in case of non-sinusoidal waveforms.		
	2.2	AC Analysis of series and parallel RLC Circuits with Resonance, Concept of Bandwidth and Q-factor,		
	2.3.	Three-phase balanced circuits, voltage and current relations in star and delta connections.		
3	**Title**	**Electromagnetic and Electro-Mechanical Energy Converters**	1,2	**9**
	3.1	Magnetically Coupled Coils, Self and Mutual Inductance and Dot Convention		
	3.1	Single Phase Transformer: Principle of Operation, Equivalent Circuits		
	3.2	Single Phase Transformer: Efficiency and Regulation		
	3.3	Introduction to Three-phase Transformers and Applications		

4	**Title**	**Electric Power Converters**	1,2	**6**
	4.1	Fundamental Principles of Buck, Boost and buck-boost DC-DC converters and their Transfer Characteristics, Duty Ratio Control		
	4.2	Single-phase voltage source inverters and PWM		
5	**Title**	**Batteries: Electrical Characteristics and Applications**		**3**
	5.1	Introduction to type of Batteries, Generalized Battery parameters such as SoC, DoD, Energy and Power Densities, Battery C-rating, etc. Comparison of Batteries, Charging and Discharging Characteristic		
	5.2	Selection and Sizing of Battery Packs for Specific Applications		
6	**Self-Study**	Fundamental Principles of Rotating Machines, Working Principle,Characteristics and applications of Induction motor, BLDC motor and Stepper Motor		
			Total	**42**

Laboratory Component (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
01	Introduction to Electrical Measuring instruments, Lamp Loads, Inductor Loads and Capacitor Bank
02	Verification of Star-Delta and Delta-star Transformation with Kirchhoff's Laws
03	Verification of Thevenin's Norton's and Maximum Power Transfer Theorem
04	Verification of DC Transient equations in RL and RC Circuits
05	Experimental study of single-phase AC circuit with R-L and R-C Load with Measurement of Power and Power factor
06	Experimental study of R-L-C series Resonance. To plot resonance curve, To compute Bandwidth and Q-factor
07	Experiment on Magnetic Circuit Fundamentals
08	Loading of a transformer: measurement of primary and secondary voltages and currents, and power. To compute efficiency and regulation.
09	Three-phase transformers: Star and Delta connections. Voltage and Current relationships (line-line voltage, phase-to-neutral voltage, line and phase currents). Phase-shifts between the primary and secondary side. Cumulative three-phase power in balanced three-phase circuits.
10	Synchronous speed of two and four-pole, three-phase induction motors. Direction reversal by change of phase-sequence of connections. Torque-Slip Characteristic of an induction motor. Generator operation of an induction machine driven at super-synchronous speed.
11	Demonstration of (a) dc-dc converters (b) dc-ac converters – PWM waveform.
12	Demonstration of V/F control of Induction motor
13	Experimental study of charge and Discharge characteristics of a Lead-acid Battery

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Basic Electrical Engineering	Third	D. P. Kothari and I. J.	Tata McGraw Hill	2010
2	Electrical Technology	Twenty Third	B. L. Theraja	S. Chand Publications	2003

Reference Books

Sr. No.	Title	Edition	Author	Publisher	Year
1	Basic Electrical Engineering	Second	D. C. Kulshreshtha	McGraw Hill	2019
2	Fundamentals of Electrical Engineering	Second	L. S. Bobrow	Oxford University Press	2011
3	Electrical and Electronics Technology	Third	E. Hughes	Pearson	2010
4	Electrical Engineering Fundamentals	Second	V. D. Toro	Prentice Hall India	1989
5	Elements of Power Electronics	Second	P. T. Krein	Oxford University Press	2015
6	Power Electronics: Converters, Application and Design	Second	Ned Mohan, T.M Undelands and W P Robbins	John Wiley and Sons. Inc.	1995
7	Electric Machinery	Sixth	A. E. Fitzgerald, C. Kingsley and S. D. Umans	McGraw-Hill	2003

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Soft Skills-I

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
SEC	Soft Skills (Communication Skills)	0	1	2	2	5	0	1	1	2
		Examination Scheme								
		Component		ISE %		MSE %		ESE		Total
AS107		Theory		--		20		--		20
		Laboratory		80		--		-		80

** MSE will be evaluated on the basis of a written test based on module 1 and 2.
** ISE will be evaluated on the basis of marks scored in practicals, out of 150.

Pre-requisite Course Codes, if any.	
Course Objective: To apply the principles of communication in a personal and professional environment.	
Learning Outcomes (LO): At *the end of the course students will be able to*	
AS107.1	Apply the principles of business writing for professional documents.
AS107.2	Develop advanced vocabulary and grammar for spoken and written communication.
AS107.3	Design the draft a formal speech.
AS107.4	Analyze received information by using active listening and reading skills.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AS107.1									2	2		2
AS107.2									2	2		2
AS107.3									2	2		2
AS107.4									2	2		2

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
AS107.1							
AS107.2							
AS107.3							
AS107.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	L Hrs.	P Hrs
1	**Title**	**Vocabulary Building & Grammar**		**2**	**4**
	1.1	Concept of word formation, the root words from foreign languages and their use in English	7,1		
	1.2	Common errors in writing, confused pair of words, redundancies, clichés	6, 2		
2	**Title**	**Writing Skills**		**7**	**14**
	2.1	Principles of Business Writing: 7Cs of communication, sentence structures, Organizing paragraph in direct and indirect style; Summarization	4		
	2.2	Practices in Writing: E-mail Etiquettes, e-mail for business purposes	3		
	2.3	Critical Reading: understanding the concept of critical reading and applying to analyze a given text.	5		
3	**Title**	**Oral Skills**		**5**	**10**
	3.1	Listening Comprehension(audio): Pronunciation, intonation, Stress and Rhythm	5		
	3.2	Speaking Practices: 1. Common everyday situation: Conversation and dialogues (group activity, ice-breaking session) 2. Public Speaking: Extempore, formal speech	3		
4	**Self-Study**	1. Basic Rules of Grammar 2. GRE Vocabulary 3. Reading a book(fiction/non-fiction) and preparing a review on it		**6***	
			Total (*Not included)	**42 hrs**	

List of activities (Graded, Non-graded)

Sr. No.	Title of the assignments	Marks
1	Skit based on a given situation	-
2	ISE 1 – Summary Writing	10
3	ISE 2 – Extempore	10
4	ISE 3 – Grammar	20
5	ISE 4 – Vocabulary	20
6	Reviewing a book (fiction/ non-fiction)	10
7	ISE 5 – Email Writing (Inquiry)	20
8	ISE 6 – Email Writing (Complaint)	20
9	ISE 6 – Speech	20
10	ISE 7 – Critical Reading	20
	Total	**150**

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Communication Skills	2013	Shirley Mathews	Technical Publication, Pune	2013
2	English Vocabulary in Use	1999	Michael McCarthy , Felicity O'Dell	Cambridge University Press, India	1999

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Oxford Practice Grammar	1999	John Eastwood	Oxford, India	1999
2	Communication Skills	2011	Meenakshi Raman, Sangeeta Sharma	Oxford, India	2011
3	Communication Skills	2010	Dr. Meera Bharwani	Synergy Knowledge ware, India	2010
4	English Grammar for Today	2005	Geoffrey Leech	Palgrave, UK	2005
5	Word Power Made Easy	1978	Norman Lewis	Anchor Books, New York	1978

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Technical Shop

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
SEC	Technical Shop	0	1	2			0	1	1	2
		Examination Scheme								
		Component		ISE %		MSE %		ESE %		Total %
AS106		Theory				--		–		–
		Laboratory		80		--		20		100

Pre-requisite Course Codes, if any.	
Course Objective: To provide essential modern skills that enable today's engineers to create our tomorrow.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
AS106.1	Assemble and disassemble computer hardware for PCs and laptops.
AS106.2	Create and troubleshoot basic computer networks.
AS106.3	Create and troubleshoot basic electrical networks.
AS106.4	Identify and troubleshoot electrical gadgets.
AS106.5	Model any object in 3d software and prepare it for 3d printing.
AS106.6	Design and fabricate PCBs.
AS106.7	Construct physical structures using acrylic and PVC materials.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AS106.1					3				3			3
AS106.2					3				3			3
AS106.3					3				3			3
AS106.4					3				3			3
AS106.5			3		3				3			3
AS106.6			3		3				3			3
AS106.7			3		3				3			3

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2
AS106.1	**2**					
AS106.2	**2**					
AS106.3	**2**					
AS106.4	**2**					
AS106.5	**2**					
AS106.6	**2**					
AS106.7	**2**					

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Lecture Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Unit1**	**Introduction to skill shop.**		
	1.1	The need for skill shop. The utility of the contents covered. Special significance to Computer and EXTC students.		**1**
2	**Unit2**	**3D modeling - ideation and creation.**		
	2.1	Introduction to the various 3d software and their specialization. AutoCAD: brief 2d toolset, establishing 3d workspaces, 2d to 3d tools, primitive based design, sections, lofting and extrusion, revolving and other solid operations, dimensionally accurate design. Blender (open source): differences in approach compared to AutoCAD, solid and surface based modeling, modifiers and symmetric modeling, advanced mesh-based and sculpting techniques.		**6**
3	**Unit3**	**3d printing - prototyping and fabrication.**		**4**
	3.1	Processing 3d models for slicing, slicing models for 3d printing, slicing software (open source). File formats and conventions, watertighting. The file pipeline from .dwg/.blend to .stl to .gcode. File optimization. 3d printing - the process and the mechanisms. FDM (fused deposition modeling) based printing, printing materials and variations, best practices, troubleshooting, alternative, non-machine based 3d printing techniques using pens.		
4	**Unit4**	**Specialized tool sets for manufacturing**		**1**
	4.1	Autodesk Inventor/Fusion 360/ANSYS		
			Total	**12**

Laboratory Component: comprises building and assembling various trade jobs such as:

Trade job 1: Computer Assembly (4 hours)

- Assemble all components of a computer from an OFF and disassembled state, installing CPU, RAM, hard drive, connecting data and power pins, turn it on, install an operating system.
- Additional training imparted during/for this trade:
  - Identification of components such as power supply, motherboard, processor, hard disk drive, RAM etc.
  - Assembly, disassembly, and component identification of a laptop.
  - Basic troubleshooting.

Trade job 2: Network Switch Installation (2 hours)

- Identify various components of a network assembly.
- Crimp LAN cables using crimping tools and patching them into a 10 port LAN switch and testing operation of the network.
- Adopt best practices in LAN network assembly.

Trade job 3: Electrical Networks (2 hours)

- Assemble different types of networks such as:
  - House wiring
  - Staircase wiring
  - Godown wiring
  - 3-phase wiring
  - Study wiring diagram
- Test, trace and troubleshoot these networks with tools such as multimeters

Trade job 4: Electrical Gadgets (2 hours)

- Repairing electrical household appliances: ceiling fan, electric iron.

Trade job 5: 3D Modeling and Printing (2 hours)

- Using the skills learned in the 3d sessions to develop a CAD file (.iges/.step/.dwg) of a 3D model and exporting it as an .stl file for the purpose of 3D printing.
- Importing the 3D .stl file in slicers to generate a .gcode file for 3D printing through slicing, using open source software

Trade job 6: Make a House (8 hours)

- Design a PCB using software such as Eagle for a simple miniature house with various points for patching LED lights.
- Fabricate the PCB using various tools.
  - Additional skills learned: soldering and desoldering, layout drawing, positive and negative film making, PCB etching and drilling, tinning.
- Fabricate an acrylic house using tools such as cutters, hacksaws, files, glue guns, sand paper, etc.
  - Additional skills learned: drilling holes at precise locations for installing LED lights using drill machines.
- Install a PVC fence around the house using PVC formed in L shaped joints at the corners.
- Create various additional models around the house - street light, chimneys and lights, etc.
- Connect the PCB to the electrical network and test proper operation of the network.

- Additional skills learned: integration of multiple design and fabrication techniques to create a finished product.

Trade job 7: Make a Product
Students may select one of the following jobs to create into a finished product. They are then required to either successfully sell the product or purchase it themselves to recover the cost of materials, resulting in a reduction in generated waste. The choices are:

- Keychain
- Simple extension board
- Liquid Level Sensor
- Water Level Controller
- Voice Recorded Message (Jan-Gan-Man)
- Audio Amplifier using IC386
- Lead Acid Battery Charger (6v/12v)
- LM723 IC power supply AI, 0-30v
- LED chaser

Trade job 8: Modern Plumbing
To better understand modern plumbing materials and tools, fabricate a simple stool made of PVC material, including cutting pipes into half pipes, L-shaped joints, and test it for sturdiness.

The above trades will be carried out in two labs in parallel per batch as follows:

Lab 1 (Room 202)	Lab 2 (Prayas Idealab)
Trade job 1: Computer Assembly **Trade job 2: Network Switch Installation** **Trade job 3: Electrical Networks** **Trade job 4: Electrical Gadgets**	**Trade job 5: 3D Modeling and Printing** **Trade job 6: Make a House** **Trade job 7: Make a Product** **Trade job 8: Modern Plumbing**

IDEALab Demonstrations: (4 hours)
Demonstrations on various IDEALab equipment, such as:

- CO2 Laser Cutter and Engraver
- Mini Lathe with Drilling
- Sewing Machine
- Vinyl Cutter and Printer (for making banners, posters, stickers, t-shirt printing)
- 3D scanner
- All-in-one PCB maker (Voltera v1) - PCB printing, heating, drilling and soldering
- PCB fabrication - reflow oven
- Computer desktop based drilling, milling and engraving machine - Roland SRM20
- Welding machines (Spot, ARC etc)
- Test and measuring equipments
- Mechanical fabrication (Wood cutter etc)

Course Evaluation:
The course is evaluated via:

- End-of-semester quiz exam. (20 marks)
- Fabrication and submission of finished jobs in all 6 included trades. (80 marks)

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Munshi Nagar, Andheri (West), Mumbai-400058

Indian Knowledge System (IKS)

Indian Knowledge System (IKS) holds a significant place in your curriculum, as recommended by NEP 2020. The Government of India has mandated a compulsory credit generic IKS course to spread the rich heritage and traditional knowledge of our country in various fields such as Arts and Literature, Agriculture, Basic Sciences, Engineering & Technology, Architecture, Management, Economics, and more. In line with this, the Government of Maharashtra has recommended a generic IKS course for UG students in the first year of your graduation.

Introduction:

The Indian knowledge system is a rich and diverse heritage of intellectual, philosophical, and scientific traditions that have evolved over thousands of years in the Indian subcontinent. It encompasses a wide range of disciplines, including philosophy, spirituality, medicine, mathematics, astronomy, and more. The Indian knowledge system reflects a deep reverence for wisdom, a holistic approach to life and well-being, and a profound understanding of the interconnectedness of various aspects of human existence. It remains a vibrant and influential part of India's cultural and intellectual heritage.

The Indian knowledge system has had a profound and lasting influence not only on the subcontinent but also on the broader world. Its teachings and practices continue to be studied, adapted, and appreciated globally, making it an integral part of human heritage.

<table>
<tr><th rowspan="2">Course (Category) Code</th><th rowspan="2">Course Name</th><th colspan="5">Teaching Scheme (Hrs/week)</th><th colspan="4">Credits Assigned</th></tr>
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<tr><td rowspan="4">AEC</td><td rowspan="4">Indian Knowledge System (IKS)</td><td>1</td><td>1</td><td>-</td><td>2</td><td>3</td><td>1</td><td>1</td><td>-</td><td>2</td></tr>
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** ISE will be evaluated on the basis of individual and group activities.
** ESE will be evaluated on overall modules of the course.

Pre-requisite Course Codes, if any.
Course Objectives: ● To create awareness amongst the students about the true history and rich culture of the country. ● To understand the scientific value of the traditional knowledge of India. ● To explain the key features of the Indian Numeral System and appreciate the key role it has played in the advancement of Science & Technology. ● To develop familiarity with the science, engineering & technology heritage of ancient and medieval India

Learning Outcomes (LO): At *the end of the course students will be able to*	
AEC01.1	Develop a deep understanding of the major philosophical traditions in India.
AEC01.2	Appreciate Indian mathematics and mathematicians along with number systems.
AEC01.3	Understand the interdisciplinary nature of Indian knowledge systems and their connections to science, technology, and other fields.
AEC01.4	Establish familiarity with the classical languages and linguistics of India and exposure to key texts.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
AEC01.1									**2**	**2**		**2**
AEC01.2									**2**	**2**		**2**
AEC01.3									**2**	**2**		**2**
AEC01.4									**2**	**2**		**2**

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	**PEO2**	**PEO3**	**PEO4**	**PSO1**	**PSO2**	**PSO3**
AEC01.1							
AEC01.2							
AEC01.3							
AEC01.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Create**

Theory Component

Module No.	Unit No.	Topics	Ref.	L (Hrs.)	P (Hrs.)
1	Title	**Indian Knowledge System – An Introduction**		1	
	1.1	What is IKS? Why do we need IKS?			
	1.2	Organization of IKS Historicity of IKS Some salient aspects of IKS			
2	Title	**The Vedic Corpus:**		2	3
	2.1	Introduction to Vedas A synopsis of the four Vedas Sub-classification of Vedas Messages in Vedas			
	2.2	Introduction to Vedāṅgas Prologue on Śikṣā and Vyākaraṇa Basics of Nirukta and Chandas			
	2.3	Introduction to Kalpa and Jyotiṣa Vedic Life: A Distinctive Features			
3	Title	**Number Systems and Units of Measurement**		3	2
	3.1	Number systems in India - Historical evidence Salient aspects of Indian Mathematics Bhūta-Saṃkhyā system			
	3.2	Kaṭapayādi system Measurements for time, distance, and weight 1. Piṅgala and the Binary system			
4	Title	**Mathematics**		3	2
	4.1	Introduction to Indian Mathematics Unique aspects of Indian Mathematics Indian Mathematicians and their Contributions			
	4.2	Algebra Geometry Trigonometry			
	4.3	Binary mathematics and combinatorial problems in Chandaḥ Śāstra Magic squares in India			
5	Title	**Linguistics**		3	3
	5.1	Introduction to Linguistics Role of Sanskrit in natural language processing Aṣṭādhyāyī Phonetics			
	5.2	Word generation Computational aspects Mnemonics			
	5.3	Recursive operations Rule based operations Sentence formation Verbs and prefixes			
			Total (*Not included)	**12 hrs**	

List of activities (Graded, Non-graded)

Sr. No.	Title of the assignments	Marks
1	ISE 1 – Group Discussion	30
2	ISE 2 – Presentation	30
3	ISE 3 – Assignment (Group or Individual)	20
5	ESE	20
	Total	**100**

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	IKS: The Knowledge System of Bhārata	2023	Bhag Chand Chauhan	Garuda Prakashan	2023
2	Introduction to Indian Knowledge System: Concepts and Applications	2022	B.Mahadevan (Author), Nagendra Pavana (Author), Vinayak Rajat Bhat (Author)	PHI Learning	2022

Reference Books

1. Raha, Sibaji, et al. Histrory of Science in India Volume-1, Part-I, Part-II, Volume VIII. National Academy of Sciences, India and The Ramkrishan Mission Institute of Culture, Kolkata (2014).
2. Kohle, Pradeep et al (edited by). Pride of India- A Glimpse of India's Scientific Heritage. Samskrit Bharati (2006).
3. Verma, Keshav Dev. Vedic Physics, Motilal Banarsidass Publishers (2012).
4. Soni, Suresh. India's Glorious Scientific Tradition, Ocean Books Pvt. Ltd. (2010).
5. Kapur K and Singh A.K (Eds) 2005). Indian Knowledge Systems, Vol. 1. Indian Institute of Advanced Study, Shimla. Tatvabodh of sankaracharya, Central chinmay mission trust, Bombay, 1995.
6. Nair, Shantha N. Echoes of Ancient Indian Wisdom. New Delhi: Hindology Books, 2008.
7. Kulkarni, R P, Glimpese of Indian Engineering and Technology (Ancient & Medieval period, Munshiram Manoharlal Publishers Pvt. Ltd. 2018.
8. Pathak, AK, Science and Technology in India, Anshika prakashan pratapgarh, 2016.
9. NVP, Unithiri, Indian Scientific Traditions (Professor K.N. Neelakantan Elayath Felicitation Volume), publication division univeristy of Calicut, 2006.
10. Biswal, S, B L ray, Vedic Science and technology, DK Print world, 2009.
11. Yadav, BS, Ancient Indian Leaps into Mathematics, Brikausher publication, 2010.

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus,
Munshi Nagar, Andheri (West), Mumbai-400058

Universal Human Values

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
AEC	Universal Human Values (UHV)	1	1	-	2	3	1	1	-	2
		Examination Scheme								
		Component		ISE %		MSE %		ESE %		Total %
AEC02		Theory		--		--		20		20
		Laboratory		80		--		--		80

** ISE will be evaluated on the basis of individual and group activities.
** ESE will be evaluated on overall modules of the course.

Pre-requisite Course Codes, if any.	
Course Objectives: ● To develop a holistic perspective based on self-exploration, family, society and nature/existence. ● To understand (or developing clarity) of the harmony in the human being, family, society and nature/existence. ● To strengthen self-reflection. ● To develop commitment and courage to act.	
Learning Outcomes (LO): At *the end of the course students will be able to*	
AEC02.1	Demonstrate a comprehensive understanding of what human values are and their significance in individual and societal contexts.
AEC02.2	Develop ethical awareness and the ability to critically evaluate moral and ethical dilemmas in various aspects of life.
AEC02.3	Foster a sense of social responsibility and awareness of the global community, including environmental and social justice issues.
AEC02.4	Promote interdisciplinary learning by exploring how values and ethics intersect with various academic disciplines and real-world challenges.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AEC02.1								2	2	2		2
AEC02.2								2	2	2		2
AEC02.3								2	2	2		2
AEC02.4								2	2	2		2

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
AEC02.1							
AEC02.2							
AEC02.3							
AEC02.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand ✔	Apply ✔	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	L Hrs.	P Hrs
1	**Title**	**Course Introduction - Need, Basic Guidelines, Content and Process for Value Education**		**2**	**2**
	1.1	1. Purpose and motivation for the course, recapitulation from Universal Human Values-I 2. Self-Exploration–what is it? - Its content and process; 'Natural Acceptance' and Experiential Validation- as the process for self-exploration 3. Continuous Happiness and Prosperity- A look at basic Human Aspirations			
	1.2	1. Right understanding, Relationship and Physical Facility- the basic requirements for fulfilment of aspirations of every human being with their correct priority 2. Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario 3. Method to fulfil the above human aspirations: understanding and living in harmony at various levels.			
2	**Title**	**Understanding Harmony in the Human Being - Harmony in Myself!**		**2**	**2**
	2.1	1. Understanding human being as a co-existence of the sentient 'I' and the material 'Body' 2. Understanding the needs of Self ('I') and 'Body' - happiness and physical facility 3. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer)			
	2.2	1. Understanding the characteristics and activities of 'I' and harmony in 'I' 2. Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail 3. Programs to ensure Sanyam and Health.			
3	**Title**	**Understanding Harmony in the Family and Society-**		**3**	**2**

		Harmony in Human-Human Relationship			
	3.1	1. Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship 2. Understanding the meaning of Trust; Difference between intention and competence 3. Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship			
	3.2	1. Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals 2. Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family. Include practice sessions to reflect on relationships in family, hostel and institute as extended family, real life examples, teacher-student relationship, goal of education etc. Gratitude as a universal value in relationships. Discuss with scenarios. Elicit examples from students' lives			
4	**Title**	**Understanding Harmony in the Nature and Existence - Whole existence as Coexistence**		**2**	**2**
	4.1	1. Understanding the harmony in the Nature 2. Interconnectedness and mutual fulfillment among the four orders of nature recyclability and self regulation in nature			
	4.2	1. Understanding Existence as Co-existence of mutually interacting units in all pervasive space 2. Holistic perception of harmony at all levels of existence. Include practice sessions to discuss human being as cause of imbalance in nature (film "Home" can be used), pollution, depletion of resources and role of technology etc.			
5	**Title**	**Implications of the above Holistic Understanding of Harmony on Professional Ethics**		**3**	**2**
	5.1	1. Natural acceptance of human values 2. Definitiveness of Ethical Human Conduct 3. Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order			
	5.2	1. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. 2. Case studies of typical holistic technologies, management models and production systems			

	5.3	1. Strategy for transition from the present state to Universal Human Order: a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers b. At the level of society: as mutually enriching institutions and organizations 2. Sum up. Include practice Exercises and Case Studies will be taken up in Practice (tutorial) Sessions eg. To discuss the conduct as an engineer or scientist etc.			
			Total	**12 hrs**	

List of activities (Graded, Non-graded)

Sr. No.	Title of the assignments	Marks
1	ISE 1 – Group Discussion	30
2	ISE 2 – Presentation	30
3	ISE 3 – Assignment (Group or Individual)	20
5	ESE	20
	Total	**100**

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Human Values and Professional Ethics	2010	R R Gaur, R Sangal, G P Bagaria	Excel Books	2010

Reference Books

1. Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.

2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.

3. The Story of Stuff (Book).

4. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi

5. Small is Beautiful - E. F Schumacher.

6. Slow is Beautiful - Cecile Andrews

7. Economy of Permanence - J C Kumarappa

8. Bharat Mein Angreji Raj - PanditSunderlal

9. Rediscovering India - by Dharampal

10. Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi

11. India Wins Freedom - Maulana Abdul Kalam Azad

12. Vivekananda - Romain Rolland (English)

13. Gandhi - Romain Rolland (English

SEMESTER III

SEM III										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic &Engg. Sciences	BSES	CS201	Discrete Structures and Graph Theory	3	0	0	5	9	3
2	Basic & Engg. Sciences *	FOM-I	MA202	Foundation of Mathematics-I*	2	1	0	0	3	3
3	Skill enhancement course	SEC	AS201	Soft Skill II--Professional Communication Skills	1	0	2	2	5	2
4	Basic &Engg. Sciences Elective	BSESE	AS101 AS102 AS103 AS104 AS105	**Course III**						3
				Engineering Physics/	2	0	2	4	8	
				Engineering Chemistry/	2	0	2	3	7	
				Biology for Engineers/	3	0	0	3	6	
				Engineering Graphics/	1	0	2+2	2	7	
				Engineering Mechanics	2	0	2	4	8	
5	Humanities	HSSM-I	HS2XX	**Course I**	2	0	0	3	5	2
7	Program Core Courses (12)	PCC	CS202	Data Structures	3	0	2	4	9	4
8		PCC	CS203	Computer Architecture and Organization	3	0	2	4	9	4
9		PCC	CS204	Database Management Systems	3	0	2	4	9	4
10	Cocurricular Courses	CC (LLC)	LLCXX	LLC--III	1	0	0	1	2	1
				Total	**18**	**0**	**10**	**27**	**57**	**23**

**Only for Lateral Entry Students*

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Discrete Structures and Graph Theory	3	0	0	4	7	3	0	0	3
CS201		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	75		75		150		300	
		Laboratory	--		--		--		--	

Pre-requisite Course Codes, if any.	
Course Objective: To teach students how to think logically and mathematically. It provides the mathematical foundation that is used in most areas of computer science.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
CS201.1	Solve problems using set theory, logic and its various proof techniques.
CS201.2	Apply the concepts of relations, functions, lattices and recurrence relations to solve problems
CS201.3	Apply the concepts of graphs, trees and their various types with their traversing techniques to solve problems.
CS201.4	Apply the basics of coding theory and cryptography to solve real world problems.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS201.1	3											
CS201.2	3	2										
CS201.3	3											
CS201.4	3	2	2									2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS201.1							
CS201.2							
CS201.3							
CS201.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply ✓	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Set Theory, Logic and Proofs**	1, 2	
	1.1	Finite and infinite set, Union, Intersection, Disjoint, and Difference of two sets. Power Set, Partition of Sets, Ordered Sets, De Morgan's Laws, Principle of Inclusion Exclusion		2
	1.2	Predicates, Propositions, Conditional Propositions, Logical Connectivity, Proposition calculus, Universal and Existential Quantifiers, First order logic, Equivalence, Normal Forms, Introduction to proofs, Mathematical Induction, Strong Induction, Well-ordering principle, Logical inference		6
2	Title	**Relations, Functions and Lattices**	1, 2	
	2.1	Product Sets and Partitions, Paths in relations and Diagraphs, Properties of Relations, Closure of Relation, Equivalence Relations, Operations on Relations, Warshall's Algorithm, Partially Ordered Sets, External Elements of Partially Ordered Sets, Hasse Diagram		8
	2.2	Composition of Functions, Invertible Functions, Recursive Functions, Hashing, Pigeon hole Principle, Extended PHP		3
	2.3	Lattice, Sub lattice, Isomorphic Lattices, Properties of Lattice, Special Types of Lattices		4
	2.4	Recurrence Relations – Introduction, Linear Recurrence Relations with constant coefficients, Homogeneous solutions, Particular Solutions, Total Solutions, Solution by the method of Generating functions, solving Recurrence Relations		5
3	Title	**Graph Theory**	1, 2	4
	3.1	Concepts and terminologies, Graphs as Model (Konigsberg Bridge Problem)		
	3.2	Matrices, Isomorphism, Bipartite Graphs, Directed Graphs		
	3.3	Minimal Spanning Trees-Prim's Algorithm and Kruskal's Algorithm		
4	Title	**Graph connectivity**	1, 2	6
	4.1	Cycles – Transport Networks, Max Flows, Matching Problems, Maximum Bipartite Matching, Perfect Matching		
	4.2	Euler Paths- Circuits, Hamiltonian Paths- Circuits		
	4.3	Coloring Graphs, Chromatic Polynomial, Planer Graphs		
5	Title	**Coding Theory**	1, 2	4
	5.1	Hamming Code, Minimum Distance		
	5.2	Number Theory, Modular Arithmetic and applications to cryptography; Diffie-Hellman Algorithm		
6	Self-Study	Algebraic Structures - Semi group, Monoids, Groups, Cyclic groups, Abelian groups, Normal Subgroups	1, 2	5*
			Total (*Not included)	**42**

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Discrete Mathematics and it's applications	Eight	Kenneth H. Rosen	Tata McGraw- Hill	2021
2	Discrete Mathematical Structures	Sixth	Bernad Kolman, Robert Busby,Sharon Cutler Ross, Nadeem-ur-Rehman	Pearson Education	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Elements of Discrete Mathematics	Fourth	C. L. Liu	Tata McGraw- Hill	2012
2	Introduction to graph Theory	Second	Douglas B. West	Pearson Education	2015
3	Discrete Mathematical Structures with Applications to Computer Science	First	Jean-Paul, Tremblay R. Manohar	Tata McGraw- Hill	1987

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Data Structures	3	0	2	5	10	3	0	1	4
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
CS202		Theory	20		30		100		150	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	Problem solving using imperative programming
Course Objective: To introduce the fundamentals and abstract concepts of Data Structures for Problem Solving.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
CS202.1	Apply various operations of linear and non-linear data structures to given problems.
CS202.2	Apply the concepts of Trees and Graphs to a given problem.
CS202.3	Apply various operations of heap data structures.
CS202.4	Apply the concepts of hashing on a given problem.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS202.1	3		3						3	3	3	
CS202.2	3		3						3	3	3	
CS202.3	3		3						3	3	3	
CS202.4	3		3						3	3	3	

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS202.1			1			1	
CS202.2			1			1	
CS202.3			1			1	
CS202.4			1			1	

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply ✓	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Introduction to Data Structures**		
	1.1	Concept of Linear and Nonlinear Data Structures	1,2	**1**
	1.2	Stack: Stack as ADT, operations on stack, Applications of Stacks	1,2	**3**
	1.3	Queue: Queue as ADT, Operations on Queue, Applications of Queue, Types of Queue-Circular and Priority Queue	1,2	**3**
	1.4	Linked List: Linked List as ADT, Operations on Singly Linked List. Types of linked list- Linear and circular linked lists, Doubly Linked List, Circular Linked List and its operations, Applications of linked List	1,2	**6**
2	**Title**	**Trees**		
	2.1	Trees as ADT, General tree v/s Binary Tree Terminology, Traversal of Binary Tree, Operations on Binary tree, Binary Search Tree and its operations, Expression Tree	1,2	**4**
	2.2	Self-balancing trees: AVL trees Properties, Rotations, Insertion, Deletion	1,2	**4**
	2.3	Introduction to B tree- Insertion, Deletion, Introduction to Tries (Prefix Tree) - Operations and Applications	1,2	**5**
3	**Title**	**Graphs**		
	3.1	Graph as ADT, Introduction to Graph, Representation of GraphAdjacency Matrix, Adjacency List, Graph Traversal Technique	1,2	**3**
4	**Title**	**Heap Structure**		
	4.1	Heap as ADT, Introduction to Heap Structures, Min Heap, Max Heap, Construction of Heap	1,2	**4**
	4.2	Binomial Heaps, Fibonacci heaps - Structure of Fibonacci heaps, Mergeable-heap, operations, decreasing a key and deleting a node	1,2	**5**
5	**Title**	**Hashing**		
	5.1	Introduction to Hash Table, Hash functions, Collision Resolution Techniques	1,2	**4**
6	**Self-Study**	Optimal Binary Search Tree, Probabilistic Data Structures, B+ trees, multi-dimensional trees (k-d trees, quadtrees)	**1,2**	**5***
			Total	**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No	Title of the Experiment
1	Implement a given problem statement using Stack
2	Implement a given problem statement using Queue
3	Implement a given problem statement using Linked List
4	Implement a given problem statement using Doubly Linked List
5	Implement a given problem statement using Binary Trees or Tries
6	Implementation of Expression Tree
7	Implement insertion of node in AVL tree
8	Apply Graph Traversal Technique on a given problem statement to solve the problem
9	Implement Operations of Heap Structures
10	Implement hash functions with different collision resolution techniques

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Algorithms	Third	Thomas H. Cormen, Charles E. Leiserson, Ronald L Rivest, Clifford Stein	MIT Press	2009
2	Fundamentals of Computer Algorithms	Second	Horowitz E, Sahni S and S. Rajasekaran	Galgotia Publications	2010
3	Data Structures and Algorithm Analysis in C++	Fourth	Mark Allen Weiss	Pearson	2014
4	Probabilistic Data Structures and Algorithms for Big Data Applications	First	Andrii Gakhov	BoD GmbH, Germany	2022

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Classic Data Structures	Second	Samanta Debasis	PHI	2009
2	Data Structures with C	First	Seymour Lipschutz	Schaum's Outline Series	2010
3	Data Structures and Algorithms in Python	First	Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser	Wiley	2013
4	The Algorithm Design Manual	Second	Steven S. Skiena	Springer	2008

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Computer Architecture & Organization	3	0	--	4	3	3	0	--	3
CS203		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	20		20		60		100	
		Laboratory	--		--		--		--	

Pre-requisite Course Codes, if any.	Digital Systems
Course Objective: Imparting concepts of each component of computer architecture thoroughly with practical aspects including memory systems and I/O communications with interfacing	
Learning Outcomes (LO): *At the End of the course students will be able to*	
LO.1	Conceptualize basic computer structure with its models and compute performance metrics.
LO.2	Design algorithms to solve ALU operations
LO.3	Comprehend processor organization with various design methods of CPU with comparative analysis
LO.4	Design memory systems with analysis of mapping techniques for cache and virtual memory
LO.5	Comprehend different types of I/O buses, compare and contrast different types of data transfer methods and arbitration techniques
LO.6	Analyze different parallel organizations that includes pipelined and parallel processors

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
LO.1	3				2							
LO.2	3	3										
LO.3	3	2	2		2							
LO.4	3											
LO.5		2	2		2							

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2
LO.1	2	2	-	-	-	-
LO.2	2	2	-	-	-	-
LO.3	2	2	-	-	-	-
LO.4	2	2	-	-	-	-
LO.5	2	2	-	-	-	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✓	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Overview of Computer Architecture and Organization**		5
	1.1	Introduction of Computer Organization and Architecture, Basic organization of computer and block level description of the functional units, Evolution of x86 Computers, Von Neumann model, Harvard Model, Embedded system	1,2	
	1.2	Performance Issues: Designing for performance, Amdahl's Law, Multi-core, GPGPU	1,2	
2	Title	**ALU, Processor Organization and Control Unit Design**		10
	2.1	ALU: Integer and Floating Point Operation CPU Architecture, Register Organization , Instruction formats, basic instruction cycle. Instruction interpretation and sequencing, Case Study of 80386 architecture and Register Organization	2,3	
	2.2	Control Unit: Soft wired (Micro-programmed) and hardwired control unit design methods. Microinstruction sequencing and execution. Micro operations	2,3	
	2.3	RISC and CISC: Introduction to RISC and CISC architectures and design issues.	2,3	
3	Title	**Memory Systems Organization**		12
	3.1	Introduction to Memory and Memory parameters. Classifications of primary and secondary memories. Types of RAM and ROM, Allocation policies, Memory hierarchy and characteristics.	2	
	3.2	Cache memory: Concept, architecture (L1, L2, L3), mapping techniques. Cache Coherency, Interleaved and Associative memory. Case study of Pentium Processor Cache Memory Model (MESI Protocol)	2,4	
	3.3	Virtual Memory: Concept, Segmentation and Paging, Page replacement policies. Case study of 80386 Virtual Memory Concepts	2,3	
4	Title	**I/O Organization**		5
	4.1	Buses: Types of Buses, Bus Arbitration, BUS standards	1,2	
	4.2	I/O Interface, I/O channels, I/O modules and IO processor, Types of data transfer techniques: Programmed I/O, Interrupt driven I/O and DMA.	1,2	
5	Title	**Parallel Processing**		11
	5.1	Advanced Processor Models: Real Model, Protected Model, Virtual Model (x86 Processors)	3	
	5.2	Superscalar Architecture: Case study of Pentium processor	4	
	5.3	Pipelined Architecture: Pipleine Stages, Pipeline Hazards, Mitigation of Hazards with branch prediction and data forwarding techniques	1,2,4	
	5.4	Introduction to parallel processing concepts, Flynn's classifications,	2	
6	Self-Study	Comparative Study of microprocessors and micro architectures with respect to their important features. Detailed analysis of Multicore and GPGPU Architectures. Vector and Array Processors with VLIW architecture. 8086 instructions set with assembler directives		
			Total	**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)
[Only for CE/CSE]

Sr. No.	Title of the Experiment
1	Installation and configure: DOS, MASM, Debug and X86 Mode
2	Implementation of various arithmetic operations through assembly language programming for 8086 using MASM and Debug.
3	Implement various String Operations in 8086 through the utilities provided by DOS and BIOS interrupts (MASM)
4	Block Transfer and Block Exchange using Index Registers
5	Drawing basic shapes like triangle, etc. using BIOS services [Use C/MASM]
6	Design Password Detection Application using BIOS and DOS interrupts along with 8086 instructions.
7	Implement file operations [DOS Interrupts in C/MASM]
8	Implement I/O interfacing using inbuilt speakers of IBM PC
9	Implement Booth's Multiplication Algorithm
10	Implement Division Algorithm (Non-Restoring and Restoring)
11	Implementation of Mapping techniques of Cache memory
12	Implementation of Page Replacement Policies

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Computer Organization	Fifth	Carl Hamacher, Zvonko Vranesic and Safwat Zaky	Tata McGraw-Hill	2002
2	Computer Organization and Architecture: Designing for Performance	Eighth	William Stallings	Pearson	2010
3	The 80386, 80486, and Pentium Microprocessor: Hardware, Software, and Interfacing	Third	Walter Triebel	Pearson	1997
4	Pentium Pro Processor System Architecture	Third	Tom Shanely	Addison Wesley	1996

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Structured Computer Organization	Sixth	Andrew S. Tanenbaum	Pearson	2013
2	Microprocessor and Interfacing: Programming & Hardware	Third	Douglas V Hall	Tata-McGraw Hill	2012
3	Computer Architecture and Organization: Design Principles and Applications	Second	B. Govindarajulu	McGraw Hill	Paperback-2017
4	Advance Computer Architecture: Parallelism, Scalability, Programmability	Third	Kai Hwang	Tata-McGraw Hill	2017
5	Programmer's reference Manual for IBM Personal Computers	First	Steven Armburst	Tata-McGraw Hill	1986

Bharatiya Vidya Bhavan's

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Database Management systems	3	0	2	5	10	3	0	1	4
CS204		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	75		75		150			300
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	-
Course Objective: To efficiently and effectively Design, develop, maintain and retrieve the Information from DBMS.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
CS204.1	Demonstrate understanding of a given system to construct a database model.
CS204.2	Apply various Relational and SQL commands on the populated database.
CS204.3	Examine the functional dependencies to make a normalized database system.
CS204.4	Examine transaction processing techniques on a database.
CS204.5	Illustrate indexing, query processing and optimization methods on a database.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS204.1	1	2	2	2	2	1		1	2			2
CS204.2	2	2	2		2							
CS204.3	1	2	2		2							
CS204.4	1	2			2							
CS204.5	2	2										

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS204.1					3	3	
CS204.2							
CS204.3							
CS204.4							
CS204.5						2	

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✓	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction: Database Concepts and ER Modeling**	1,2	08
	1.1	Introduction to basic concept of Database, Characteristics of databases, File system V/s Database system, Users of Database system, Database Administrator, Data Independence, Codd's Rule, DBMS system architecture.		
	1.2	Introduction to ER model, Benefits of Data Modeling, Types of data Models, Phases of Database Modeling, The Entity-Relationship (ER) Model, Extended Entity-Relationship (EER) Model		
2	Title	**Relational Algebra and SQL**	1,2	15
	2.1	Introduction, Mapping the ER and EER Model to the Relational Model, Relational Algebra: Overview, Basic Operators, Extended Operators.		
	2.2	Overview of SQL, Data Definition Commands, Data Manipulation commands, Data Control commands, Set operations, aggregate function, null values, Views in SQL, Subquery, Trigger, stored procedure		
3	Title	**Normalization**	1,2	06
	3.1	Design guidelines for relational schema, Functional dependencies		
	3.2	Normal Forms- 1NF, 2 NF, 3NF, BCNF and 4NF,5NF		

4	**Title**	**Transaction Processing and Recovery**	1,2	**07**
	4.1	Transaction concept, Transaction states, ACID properties, Implementation of atomicity and durability, Concurrent Executions, Serializability, Recoverability, Lock-based, Timestamp-based, Validation-based protocols.		
	4.2	Shadow paging, Deadlock handling.		
5	**Title**	**Indexing and Query Optimization:**	1,2	**06**
	5.1	Indexing techniques: B-trees, Hashing.		
	5.2	Query processing and optimization, Query execution plans.		
6	**Self-Study**	1. NO SQL-Data type, Database creation, Basic command for creation, updating and querying the database, Mongo dB, 2. Real-world database applications and case studies	1,2	**5***
			Total (*Not included)	**42**

Laboratory Component:
Assign a case study for a group of 2/3 students and each group will perform the following experiments on the case study.

Exp. No.	Title of the Experiment
1	Formulate a case study and create an E-R Diagram. Mapping of E-R model to Relational Model.
2	To create a database and populate using SQL commands (With constraints) Data Definition Language- Create, Alter, Drop, Rename, Truncate Data Manipulation Language- Insert, Update, Delete, Select Constraints-Not Null, Unique Key, Primary Key, Foreign Key, Check, Dropping a constraint.
3	To perform DCL, TCL commands Data Control Language: Grant, Revoke, Roles Transaction Control Language: Commit, Rollback, Save point
4	To perform Date, Time, Arithmetic and Set operation on database.
5	To perform Aggregate function and Group by- Having clause on database
6	To perform Join operations on database. Equijoins, Non-Equijoins, Self Joins, Outer Join, cross Join
7	To retrieve a data using Subquery.
8	To Create a different view of database.
9	To examine integrity of database using Triggers.
10	To Create a application using above concept using front end and back end.

Textbooks

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Database System Concepts	Seventh	Korth, Slberchatz, Sudarshan	McGraw Hill	2019
2	Fundamentals of Database Systems	Sixth	Elmasri and Navathe	PEARSON Education	2011

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Database Management Systems	Third	Raghu Ramkrishnan and Johannes Gehrke	TMH	2003
2	Database Management Systems	First	G. K. Gupta	McGraw Hill.	2018
3	SQL, PL/SQL programming language of ORACLE	Forth	Ivan Bayross	BPB	2010

SEMESTER IV

SEM IV										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic & Engg. Sciences	BSES	CS205	Statical methods in Computer Science (Linear Algebra, Probability and Statistics)	3	0	0	6	9	3
2	Basic & Engg. Sciences *	FOM-II	MA204	Foundation of Mathematics-II*	2	1	0	0	3	3
3	Skill enhancement course	SEC	AS202	Python programming for data science	0	1	2	4	7	2
4	Humanities	HSSM-II	HS2XX	**Course II**	2	0	0	3	5	2
5	Program Core Courses (12 credits)	PCC	CS206	Operating Systems	3	0	2	4	7	4
6		PCC	CS207	Design and Analysis of Algorithms	3	0	2	4	7	4
7		PCC	CS208	Computer Communications and Networks	3	0	2	4	7	4
8	Cocurricular Courses	CC (LLC)	LLCXX	LLC--IV	1	0	0	1	2	1
9	Multidisciplinary Minor	MDM	MDEC 1X	MDM-I	To be defined by others					3
				Total	**15**	**1**	**8**	**26**	**49**	**23**

Summer term (For Lateral Entry Students)										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Basic &Engg. Sciences	BSES	CS201	Discrete Structures and Graph Theory	3	0	0	5	9	3
2	Basic &Engg. Sciences	BSES	CS205	Statical methods in Computer Science (Linear Algebra, Probability and Statistics)	3	0	0	6	9	3

- **Students are expected to start working for the Mini Project I during the summer.**
- **Summer Research Phase I of minimum 2 months and Course on Research Methodology for the "Honors by Research" for 7(4+3) credits- HR21**
- **For Enrollment to Honors by research, Minimum CGPA must be 8.25**

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(BSCS)	Statistical methods in Computer Science (Linear Algebra, Probability and Statistics)	3	0	0	5	9	3	0	0	3
CS205		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory			--					

Pre-requisite Course Codes, if any.	Engineering Calculus/Foundations of Mathematics-I and Differential Equations and Complex Analysis/Foundations of Mathematics-II
Course Objective: To develop mathematical skills for solving engineering problems.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
CS205.1	Solve a homogeneous and non-homogeneous system of linear equations using rank of a matrix
CS205.2	Apply concepts of eigenvalues and eigenvectors to calculate functions of a square matrix, google rank vector and solve systems of differential equations using diagonalisation of matrices
CS205.3	Familiarize with basic probability axioms, rules and their applicability
CS205.4	Identify the characteristics of various discrete and continuous distributions.
CS205.5	Test the hypothesis for means and variances using t and F; chi-square distribution tests
CS205.6	Find correlation and regression and fit different types of curves.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS205.1	3											
CS205.2	3											
CS205.3	3	1										
CS205.4	2											
CS205.5		2										
CS205.6		2										

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS205.1							
CS205.2							
CS205.3							
CS205.4	2						
CS205.5	2						
CS205.6	2						

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create
✔	✔	✔			

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Linear Algebra**	3,4,6,7	08
	1.1	Review of basic matrices, row echelon form, reduced row echelon form, rank of a matrix		
	1.2	Consistency and solution of simultaneous linear homogeneous and non-homogeneous equations, Applications of solving systems of equations in traffic control.		
2	Title	**Eigenvalues and Eigenvectors**	3,4,6,7	07
	2.1	Eigenvalues, Eigenvectors and its properties, Cayley Hamilton Theorem and its applications, Diagonalisation of matrices.		
	2.2	Application to find Google Page Rank, Function of a square matrix, Solving system of differential equations using diagonalization		
3	Title	**Probability and Random Variables**	1,2	14
	3.1	Classical, relative frequency and axiomatic definitions of probability, addition rule and multiplication rule, Conditional Probability, Bayes' Theorem and Independence		
	3.2	Discrete, continuous and mixed random variables, Probability Mass Function , Probability Density Function		
	3.3	Mathematical expectation, moments, moment generating function		
	3.4	Standard discrete distributions: Binomial, Poisson and Normal		
4	Title	**Hypothesis Testing**	1,2	9
	4.1	Statistical hypothesis, Null and Alternate Hypothesis, test of hypothesis and significance, Type I and Type II errors, Level of significance.		
	4.2	Special tests of significance for large samples and small samples- F, chi-square,z,t-test.		
5	Title	**Correlation and Regression**	1,2	4
	5.1	Spearman and Karl-Pearson Rank Correlation, Regression Analysis, Curve Fitting by method of least squares.		
6	Self Study	1. Applicability of Bayes Theorem 2. Examples to test goodness of fit using Chi-square 3. Normal form 4. Singular Value Decomposition 5. Derogatory and non-derogatory matrices		5
			Total	**42***

***Total of 42 hours does not include the self-study hours.**

Text Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Introduction to Probability and Statistics for Engineers and Scientists	Fourth	Sheldon M. Ross	Academic Foundation	2011
2	Probability and Statistics for Engineers and Scientists	Eighth	E.Walpole, R.H.Mayers, S.L.Mayers, K.Ye	Pearson Education	2007
3	Linear Algebra and its Applications	Fourth	Gilbert Strang	Cengage	2014
4	Higher Engineering Mathematics	Forty Fourth	Dr. B.S.Grewal	Khanna Publications	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Elementary Linear Algebra with Applications	Sixth	H Anton and C Rorres	John Wiley and Sons	2010
2	Advanced Engineering Mathematics	Twenty Eighth	H.K.Das	S.Chand	2014
3	Advanced Engineering Mathematics	Tenth	Erwin Kreysizg	John Wiley and Sons	2011

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Course (Category) Code	Course Name	Teaching Scheme (hours/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Python programming for Data Science Lab	0	1	2	4	7	0	1	2	2
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
AS202		Theory								
		Laboratory	100		--		100		200	

Pre-requisite Course Codes, if any.	Linear Algebra, Probability and dStatistics, Programming Lab-I,II

Course Objective: To provide students with a comprehensive understanding of the fundamental concepts, tools, and techniques used in data science and data visualization. This course is designed to introduce students to the basic principles of data science and data visualization, including libraries used for Data Science, data exploration, data preprocessing, EDA, data visualization and basic model building.

Learning Outcomes (LO): ***At the End of the course, students will be able to***	
CS403.1	Demonstrate the need of libraries used in Data Science.
CS403.2	Make use of different tools and techniques for Data Visualization
CS403.3	Analyze the data performance using the EDA process.
CS403.4	Develop Linear Regression and Logistic Regression Models

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS403.1	3	3	3	3	3				2	2	2	
CS403.2	3	3	3	3	3				2	2	2	
CS403.3	3	3	3	3	3				2	2	2	
CS403.4	3	3	3	3	3				2	2	2	

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2
CS403.1					1	1
CS403.2					1	1
CS403.3					1	1
CS403.4					1	1

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply✔	Analyze✔	Evaluate✔	Create

Lab Experiments

Exp No.	Topics	Ref.
1	**Case Study on Python Programming Fundamentals** Your first program, Types, Expressions, Variables, String Operations, Conditions and Branching, Loops, Functions	1,2
2	**Case Study on Python Data Structures:** Lists and Tuples, Sets, and Dictionaries	1,2
3	**Case Study on Python for Data Science: Numpy Library** Basics of Numpy, Creating NumPy Arrays: Mathematical Operations on NumPy Arrays	1,2
4	**Case Study on Python for Data Science: Pandas Library** Basics of Pandas, Data Frames, Indexing and Slicing, Operations on Data Frames, GroupBy and Aggregate Functions, Merging Data Frames, and Pivot Tables	1,2,4
5	**Case Study on Data Visualization** Univariate Analysis using Matplotlib and Seaborn Categorical Ordered and Unordered Univariate Analysis	3,4
6	**Case Study on Data Visualization** Bivariate and Multivariate Analysis using Matplotlib and Seaborn Numerical-Numerical Analysis, Numerical-Categorical Analysis, Categorical-Categorical Analysis	3,4
7	**Case Study on Exploratory Data Analysis** Fixing the rows and columns, Overfitting and Underfitting, Impute/Remove missing values, Feature Engineering Techniques,	3,4,5
8	**Case Study on Exploratory Data Analysis** Feature Reduction Techniques, Handling Outliers, Normalization, and Standardizing values	3,4,5
9	**Case Study on Linear Regression** Simple Linear Regression, and Multiple Linear Regression	3,4,5
10	**Case Study on Logistic Regression** Univariate Logistic Regression, Multivariate Logistic Regression	3,4,5

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Python for Data Science Handbook: Essentials Tools for Working with Data	Second	Jake VanderPlas	O'Reilly	2022
2	Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter	Third	Wes McKinney	O'Reilly	2022
3	Hands-On Exploratory Data Analysis with Python	First	Suresh Kumar Mukhiya, Usman Ahmed	Packt	2020
4	Python Data Science Essentials	Third	Alberto Boschetti, Luca Massaron	Packt	2018
5	Mastering Python for data science,	First	Samir Madhavan	PACKT	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1.	Visualization Analysis and Design	1st	Tamara Munzner	A K Peters/CRC Press	2014
2.	Python Data Visualization Essentials Guide	First	Kalilur Rahman	BPB	2021
3.	Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython.		McKinney, W.	O'Reilly Media, Inc.	2012

Web References

1. https://www.analyticsvidhya.com/
2. https://www.simplilearn.com/
3. https://www.kaggle.com
4. https://matplotlib.org/
5. https://seaborn.pydata.org/

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Operating Systems	3	0	2	5	10	3	0	1	4
		Examination Scheme								
		Component		ISE		MSE		ESE		Total
CE/CSE 206		Theory		50		50		100		200
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	
Course Objective:	
Learning Outcomes (LO)): *At the End of the course students will be able to*	
CE206.1	Comprehend the primitive concepts of Operating System services and System Programming functionality.
CE206.2	Articulate process scheduling algorithms in effective execution of processes.
CE206.3	Acquaint with efficient process synchronization techniques in effective execution of programs.
CE206.4	Analyze virtual memory management algorithms in effective allocation of main memory usage.
CE206.5	Evaluates various algorithms of File Storage & I/O management for performance and quality criterion.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CE206.1	1											1
CE206.2	2	2	2	2								2
CE206.3	2	2	2	2								2
CE206.4	2	2	2	2								2
CE206.5	2	2	2	2								2

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CE206.1							
CE206.2							
CE206.3							
CE206.4							
CE206.5							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply ✔	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	Introduction to Operating Systems and system software	1,2	6
	1.1	**Operating Systems** – Introduction, Structure and Principles of Operations of Operating Systems, Classes of Operating Systems, Batch Processing Systems, Multiprogramming Systems, Time Sharing Systems, Real Time Operating Systems, linker,loader, static and dynamic linking , virtual machine		
2	Title	**Process and Threads Management**	1,2	9
	2.1	Processes – Process abstraction, System calls for process management, Process execution mechanisms, Process Scheduling, Inter-process communication		
	2.2	Introduction to threads and concurrency – Single threaded process, Multithreaded process, process vs thread, Why threads, scheduling threads, creating threads, threads with shared data, race condition and synchronization .		
3	Title	**Process Coordination**	1,2	12
	3.1	Process Synchronization - Critical-Section Problem, Peterson's Solution. Locks- basic idea, building a lock, hardware atomic instructions, test- and-set , Spin lock using compare - and – swap, sleeping mutex, Semaphores, Classic Problems of Synchronization, Monitors, Synchronization Examples		
	3.2	**Deadlock** - Characterization, Methods for Handling Deadlocks, Detection, Prevention, Avoidance, Recovery methods Deadlock.		
4	Title	**Memory Management**	1,2	8
	4.1	**Memory Management Strategies** - Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table.		
	4.2	**Virtual Memory Management** - Demand Paging, Allocation of Frames, Thrashing, Memory-Mapped Files, Allocating Kernel Memory, Operating System Examples.		
5	Title	**File Management**		7
	5.1	**Storage Management** - Disk Structure, Time taken for I/O operation, Disk Scheduling, Disk Management, RAID, introduction to (solid-state storage) SSD		
	5.2	**File-System** - File-System Structure. Allocation Methods, Free Space Management		
	5.3	**UNIX Internal File Representation – Files and directories , file abstraction, directory tree, operations on files and directories, hard links , soft links** Inodes, .		
6	Self Study	1) Explore Features, characteristics and CPU scheduling of Real Time Operating System along an example 2) Explore the requirements of Kernel, CPU Scheduling, Disk Scheduling for Multimedia Systems 3) Explore all UNIX System Calls for File System.	1,2, 3	
Total (*Not included)				**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Introduction to Linux tools
2	Write a program for creating a static/dynamic link library
3	Write a program to implement various systems call of a process
4	Write a multithreaded program to show race condition
5	Implement different CPU scheduling algorithms
6	Program to implement shared memory and perform read and write operations on it for a given application
7	Program on process synchronization .
8	Write a program for memory management by building a custom memory manager to allocate memory dynamically .
9	Write a program to simulate disk scheduling .
10	Write a program to prevent destructive update of files

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Operating System Concepts	Ninth	Abraham Silberschatz, Peter B Galvin, Greg Gagne	Wiley	2012
2	Operating Systems: Three Easy Pieces		REMZI H. ARPACI-DUSSEAU , ANDREA C. ARPACI-DUSSEAU .	(University of Wisconsin-Madison)	2014
3	UNIX Internals: The New Frontiers	First	Uresh Vahalia	Prentice Hall	1995
4	Design of the UNIX Operating Systems	First	Maurice J. Bach	Prentice-Hall	1990

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Operating Systems: Internals and Design Principles	Eighth	William Stallings	Pearson	2014
2	Modern Operating Systems	Fourth	Andrew S. Tanenbaum, Herbert Bos	Pearson	2014

Course(Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Design and Analysis of Algorithms	3	0	2	5	10	3	0	1	4
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
CS205		Theory	75		75		150		300	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	
Course Objective: Apply various algorithmic design paradigms, analyze their asymptotic performance, and synthesize efficient algorithms for common engineering design scenarios.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
CS205.1	Analyze the complexity of algorithms and apply divide-and-conquer strategy to solve problems.
CS205.2	Make use of string matching algorithms to solve problems .
CS205.3	Apply the concept of greedy method, dynamic programming and Maximum Flow.
CS205.4	Apply the idea of backtracking, branch and bound strategy to solve problems.
CS205.5	Make use of NP-completeness and Approximation Algorithms

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS205.1	3	3	2	1	-	-	1	-	-	-	-	-
CS205.2	3	2	2	1	-	-	1	-	-	-	-	-
CS205.3	3	3	2	1	-	-	1	-	-	-	-	-
CS205.4	3	3	2	1	-	-	1	-	-	-	-	-
CS205.5	2	2	3	1	-	-	1	-	-	-	-	-

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS205.1	-	-	-	-	-	-	-
CS205.2	-	-	-	-	-	-	-
CS205.3	-	-	-	-	-	-	-
CS205.4	-	-	-	-	-	-	-
CS205.5	-	-	-	-	-	-	-

BLOOM'S Levels Targeted

Remember	Understand	Apply	Analyze ✓	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Algorithm**		8
	1.1	**Role of Algorithms** – Algorithm as Technology, Introduction to design and analysis of algorithm e.g. Insertion Sort and Selection Sort		
	1.2	**Asymptotic Notations** – All asymptotic (O, Θ, Ω, o, θ and ω) notations, Common Functions, Recurrence Relation and Solutions Methods: Substitution, Recursion-tree and Master Theorem.		
	1.3	**Divide and Conquer** – Concepts, Maximum Subarray Problem, Closest Pair, Strassen's matrix multiplication, Quick-Sort, Merge-Sort, Selection in Linear Time		
2	Title	**Greedy Methods and String Matching**		10
	2.1	**Disjoint Sets DS** – Disjoint set operations, Linked List Representation, Disjoint Set Forests		
	2.2	**Greedy Approach** – Prim's and Kruskal's MST algorithms, Dijkstra's and Bellman Ford's SSSP algorithms, Activity selection, Fractional knapsack problem, Job sequencing with deadlines problem.		
	2.3	**String Matching algorithms** – Naïve string matching Algorithm, Rabin Karp algorithm, String matching with finite automata, Knuth-Morris-Pratt algorithm		
3	Title	**Dynamic Programming and Maximum Flow Problem**		10
	3.1	**Dynamic Programming** – Concepts, Longest Common Subsequence, Matrix Chain Multiplication, Optimal Binary Search Trees, 0/1 knapsack problem,		
	3.2	**All Pairs Shortest Paths** – Shortest Paths and Matrix Multiplications, Floyd-Warshall algorithm, Johnson's algorithm for sparse graphs.		
	3.3	**Flow Networks** – Graph-theoretic definition of Flow networks, Ford-Fulkerson and Maximum Bipartite Matching		
4	Title	**Backtracking and Branch-and-bound**		8
	4.1	**Backtracking** – General method, 8-queen problem (N-queen problem), Sum of subsets, Graph coloring, 0/1 Knapsack problem		
	4.2	**Branch-and-bound** – General method, 0/1 knapsack problem, Travelling Salesman problem, 15 puzzle problem		
5	Title	**Approximation and NP-Completeness**		6
	5.1	**NP-Completeness** – Polynomial time, Polynomial-time verification, NP-completeness reducibility and proof.		
	5.2	**Approximation** – Introduction, Vertex-cover, Travelling-salesman problem and set-covering.		
6	Self Study	**Amortized analysis** – Aggregate analysis, accounting and Potential Method, Dynamic Table.		
			Total	**42**

Laboratory Component (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Experiment based on common mathematical functions.
2	Experiment based on divide and conquers approach.
3	Experiment based on greedy approach
4	Experiment using dynamic programming approach
5	Experiment based on maximum flow
6	Experiment based on graph Algorithms
7	Experiment using Backtracking strategy
8	Experiment using branch and bound strategy
9	Experiment based on Approximation Algorithms
10	Experiment based on string matching/amortized analysis.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Algorithms	Third	Thomas H. Cormen, Charles E. Leiserson, Ronald L Rivest, Clifford Stein	MIT Press	2009
2	Fundamentals of Computer Algorithms	Second	Horowitz E, Sahni S and S. Rajasekaran	Galgotia Publications	2010

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	The Design and analysis of algorithms	First	Alfred V. Aho**,** John E. Hopcroft and Jeffrey D. Ullman	Pearson Education India	2006
2	Fundamentals of Algorithmics	First	Gilles Brassard, Paul Bratley	Prentice Hall	1995

Bharatiya Vidya Bhavan's

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Computer Communications and Networks	3	0	2	5	10	3	0	1	4
CS207		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		50		50		100		200
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	

Course Objective: The objective of the Computer Communication Networks course is to equip students with comprehensive knowledge and practical skills in the field of network engineering. The course aims to prepare individuals for various specialized roles within the networking industry, enabling them to proficiently design, implement, and manage computer communication networks. Throughout the course, students will develop expertise in areas such as network analysis, administration, architecture, and specialized technical roles. By the end of the program, graduates should be capable of seamlessly navigating between the technical and business aspects of network management, and be well-prepared for roles such as network engineer, network manager, network analyst, network administrator, network specialist, network technician, and network architect. The curriculum will emphasize hands-on experience, problem-solving, and staying abreast of the latest advancements in networking technologies, ensuring that students are well-positioned to contribute effectively in the dynamic and evolving field of computer communication networks.

Learning Outcomes (LO): At the End of the course students will be able to

LO1	Describe the fundamental concepts of computer networking and Internet.
LO2	Evaluate the effectiveness of different protocols in diverse network applications.
LO3	Apply knowledge of protocols to configure transport layer settings for optimal data transmission.
LO4	Apply the knowledge of subnetting, routing mechanisms and Software Defined Networking.
LO5	Analyze the performance and security of different access network technologies.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
LO1	2	2			2							1
LO2	2	2			2							1
LO3	2	2			2							1
LO4	2	2			2							1
LO5	2	2			2							1

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
LO1	2						
LO2	2						
LO3	2						
LO4	2						
LO5	2						

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply ✔	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Computer Communication and The Internet**	**1,2**	**06**
	1.1	**Internet:** A Nut and Bolts Description, A Services Description, Protocol. The Network Edge: Access Network, The Network Core: Packet Switching, Circuit Switching, A Network of Networks	**1,2**	
	1.2	**Delay, Loss, Throughput in Packet Switched Networks:** Overview of Delay in Packet Switched Networks, Queuing Delay and Packet Loss, End to End Delay, Throughput in Computer Networks.	**1,2**	
	1.3	**Protocol Layers and their Service Models:** Layered Architecture and their Encapsulation.	**1,2**	
	1.4	**Data and Signals:** Analog and Digital, Periodic analog signals, Digital signals, Transmission impairment.	**2**	
2	**Title**	**Application Layer**	**1,2**	**06**
	2.1	**Principles of Network Applications:** Network Applications Architecture, Processes Communicating, Transport Services Available to Applications, Transport Services Provided by the Internet, Application Layer Protocols.	**1,2**	
	2.2	**The Web and HTTP:** Overview of HTTP, Non Persistent and Persistent Connections, HTTP Message Format, User Server Interaction: Cookies, Web Catching, The Conditional Get.	**1,2**	
	2.3	**File Transfer Ftp:** Ftp Commands and Replies. FTP, SMTP, Mail Access Protocol (IMAP, POP), DNS	**1,2**	
3	**Title**	**Transport Layer**	**1,2**	**10**
	3.1	**Introduction and Transport-Layer Services:** Relationship Between Transport and Network Layers, Overview of the Transport Layer in the Internet	**1,2**	
	3.2	**Multiplexing and Demultiplexing**	**1,2**	
	3.3	**Connection less Transport - UDP:** UDP Segment Structure, UDP Checksum	**1,2**	

	3.4	**Principles of Reliable Data Transfer:** Building a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocols, Go Back-N (GBN), Selective Repeat (SR),	**1,2**	
	3.5	**Connection-Oriented Transport - TCP:** The TCP Connection, TCP Segment Structure, Round-Trip Time Estimation and Timeout, Reliable Data Transfer, Flow Control, TCP Connection Management	**1,2**	
	3.6	**Principles of Congestion Control:** The Causes and the Costs of Congestion, Approaches to Congestion Control. additive-increase, multiplicative-decrease (AIMD)	**1,2**	
4	**Title**	**The Network Layer**	**1,2**	**10**
	4.1	**Introduction:** Forwarding and Routing, Network Service Models.	**1,2**	
	4.2	**Virtual Circuit and Datagram Networks:** Virtual-Circuit Networks, Datagram Networks, Origins of VC and Datagram Networks.	**1,2**	
	4.3	**Router:** Input Processing, Switching, Output Processing, Queuing, The Routing Control Plane.	**1,2**	
	4.4	**The Internet Protocol (IP):** Forwarding and Addressing in the Internet, Datagram Format, IPv4 Addressing, Internet Control Message Protocol (ICMP), IPv6	**1,2**	
	4.5	**Routing Algorithms:** The Link-State (LS) Routing Algorithm, The Distance-Vector (DV) Routing Algorithm, Hierarchical Routing	**1,2**	
	4.6	**Software Defined Networking:** Introduction and Overview	**R3**	
5	**Title**	**The Link Layer:** Links, Access Networks, and LANs	**1,2**	**10**
	5.1	**Introduction to the Link Layer:** The Services Provided by the Link Layer, Implementation of the Link Layer	**1,2**	
	5.2	**Error-Detection and Correction Techniques:** Parity Checks, Check summing Methods, Cyclic Redundancy Check (CRC)	**1,2**	
	5.3	**Multiple Access Links and Protocols:** Channel Partitioning Protocols, Random Access Protocols, Taking-Turns Protocols.	**1,2**	
	5.4	**WiFi:** 802.11 Wireless LANs:The 802.11 Wireless LAN Architecture,The 802.11 MAC Protocol,The IEEE 802.11 Frame, Advanced Features in 802.11, Personal Area Networks: Bluetooth	**1,2**	
6*	**Title**	**Self Study Topic**	**1,2**	**5***
	6.1	**Transmission Media:** Guided media, Unguided media: Wireless	**1,2**	
	6.2	ARP and RARP usage	**1,2**	
	6.3	Multicast routing and Broadcast routing	**1,2**	
	6.4	**Routing in the Internet:** Intra-AS Routing in the Internet: RIP, Intra-AS Routing in the Internet: OSPF, Inter-AS Routing: BGP	**1,2**	
	6.5	Network Function Virtualization	**R3**	
			Total	**42**

***This module hrs. not included in Total 42 hrs

Laboratory Component (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Network Setup: Network configuration, commands and troubleshooting
2	Network Socket Programming
3	Network Socket Programming using Python Scapy
4	Server Configuration-I
5	Server Configuration-II
6	Network Protocol Analysis
7	Network Simulation using Cisco Packet Tracer
8	Software Defined Networking using Mininet
9	Switch and WLAN Configuration
10	Installation and configuration of network traffic analyzer NTOP
11	Network Server Automation using Ansible
12	Network mapping techniques using open source tools

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Computer Networking: Top Down Approach	8th	James Kurose and Kieth Ross	Pearson	2020
2	Data Communication and Networking	5th	Behrouz Forouzan	McGraw Hill	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Computer Networks	Fifth	Andrew Tanenbaum	Pearson	2013
2	Computer Networks	Third	Larry L. Peterson and Bruce Davie	Morgon Huffman	2003
3	SDN and NFV Simplified	First	Jim Doherty	Addison Wesley	2016

SEMESTER V

SEM V										
Sr. No.	**Course Category**	**Abbre-viation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Experiential Learning	ELC	PR1	Mini Project	0	0	4	4	8	2
2	Program Core Courses (17 credits)	PCC	CS301	Distributed Computing	3	0	2	6	10	4
3		PCC	CS302	Software Engineering	3	0	2	6	10	4
4		PCC	CS303	Artificial Intelligence and Soft Computing	3	0	2	6	11	4
5		PCC	CS304	Theory of Computation	3	0	0	5	8	4
6		PCC	CS305	Cryptography and Network Security	3	0	2	5	10	4
7	Multidisciplinary Minor	MDM	MDEC2X	MDM-II	To be defined by others					4
				Total	**13**	**0**	**12**	**32**	**50**	**25**

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
ELC	Mini Project	0	0	4	4	8	0	0	0	2
		Examination Scheme								
PR1		Component	ISE		MSE		ESE		Total	
		Theory	--		--		--		--	
		Laboratory	100		--		100		200	

Pre-requisite Course Codes, if any.	
Course Objective: This course inculcates self-learning, research and entrepreneurship attitude in students. Students will be able to understand the formal project development process to complete a project in a team. It will help students to develop communication, organizational skills and maturity through discussion, presentation etc.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
PR1.1	Conduct a survey of basic and contemporary literature in the preferred field by identifying problems based on societal /research needs.
PR1.2	Formulate the problem statement by making judgments on validity of ideas.
PR1.3	Apply Knowledge and skill to solve identified problem.
PR1.4	Analyze the impact of solutions in societal and environmental context for sustainable development.
PR1.5	Demonstrate capabilities of self-learning, which leads to lifelong learning.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PR1.1	2	2				2	1	3	3	3		2
PR1.2	2	3	2	2			1	3	3	3		2
PR1.3	2	2	2	2	2			3	3	3		2
PR1.4								3	3	3	3	2
PR1.5	2	2						3	3	3	2	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
PR1.1							
PR1.2							
PR1.3							
PR1.4							
PR1.5							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	✔ Create

The **Mini Project** serves as an opportunity to cultivate research aptitude among students, encouraging them to explore, identify, and bridge research gaps while developing solutions. This hands-on experience allows students to apply theoretical knowledge to real-world issues, fostering critical thinking and problem-solving skills. By engaging in this process, students learn to assess the feasibility of various solutions, considering technical, societal, and environmental aspects.

Steps:

Phase 1-

1) **Project Selection and Idea Generation**- Choose a project topic that is relevant, feasible, and aligns with your interests or course requirements.
2) **Research and Literature Review** - Understand the existing solutions, theories, and technologies related to your project topic.
3) **Problem Definition, Objective Setting and Planning** - Clearly define the problem you're addressing and set specific, achievable project objectives. Create a plan for how you will approach the project

Phase 2-

1) **Design**- Design the solution.
2) **Development and Implementation**- Build and implement the solution based on the design.
3) **Testing** - Test the solution to ensure it meets the objectives and solves the problem effectively.

Students can seek guidance from teachers, other experts and make effective use of other sources of information available around them. Students must ensure that the problem be manageable in one semester.

Criteria of a good project:

✓ Appropriate idea, clear understanding and proper presentation of the concept
✓ Quality of work
✓ Project plan and its execution
✓ Credibility of the work
✓ Probable impact of the work on the attitude of students and society
✓ Scientific attitude, creativity and novelty reflected in project work and analysis of the situation
✓ Utility and innovation of the remedial measures
✓ Desirability, Feasibility and Viability in real life

Evaluation:

Phase 1-
Rubric for Phase 1 -Total 50 marks

Field/Market Survey & Literature Survey (15)	Problem Definition (10)	Presentation (15)	Project Planning (10)	Total Marks out of (50)

Phase 2-
Rubric for Phase 2 -Total 50 marks

P1: Technical Competency (Understand project technicality/Design exceeds project objectives)
P2: Work done (100 % Implementation/ Meets project objectives/result analysis/test cases/solution shows originality)
P3: Presentation (Coverage of topic/use of graphics/Delivery & Speaking skills/Presentation length) and Ethics (Attitude/Deadline/Distribution of Work)
P4: Report with appropriate citations

P1 (10)	P2 (20)	P3 (10)	P4 (10)	Total Marks (Max=50)

ESE-
Rubric for ESE -Total 100 marks

P1: Problem Definition (Idea/Originality/Novelty/Feasibility/Application in Society)
P2: Presentation (Demo/Oral/Written) and Ethics (Attitude/Deadline/Distribution of Work)
P3: Work Done (Survey/Design/Simulation/Experimentation/Tools Usage/Budget Plan/Execution Plan)
P4: Knowledge (Question-answer)
P5: Poster

P1 (20)	P2 (20)	P3 (30)	P4 (20)	P5 (10)	Total Marks (Max=100) (Passing=40)

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Distributed Computing	3	0	2	5	10	3	0	1	4
CE301		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	50		50		100		200	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	
Course Objective:	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CS301/CE301 .1	Understand the fundamental concepts and design principles of distributed systems.
CS301/CE301 .2	Analyze and apply synchronization techniques and algorithms in distributed environments.
CS301/CE301 .3	Evaluate consistency models and replication strategies for distributed data management.
CS301/CE301 .4	Design and assess distributed file systems and understand their real-world applications.
CS301/CE301 .5	Explore emerging trends and technologies in distributed computing.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS301/CE301 .1	3	2	2	2	1	-	-	-	-	-	-	-
CS301/CE301 .2	3	3	3	2	2	-	-	-	1	-	-	-
CS301/CE301 .3	3	3	3	2	2	-	-	-	1	-	-	-
CS301/CE301 .4	3	2	3	2	2	-	-	-	1	-	-	-
CS301/CE301 .5	3	2	2	2	2	-	-	-	1	-	-	-

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS301/CE301 .1							
CS301/CE301 .2							
CS301/CE301 .3							
CS301/CE301 .4							
CS301/CE301 .5							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze √	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Distributed Systems**		8
	1.1	Definition, Goals, Types of Distributed Computing Models, Issues in Distributed Systems.	T1,T2	
	1.2	Hardware Concepts, Software Concepts, The Client-Server Model, Positioning Middleware, Models of Middleware, Services offered by Middleware.	T1	
2	Title	**Communication in Distributed Systems**		10
	2.1	Interprocess Communication: Message passing, RPC, RMI, Message Oriented Communication: Persistence and synchronicity in communication, Message Oriented Transient and Persistent Communications sockets.	T1,T2	
	2.2	Desirable features of message passing, message structure, Issues in IPC by message passing, message synchronization, Buffering, Multi datagram message, failure handling, Group communication and multicast.	T1,T2	
3	Title	**Process and Clock Synchronization and Coordination**		10
	3.1	Introduction to Threads, Threads in Distributed Systems, Clients, Server, Code Migration: Approaches to Code Migration, Models, Migration and Local Resources.	T1	
	3.2	Clock Synchronization: Cristian's and Berkeley algorithms, Logical Clocks: Lamport's Timestamps, Vector Clocks, Mutual Exclusion: Token based, Non token based and Quorum based. Election Algorithms: Bully and Ring Algorithm.	T1,T2	
4	Title	**Consistency and Replication**		6
	4.1	Data-centric and client-centric consistency models, Replication strategies, quorum consensus, and consistency protocols.	T1,T2	
	4.2	Types of Failures and Replication Techniques, Fault Tolerance.	T1,T2	
5	Title	**Distributed File Systems & Case Studies**		8
	5.1	Design and implementation of DFS: NFS, AFS, HDFS, Google File System (GFS), Hadoop, Apache Kafka.	T1, R1,2,3,4	

	5.2	Emerging trends: Blockchain, types of Blockchain, Bitcoin, High performance computing, Microservices, Distributed AI & Machine Learning.	R1,2,3,4	
6	**Self Study**	Cloud Computing, Edge Computing and Fog Computing, Need for Edge/Fog in IoT, Architectures and real-world applications, Open MPI.	R1,2,3,4	**6***
			Total	**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Client-Server Communication using RPC or Java RMI
2	Implementation of Multithreading in Distributed System
3	Implementation of Clock Synchronization (logical/physical).
4	Implement Bully and Ring Election Algorithms
5	Implementation of data consistency and replication models
6	Implementation Load Balancing Algorithm.
7	Implementation a basic MapReduce job using Hadoop or Apache Spark
8	Simulate simple fault tolerance with primary-backup replication
9	MPI Collective Communication: Broadcast, Scatter, Gather
10	Parallel Matrix Multiplication using MPI

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Distributed Systems: Principles and Paradigms	2nd Edition	Andrew S. Tanenbaum, Maarten Van Steen	Pearson Education	2007
2	Distributed Operating Systems Concepts and Design	Second Edition	P. K. Sinha	PHI	2010

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Distributed Systems: Concepts and Design	5th Edition	George Coulouris, Jean Dollimore, Tim Kindberg, Gordon Blair	Pearson	2012
2	Distributed Computing: Principles, Algorithms, and Systems	-	Ajay D. Kshemkalyani, Mukesh Singhal	Cambridge University Press	2008

3	Using MPI: Portable Parallel Programming with the Message-Passing Interface	2nd Edition	William Gropp, Ewing Lusk, Anthony Skjellum	MIT Press	1999
4	Distributed VOD Systems	First Edition	Sudhir D. & Bandu B.M	Research India Publication	2011

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus,
Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Software Engineering	3	0	2	5	10	3	0	1	4
CE-302		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	50		50		100		200	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	CE102 :Problem Solving using Object Oriented Programming Lab, CE204: Database Management Systems.
Course Objective: To understand the best practices in software engineering and gain knowledge to analyze, design, implement and test software project.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CE-302.1	**Illustrates** software development lifecycle models.
CE-302.2	**Analyze** user requirements and specifications to identify functional and non-functional requirements of a software system.
CE-302.3	**Design** structured and object-oriented system using appropriate design patterns.
CE-302.4	**Apply** project management principles to schedule tasks, estimate development costs, and manage risks in software system.
CE-302.5	**Apply** software testing methods to create test cases that validate system functionality

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CE-302.1	2	2										
CE-302.2	3	3								1		1
CE-302.3	2		3		3				2			2
CE-302.4	2		2	1	1				2	1	3	2
CE-302.5	2			3	2							2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
CE-302.1					
CE-302.2					
CE-302.3					
CE-302.4					

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember✔	Understand ✔	Apply✔	Analyze✔	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1		**Introduction**		**06**
	1.1	Introduction to Software Engineering, The nature of software, the software process, software Engineering Practice.	1,2	
	1.2	Software Methodologies and Related Process Models with applications, Traditional Life Cycle Models, Waterfall, Incremental, Iterative models, Spiral, V-Model, Agile Software Engineering Process Models, SCRUM, Extreme Programming	1,2	
2		**Requirements Management and Project Management**		**10**
	2.1	Requirement Gathering and Analysis, Software Requirement Specification, Functional Specifications with Use Cases, Defining User Requirements, Validating Requirements, Achieving Requirements Traceability, Managing Changing Requirements, Agile Requirements Engineering.	1,2	
	2.2	Scheduling, Work Breakdown Structure, Gantt Chart, Pert Chart, Critical Path, Schedule and Cost slippage, Estimation, Decomposition techniques, Empirical estimation models, Software Risk Management: Risk Identification, Risk Projection, Risk Refinement, RMMM Plan.	1,2,3	
3		**Software Analysis and Design**		**10**
	3.1	Difference between Structured and Object-Oriented analysis, Structured Analysis, Data Flow Diagrams Structured Design Structure Chart.	4,5	
	3.2	Coupling and Cohesion, Software Architecture, Architecture view, Architecture Styles for Component and connector view: Pipe and Filter, Shared-Data Style, Client-Server Styles	2,3	
	3.3	Object Oriented Analysis and Design, Use Case Diagram, Class diagram, Interaction diagrams, Activity diagram, State Chart diagram, Component and Deployment diagram	5,6	
4		**Object Oriented Software Development**		**06**
	4.1	Design Patterns, types of Patterns, Architectural Patterns: MVC, GoF Design Pattern: Singleton, Observer	2,3	
	4.2	GRASP(General Responsibility Assignment Patterns): Low coupling/high cohesion, Controller, Law of Demeter, Expert, Creator.	2,3	

5		**Software Testing and Quality Assurance**		**10**
	5.1	Software Testing, Unit Testing, Integration Testing, System Testing Black-Box Testing: Equivalence Class Partitioning, Boundary Value Analysis, White-Box Testing: Statement Coverage, Branch Coverage, Condition Coverage, Condition and Decision Coverage, Multiple Condition Coverage, Multiple Condition Coverage/Decision Coverage (MC/DC), Path Coverage, McCabe's Cyclomatic Complexity Metric, Data Flow-based Testing, Mutation Testing	1,2	
	5.2	Software Metrics, Product Metrics – McCall's Quality Factor, Metrics for Analysis Model and Design Model, Project Metrics, Process Metrics, Metrics for Source Code	1,2	
	5.3	Software Quality Concepts, Quality Assurance, Quality Control, Formal Technical Reviews	1,2	
6	Self Study	Software Engineering with Open Source and GenAI		**05**
			Total	**42 5***

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Gather requirements and write a project proposal for case study. Prepare SRS document. (Use IEEE template)
2	Design UML diagram -Use Case
3	Design UML diagram - Class diagram
4	Design UML diagram -Interaction diagrams
5	Design UML diagram -Activity diagram
6	Design Data flow diagram (level 0 and1) for the case study.
7	Create work breakdown structure and schedule the activities
8	Develop Risk Mitigation, Monitoring and Management Plan for the case study.
9	Implement any one Module from chosen case study.
10	Prepare test cases and perform Unit Testing (test scenario, test cases, test data)

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Software Engineering: A Practitioner's Approach	8th Edition	Roger S. Pressman and Bruce Maxim	McGraw-Hill	2021
2	Fundamentals of Software Engineering	5th Edition	Rajib Mall	PHI Learning	2018
3	An Integrated Approach to Software Engineering	3rd Edition	Pankaj Jalote	Narosa Publishing House	2019

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Software Engineering	9th Edition	Ian Sommerville	Pearson	2023
2	UML for Java Programmers	---	Robert C. Martin	Pearson	2006
3	The Unified Modeling Language User Guide	2nd Edition	Martin Fowler	Pearson	2010
4	Pankaj Jalote's Software Engineering: With Open Source and GenAI	2nd Edition	Pankaj Jalote	WILEY	2024

Bharatiya Vidya Bhavan's

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PC)	Artificial Intelligence and Soft Computing	3	0	2	5	10	2	0	1	4
CS303		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	20		20		60			100
		Laboratory	80		--		20			100

Pre-requisite Course Codes, if any. Mathematics	
Course Objective: The objective of this course is to introduce students to the foundational concepts of Artificial Intelligence (AI) and Soft Computing, which are essential components of modern intelligent systems. The course aims to provide students with both a solid theoretical background and practical understanding necessary to design, analyze, and implement intelligent systems. Emphasis is placed on key AI paradigms and soft computing techniques, including neural networks and fuzzy logic, enabling students to develop systems that can learn, adapt, and make decisions in complex and uncertain environments.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CS303.1	Identify the various characteristics of Artificial Intelligence and soft computing techniques in building intelligent machines
CS303.2	Apply the supervised and unsupervised Neural Network Learning algorithm to solve real world engineering problems.
CS303.3	Design an associative memory network for real world problems
CS303.4	Design Fuzzy Logic Controller System
CS303.5	Analysis Working of NN-FL Hybrid System

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS303.1	2	2	--	2	--	--	--	--	--	--	--	--
CS303.2	2	2	2	2	2	--	--	--	1	1	--	--
CS303.3	2	2	2	2	2	--	--	--	1	1	--	--

CS303.4		**2**	**2**	**2**	**2**	--	--	--	--	--	--	--
CS303.5		**2**	**2**	**2**								

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	**PEO2**	**PEO3**	**PEO4**	**PSO1**	**PSO2**
CS303.1	**2**	**2**	--	--	--	--
CS303.2	**3**	**2**	--	--	**2**	--
CS303.3	**3**	**2**	--	--	**2**	--
CS303.4	**3**	**2**	**3**	--	--	--
CS303.5	**3**	**2**	--	--	--	--

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	**Create**

Theory Component

Module No.	**Unit No.**	**Topics**	**Ref.**	**Hrs.**
1	**Title**	**Introduction to AI and Soft computing technique**	T1,T2, R3	**06**
	1.1	Definition of AI, History and Future of AI. Intelligent Agents, Agents and Environments ,Rationality, Nature of Environment, Structure of Agent, Problem solving Approach to Typical AI problem, Introduction to Expert System Design		
	1.2	Soft computing Constituents, Difference between Hard Computing and Soft Computing.		
2	**Title**	**Neural Networks**	T1,T2 T3,R1	**12**
	2.1	**Basics of Neural Networks:** Introduction to Neural Networks, Biological Neural Networks, Models of ANN, terminologies in ANN, Activation functions and its types, McCulloch Pitt model, Linear separability, Hebb Network		
	2.2	**Supervised Learning algorithms:** Perceptron (Single Layer, Multi layer),, Adaline, Delta learning rule, Back Propagation algorithm.		
	2.3	**Un-Supervised Learning algorithms:** Hebbian Learning, Winner take all, Self-Organizing Maps KSOFMN , Learning Vector Quantization.		
3	**Title**	**Associative Memory Networks**	T2, T3,T4	**4**
	3.1	Auto associative and Hetero associative Memory Network, BAM, Hopfield network.		
4	**Title**	**Fuzzy Logic, Classical Set and Fuzzy Relations**	T2,T3,	**12**

	4.1	Introduction to Fuzzy Logic, Classical and Fuzzy Sets, Membership Functions, Classical and Fuzzy set operations, and properties of classical and Fuzzy sets.	T4,R1, R3	
	4.2	Classical and Fuzzy Relations**:** Cartesian product of relation, Fuzzy Max-Min and Max-Product Composition, Fuzzy extension principle		
5	**Title**	**Fuzzy control system design and Hybrid Model**	T2, T3 T4, R2	**8**
	5.1	Fuzzy Inference System (FIS), Types of FIS, fuzzification, defuzzification methods, and design of fuzzy controllers. Introduction of Neuro-Fuzzy Systems, Architecture of Neuro-Fuzzy Networks(ANFIS)		
	Self Study	Adaptive Resonance Theory, Genetic Algorithms, CNN, Transfer Learning,	-	**5**
			Total	**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No	**Title of the Experiment**
1	To study and write a program using Prolog for simple real time problems like temperature conversion, Monkey banana problem, Fibonacci series etc
2	Implement a simple intelligent agent using Python
3	To implement Transfer/Activation Functions for a given problem statement and design ANN for a given problem statement.
4	To design and implement SLP for a given problem statement. 1. Design a perceptron using Joone Editor. 2. Write a program using Single Layer perceptron
5	To implement the Supervised Learning algorithm.
6	To implement the Unsupervised Learning algorithm
7	To implement Associative Memory Network for a given problem statement
8	To implement Fuzzy Sets and Fuzzy Relations for a given problem statement
9	To design and implement Fuzzy Logic controller for a given problem statement
10	To design and implement ANFIS model for a given problem statement

Text Books

Sr. No	**Title**	**Edition**	**Authors**	**Publisher**	**Year**
T1	Artificial Intelligence A Modern Approach	2nd	Stuart J. Russell and Peter Norvig	Pearson Publication	1995
T2	Introduction to Artificial Neural Systems	1st	Jacek M. Zurada	Jaico Publisher	1994
T3	Principles of Soft Computing	3rd	Sivanandan and Deepa	Pearson Edition	2019

T4	Fuzzy logic with engineering applications	3rd	Ross, Timothy J	John Wiley & Sons	2011
T5	Neural Networks, Fuzzy Logic and Genetic Algorithms	Kindle	S.Rajasekaran and G.A.VijayalakshmiPai	PHI Learning	2013

Stuart J. Russell and Peter Norvig, "Artificial Intelligence A Modern Approach “Second Edition" Pearson Education.

Reference Books

Sr. No	Title	Edition	Authors	Publisher	Year
R1	Neural Network Design	2nd	Hagan, Demuth, Beale	CENGAGE Learning	2014
R2	Neuro-Fuzzy and Soft Computing	1st	J.-S.R.Jang .	Pearson	1996
R3	Introduction to Soft Computing	1st	Sameer Roy	Pearson	2013

Bharatiya Vidya Bhavan's

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PCC)	Theory of Computation	3	0	0	5	8	3	0	0	3
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
CE/CS304		Theory	20		20		60		100	
		Laboratory	--		--		--		--	

Pre-requisite Course Codes, if any.	CE/CS201: Discrete Structures and Graph Theory
Course Objective: To give an overview of the theoretical foundations of computer science from the perspective of formal languages which provides the mathematical foundation of formal models of computation, and fundamentals of formal grammars and languages that are used in most areas of computer science.	
Course Outcomes (CO): ***At the end of the course students will be able to***	
CE/CS304.1	Design finite automata for regular expressions and languages.
CE/CS304.2	Understand and apply properties of regular languages.
CE/CS304.3	Design context-free grammar for a language and convert it into normal forms.
CE/CS304.4	Design Pushdown Automata and Turing Machine for a language.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO 8	PO 9	PO10	PO11	PO1 2
CE/CS304.1	3	3	2		1				1	1		
CE/CS304.2	3	2							1	1		
CE/CS304.3	2	3			1				1	1		
CE/CS304.4	2	2	2		1				1	1		

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CE/CS304.1							
CE/CS304.2							
CE/CS304.3							
CE/CS304.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

✔	✔	✔	✔		

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Sets, Relations and Languages**	1,5	**3**
	1.1	Relations and functions		
	1.2	Alphabets and languages		
	1.3	Types of proof		
2	**Title**	**Finite Automata**	1,3,5	**7**
	2.1	Regular languages and regular expressions		
	2.2	Finite Automata, Nondeterministic Finite Automata, Nondeterministic Finite Automata with **ϵ**-transitions		
	2.3	Kleene's theorem		
	2.4	NFA to DFA Conversion		
	2.5	Finite Automata with output (Moore and Mealy Machine)		
3	**Title**	**Regular Languages**	1,4	**6**
	3.1	The pumping lemma for regular languages, Applications of the pumping lemma		
	3.2	Closure properties for regular languages		
	3.3	Equivalence and minimization of automata: Testing equivalence of states, Minimization of DFA's		
	3.4	Myhill-Nerode theorem		
4	**Title**	**Context-Free Grammars and Languages**	1,5	**5**
	4.1	Context free grammars: Definition of context free grammars, Derivations using a grammar, The language of a grammar, Sentential forms		
	4.2	Parse trees: Constructing parse trees, From inferences to trees, From trees to derivations, From derivations to recursive inferences		
	4.3	Ambiguity in grammars and languages: Ambiguous grammars, Removing ambiguity from grammars		
5	**Title**	**Pushdown Automata**	1,2	**6**
	5.1	Definition of the pushdown automaton: The formal definition of pushdown automata, A graphical notation for PDA's, Instantaneous descriptions of a PDA		
	5.2	The languages of a PDA: Acceptance by final state, Acceptance by empty stack, From empty stack to final state, From final state to empty stack		
	5.3	Equivalence of PDA's and CFG's: From grammars to pushdown automata, From PDA's to Grammar		
	5.4	Deterministic pushdown automata: Definition of a deterministic PDA, Regular languages and deterministic PDA's, DPDA's and context free languages		
6	**Title**	**Properties of Context-Free Languages**	1,2,3	**5**
	6.1	Eliminating useless symbols, Computing the generating and reachable symbols, Chomsky normal form, Greibach normal form		

	6.2	The Pumping lemma for context free languages: Applications of the pumping lemma for CFL's		
7	**Title**	**Introduction to Turing Machines**	1,2,	**6**
	7.1	Turing machines: Formal definition of a Turing machine, Examples of Turing machines		
	7.2	Halting Problem, Post Correspondence Problem (PCP)		
	7.3	Variants of Turing machines: Multitape Turing Machines		
	7.4	Church-Turing hypothesis		
8	**Title**	**Recursively Enumerable Languages**	3	
	8.1	Recursively Enumerable and recursive		**4**
	8.2	Enumerating a language		
	8.3	Context sensitive languages and the Chomsky hierarchy		
	Self Study	**Tractable and Intractable Problems:** Tractable and Possibly Intractable Problems: P and NP, Polynomial-Time Reductions and NP-Completeness, Cook's Theorem	3	**5***
Total (*Not included)				**42**

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Automata Theory, Languages, and Computation	Third	John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman	Pearson	2008
2	Introduction to the Theory of computation	Third	Michael Sipser	Cengage	2013

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Languages and the Theory of Computation	Fourth	John C. Martin	McGraw-Hill	2010
2	Elements of the Theory of Computation	Second	Harry R. Lewis, Christos H. Papadimitriou	Pearson	2015
3	Automata and Computability	--	Dexter C. Kozen	Springer	1997

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PC	Cryptography and Network Security	3	0	2	5	10	3	0	1	4
CS305		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	15		15		45		75	
		Laboratory	15		--		10		25	

Pre-requisite Course Codes, if any.	CS207: Computer Communications and Networks
Course Objective: To apply cryptography algorithms and use of different network security protocol/techniques.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CS305.1	Understand the fundamental principles of security and cryptography e.g. number theory, finite fields etc.
CS305.2	Make use of classical and modern encryption and decryption algorithms to secure data, ensuring confidentiality, authenticity, and efficiency in cryptographic systems.
CS305.3	Develop a deep understanding of cryptographic data integrity algorithms, including hash functions, message authentication codes (MACs), and digital signatures.
CS305.4	Apply key management, authentication mechanisms, intrusion detection systems (IDS), and firewalls to enhance security, protect data, and mitigate cyber threats.
CS305.5	Understand and implement PGP, SSL, TLS, and IPSec protocols to enhance secure communication, ensure data integrity, and support authentication in networked systems.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS305.1	3	-	-	-	-	-	-	-	-	-	-	-
CS305.2	3	1	3	-	-	-	-	-	-	-	-	-
CS305.3	3	2	3	-	2	-	-	-	-	-	-	-
CS305.4	3	2	3	-	2	-	-	-	-	-	-	-
CS305.5	3	-	3	-	-	-	-	-	-	-	-	-

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

CO-PEO/PSO	PEO1	PEO2	PEO3	PSO1	PSO2
CS305.1	-	-	-	2	-
CS305.2	-	-	-	2	-
CS305.3	-	-	-	2	-
CS305.4	2	-	-	2	-
CS305.5	-	-	-	2	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✓	Evaluate	Create

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Fundamentals for Security and Cryptography**		9
	1.1	**Network Security Concepts:** OSI Security Architecture, Security Attacks and Malware, Security Services, Security Mechanisms, Network Security Model	1,2	
	1.2	**Number Theory:** Divisibility and the Division Algorithm, Euclidean Algorithm, Fermat's and Euler's Theorems, Primality Test, Chinese Remainder Algorithm.	1,2	
	1.3	**Finite Fields:** Groups, Rings, Fields, Finite Fields of the Form GF(p), Polynomial Arithmetic, Finite Fields of the Form GF(2^n)	3	
2	Title	**Symmetric Ciphers and Asymmetric Ciphers**		9
	2.1	**Classical Encryption Techniques:** Symmetric Cipher Models, Substitution and Transposition Techniques, Rotor Machines, Steganography	1,2	
	2.2	**Data Encryption Standard:** Traditional Block Cipher Structure, The Data Encryption Standard, Strength of DES. **Advanced Encryption Standard:** Structure, Transformation Functions, Key Expansion, Implementation, **Block Cipher Operation:** Multiple Encryption and Triple DES, ECB, CBC, CFB, OFB, CTR	1,2	
	2.3	**Asymmetric Ciphers**: Public-Key Cryptosystems, RSA Algorithm, Diffie-Hellman Key Exchange, Elgamal Cryptographic System, Elliptic Curve Arithmetic, Elliptic Curve Cryptography.	1,2,3	
3	Title	**Cryptographic Data Integrity Algorithms**		8
	3.1	**Cryptographic Hash Functions:** Simple Hash Functions, Requirements and Security, Hash Functions Based on Cipher Block Chaining, Secure Hash Algorithm (SHA)	1,2	
	3.2	**Message Authentication Codes:** Requirements, Functions, Security of MACs, MACs Based on Hash Functions: HMAC, MACs Based on Block Ciphers: DAA and CMAC, Authenticated Encryption: CCM and GCM, Pseudorandom Number Generation Using Hash Functions and MACs	1,2	
	3.3	**Digital Signatures:** Elgamal Digital Signature Scheme, Schnorr Digital Signature Scheme, NIST Digital Signature Algorithm, Elliptic Curve Digital Signature Algorithm, RSA-PSS Digital Signature Algorithm.	1,2	
4	Title	**Key Management and System Security**		8
	4.1	Symmetric Key Distribution Using Symmetric Encryption and Asymmetric Encryption, Distribution of Public Keys, X.509 Certificates, Public-Key Infrastructure	1,2,3	

	4.2	User Authentication: Remote User-Authentication Principles, Remote User-Authentication Using Symmetric Encryption and Asymmetric Encryption, Federated Identity Management, Kerberos	1,2,3	
	4.3	IDS and Firewalls: Intrusion Detection System, Types of IDS, Firewalls Characteristics, Types of Firewalls	2,3	
5	**Title**	**Network and Internet Security**		**8**
	5.1	Application Layer Security: Email System Architecture, Pretty Good Privacy (PGP) – Key Rings, PGP Certificates, Key Revocation, PGP Packets, PGP Messages, S/MIME, and Web Application Security.	1,2	
	5.2	Transport Layer Security: Secure Sockets Layer (SSL) – Architecture, Protocols, Message Formats. TLS – Cipher Suite, Protocols.	1,2	
	5.3	Network Layer Security : IPSec – Two Modes, Two Protocols (AH, ESP), Security Associations, Security Policy, Internet Key Exchange, ISAKMP,	1,2	
6	**Self Study**	ACM, Access Control Mechanisms – MAC, DAC, RBAC, TGP Model, HRU Model, BLP Model, Biba Model, Constraint and State based Models.	1,2	**5***
Total (*Not included)				**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Experiment on Number Theory
2	Experiment on Symmetric Ciphers
3	Experiment on Symmetric Asymmetric Ciphers
4	Experiment on Cryptographic Hash Functions
5	Experiment on Message Authentication Codes/Digital Signatures
6	Experiment on X.509 Certificates
7	Experiment on Programming Security – Web/Mobile Application
8	Experiment on User/Service Authentication (e.g. Kerberos) /IDS/Firewall
9	Experiment on PGP/S-MIME
10	Experiment on SSL/IPSec
11	Experiment on Access Control Mechanisms - MAC/DAC/RBAC

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Cryptography and Network Security: Principles and Practice	Seventh Edition	William Stallings	Pearson	2017
2	Introduction to Cryptography and Network Security	First Edition	Behrouz A Fourouzan	TMH	2008
3	Computer Security [Art and Science]	Second Edition	Matt Bishop	Addison-Wesley	2019

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Network Security and Cryptography	Second Edition	Bernard Menezes	Cengage Learning	2011
2	Security in Computing	Fifth Edition	Charles P. Pfleeger	Pearson Education	2015
3	Security Engineering	Third Edition	Ross Anderson	Wiley Publication	2021

SEMESTER VI

SEM VI										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Program Core Courses (7credits)	PCC	CS306	Machine Learning	2	0	2	8	3	3
2		PCC	CS307	Human Machine Interaction	3	0	2	4	7	4
3	Multidisciplinary Minor	MDM	MDEC3X	MDM-III	To be defined by others					3
4	Experiential Learning	ELC	PR3-I	Main Project Stage I	0	0	2	4	6	1
5	Program Elective Courses	PEC	CS3X1	PE-I	2	0	1	4	7	3
6	Program Elective Courses	PEC	CS3X2	PE-II	2	0	1	4	7	3
7	Skill enhancement course	SEC	CS308	DevOps Lab	0	1	2	2	5	2
				Total	**9**	**1**	**12**	**26**	**37**	**19**

- **Summer Research Phase II of minimum 2 month and A course on Optimization for the "Honors by Research" for 7(4+3) credits HR31**
- **For Enrollment to Honors by research, Minimum CGPA must be 8.25**

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PCC)	Machine Learning	2	0	2	4	8	2	0	1	3
CS307		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		50		50		100		200
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	
Course Objective: To learn methodology and tools to apply machine learning algorithms to real data and evaluate their performance.	
Course Outcomes (CO): ***At the End of the course, students will be able to***	
CS307.1	Define the fundamental principles of Machine learning.
CS307.2	Apply techniques, mathematical concepts, and algorithms used in machine learning.
CS307.3	Interpret the limitations of various machine learning algorithms and the way to evaluate the performance of machine learning algorithms
CS307.4	Design an application through software implementation of different concepts and algorithms covered in the course.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS307.1	2	2	--	--	--	--	--	--	2	2	2	2
CS307.2	2	2	2	--	2	--	--	--	2	2	2	2
CS307.3	2	2	2	--	2	--	--	--	2	2	2	2
CS307.4	2	2	2	2	2	--	--	--	2	2	2	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS307.1	2	2	1	1	1	--	2
CS307.2	2	2	1	1	2	--	2
CS307.3	2	2	1	1	2	--	2
CS307.4	2	2	1	1	2	1	2

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create✔

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	Introduction to Machine Learning	T1	04
	1.1	Machine Learning, Key terminology, Key tasks of machine learning, How to choose the right algorithm, Types of Machine Learning, Issues in Machine Learning, Application of Machine Learning, Steps in developing a Machine Learning Application.		
2	Title	Learning with Trees	T1, T2, T3	06
	2.2	Learning with Trees: Decision Trees, Constructing Decision Trees using Gini Index (Regression), Classification and Regression Trees (CART)		
	2.3	Performance Metrics: Confusion Matrix, [Kappa Statistics], Sensitivity, Specificity, Precision, Recall, F-measure, ROC curve		
3	Title	Learning with Support Vector Machine	T1, R2	05
	3.1	Support Vector Machine Constrained Optimization, Optimal decision boundary, Margins and support vectors, SVM as a constrained optimization problem, Quadratic Programming, SVM for linear and nonlinear classification, Basics of Kernel trick.		
	3.2	Support Vector Regression, Multiclass Classification		
4	Title	Ensemble Learning		06
	4.1	Understanding Ensembles, K-fold cross-validation, Boosting, Stumping, XGBoost	T2	
	4.2	Bagging, Subagging, Random Forest, Comparison with Boosting, Different ways to combine classifiers	T2	
5	Title	Learning with Clustering and Dimensionality Reduction		7
	5.1	Introduction to clustering with an overview of distance metrics and major clustering approaches.	T1,T2	
	5.1	Dimensionality Reduction Techniques, Principal Component Analysis, Linear Discriminant Analysis, Singular Valued Decomposition.	T1, T2	
6	Self Study	Graph Based Clustering, Reinforcement Learning	T2	5*
			Total	28

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

All experiments should be performed through PYTHON Language

Sr. No.	Title of the Experiment
1	Implement Linear Regression of Prediction/ Classification
2	Implementation on Decision Tree learning
3	Implement SVM for linear data
4	Implement SVM for non-linear data using kernel function
5	Implementation on Ensemble learning: Bagging using Random Forest
6	Implementation on Ensemble learning: Boosting using XGBoost / AdaBoost
7	Implement K-means/ K-Modes Clustering
8	Implement Expectation Maximization(EM) algorithm to Find Natural Patterns in Data
9	Implement PCA/SVD for Dimensionality Reduction
10	Implement LDA for Dimensionality Reduction

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Machine Learning In Action	1st	Peter Harrington	DreamTech Press	2001
2	Introduction to Machine Learning	3rd	Ethem Alpaydın	MIT , PRESS	2012
3	Machine Learning	1st Edition	Tom Mitchell	Mc-Grawhill	1997

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Pattern Recognition and Machine Learning,	1st Edition	C. M. Bishop	Springer	2013
2	Elements of Statistical Learning	2nd Edition	Trevor Hastie Robert Tibshirani Jerome Friedman	Springer	2001
3	Machine Learning An Algorithmic Perspective		Stephen Marsland	CRC Press	

Useful Digital Links	
1	Data sets for Machine Learning algorithms: https://www.kaggle.com/datasets
2	Machine Learning repository- https://archive.ics.uci.edu/ml/index.php
3	Machine Learning from Coursera
4	https://towardsdatascience.com/machine-learning/home
5	https://onlinecourses.nptel.ac.in/noc21_cs85/preview

Bharatiya Vidya Bhavan's

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
CS306	Human Machine Interaction (HMI)	3	0	2	4	9	3	0	1	4
PC		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	50		50		100		200	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes	Software Engineering,
Course Objectives	This course provides an opportunity to learn and apply the design principles of Human-Machine Interaction. Learners will learn the basic human psychology of everyday actions and will be able to design a UI prototype of an application. This course covers the discussion of various interaction design concepts. The laboratory experiments are designed to practice the concepts and to adopt the systematic approach for interface design using various HMI-tools.
Course Outcomes (CO): ***At the End of the course students will be able to***	
CS306.1	Identify the various design principles used for interacting between humans and machines.
CS306.2	Apply human psychology of everyday actions and UI design process for real-world applications.
CS306.3	Implement mobile, windows, and web-based application
CS306.4	Evaluate and justify UI design
CS306.5	Create applications for social and technical tasks.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO5
CS306.1	2	-	2	-
CS306.2	2	-	2	-
CS306.3	2	3	-	2
CS306.4	2	-	2	-
CS306.5	2	3	-	-

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
CS306.1					
CS306.2					
CS306.3					
CS306.4					
CS306.5					

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply ✓	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Introduction**	T1-T4	**06**
	1.1	Introduction to Human Machine Interface, Hardware, software and operating environment to use HMI in various fields.		
	1.2	**The psychopathology of everyday things** – complexity of modern devices; human-centered design; fundamental principles of interaction		
	1.3	**The psychopathology of everyday things** – complexity of modern devices; human-centered design; fundamental principles of interaction		
2	**Title**	**Graphical User Interface and Web Interface**	T1,T2, T4	08
	2.1	The Graphical User Interface: Popularity of graphics, the concept of direct manipulation, characteristics of GUI,		
	2.2	Web user Interface: Interface popularity, characteristics. Principles of user interface design.		
3	**Title**	**Understanding Goal-Directed Design:**	T1,T2, T4	06
	3.1	Goal-directed design; Implementation models and mental models; Beginners, experts, and intermediates – designing for different experience levels		
	3.2	Understanding users; Modeling users – personas and goals.		
4	**Title**	**Design Guidelines**	T1,T2, T3,T4	10
	4.1	perception, Gestalt principles, visual structure, reading is unnatural		
	4.2	color, vision, memory, six behavioral patterns, recognition and recall, learning, factors affecting learning, time.		
5	**Title**	**Interaction Styles and Communication**	T2,T4	12
	5.1	**Interaction Styles 1:** Menus, Windows,		
	5.2	**Interaction Styles 2:**Device-based and Screen-based Controls.		
	5.3	**Communication:** Text messages, Feedback, and Guidance, Icons, Multimedia and colors.		
6	**Self Study**	UX tools: Figma, Just In Mind, and any open-source tool for prototype designing, Mobile Ecosystem: Platforms, Application frameworks: Types of Mobile Applications: Widgets, Applications.		04
			Total	**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	To Study of open-source UX tools (Justinmind Prototype, Pidoco, Marvel, Figma Prototype) and create a simple design for a given problem definition.
2	**Know your client.** a. Design an app that can teach mathematics to children of 4-5 years age in schools in Rural Sector. b. Design an app that can teach mathematics to children of 4-5 years age in schools in Urban Sector. c. Design a site that can help people to sell their handmade products in metro cities. d. Design a site that can connect housewives and keep them engaged. **Note : Students should be able to do the following for any given problem statement** i. Analysis of user's/client's behavior eg their preferences, interests etc ii. What kind of interfaces will they like and why? iii. Existing apps - analyze and rate them. iv. What will be your choice of screen elements? v. How will your app/web design be better than the existing one?
3	**Goal-oriented design** - Design an experience for passengers whose flight /train is delayed.
4	**Design Principles** - Understand principles of good UI design by heuristic evaluation. Design UI for a given problem statement.
5	**Menus & Navigation** – Redesign of a user interface (Suggest and implement changes in Existing User Interface) for a given problem statement.
6	**Windows & Screen controls** – Design UI for a given problem statement. **a.** Design a navigator for a student new in your Institute. **b.** Design a navigator for a person new in tourist city/ village. **c.** Motor paralysis for differently able people. **d.** Vaccination App design with localization
7	**Icons** - Design appropriate icons pertaining to a given domain. (Eg. Greeting cards, Travelling, restaurants, Education, Medical, security at Airport, Malls etc)
8	**Colors** – Design a personal website for any socio-technical problem. Use color guidelines with statistical graphics for better visualization.
9	Design a Map-based UI(Web User) for the given problem statement. Example: Mumbai Dabbawallas with localization feature. Pet Care New Visitors to Hospital
10	To calculate the screen complexity of the existing Graphical User Interface and redesign the interface to minimize the screen complexity.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Human Computer Interaction	3rd	Alan Dix, J. E. Finlay, G. D. Abowd, R. Beale	Peason,Prentice Hall	2003
2	The Essential Guide to User Interface Design	3rd	Wilbert O. Galitz,	Wiley publication	2007
3	Design of everyday things	2nd	Donald A. Normann	Basic Books; Reprint edition	2013
4	Galitz's Human Machine Interaction	1st	Kalbande,Kanade,Iyer	Wiley Publications	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Interaction Design: Beyond Human Computer Interaction	5th	Rogers Sharp Preece	Wiley publications	2019
2	Mobile Design and Development	1st	Brian Fling	O'Reilly Media Inc.	2009.

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PR3-I	Main Project Stage I	0	0	2	4	6	0	0	2	1
		Examination Scheme								
		Component	Phase 1		Phase 2		ESE			Total
ELC		Major Project	40		40		20			100

Pre-requisite Course Codes	All courses
Course Objectives	To provide an opportunity for students to apply theoretical knowledge to real-world or research-based problems. It emphasizes project planning, technical implementation, documentation, and evaluation with real-world relevance
Course Outcomes (CO): ***At the End of the course students will be able to***	
PR3-I.1	Identify and analyze real-world problems to define clear, feasible project objectives.
PR3-I.2	Apply knowledge of engineering fundamentals and modern tools to design and develop effective solutions.
PR3-I.3	Manage project activities including teamwork, communication, and time management.
PR3-I.4	Test, validate, and evaluate the developed system.
PR3-I.5	Present the project work through structured documentation and presentation.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PR3-I.1	3	3	2	2	2	2	1	1	2	2	1	2
PR3-I.2	3	3	3	3	3	2	1	1	2	2	2	3
PR3-I.3	2	2	2	2	2	1	1	2	3	3	3	2
PR3-I.4	3	3	3	3	3	2	1	2	2	2	2	2
PR3-I.5	2	2	2	2	2	2	2	3	3	3	2	3

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
PR3-I.1	3	2	2	2	2
PR3-I.2	3	3	2	3	3
PR3-I.3	2	3	3	2	2
PR3-I.4	3	3	2	3	3
PR3-I.5	2	2	3	2	3

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create✓

Phase I Evaluation Rubric

P1 (15)	P2 (05)	P3 (07)	P4 (03)	P4 (05)	P5 (05)	Total Marks (Max=40) (Passing= 16)
Problem Identification & Literature Survey (Problem Definition/ Idea/Originality/ Scope & objectives/ Feasibility/ Application in Society/ Gap identification /Relevance of references/ Adequate references)	Analysis & Design (UML diagrams: Use case, Class, and Sequence diagram/Block diagram/ Algorithms)	Prototype and impleme-ntation	Technical Competency and Planning (Gantt chart / Work Division)	Presentation (Coverage of topic/use of graphics/Delivery & Speaking skills) and Ethics (Attitude/Deadline/ Distribution of Work)	Societal/ Industrial Relevance	

Phase II Evaluation Rubric

P1 (05)	P2 (20)	P3 (10)	P4 (05)	Total Marks (Max=40) (Passing= 16)
Algorithm/Technique Justification	Work done (Total 50% Implementation/ Meets project objectives/result analysis/test cases/solution shows originality)	Presentation (Coverage of topic/use of graphics/Delivery & Speaking skills/Presentation length) and Ethics (Attitude/Deadline/ Distribution of Work)	Quality of Documentation/ Report (in LaTeX)	

ESE Evaluation Rubric

P1 (02)	P2 (05)	P3 (10)	P4 (03)	Total Marks (Max=20) (Passing= 08)
Problem Definition (Idea/Originality/ Novelty/Feasibility/ Application in Society)	Presentation (Demo/Oral/Written) and Ethics (Attitude/Deadline/ Distribution of Work)	Work Done (Survey/Design/Simulation/ Experimentation/Tools Usage/Budget Plan/ Execution Plan)	Knowledge (Question-Answer)	

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)						Credits Assigned		
		L	T	P	O	E	L	T	P	Total
PE1	Digital Forensic	2	0	2	2	6	2	0	1	3
CS311		Examination Scheme								
		Component		ISE		MSE		ESE	Total	
		Theory		50		50		100	200	
		Laboratory		50		--		50	100	

Pre-requisite Course Codes		OS, CCN
At the End of the course students will be able to		
Course Outcomes	CO1	Apply forensic acquisition and analysis techniques using industry-standard tools on diverse platforms including mobile devices and cloud services.
	CO2	Analyze hardware, software, and firmware components involved in forensic investigations to extract and interpret relevant digital evidence.
	CO3	Evaluate forensic tools, scripting solutions (e.g., Python), and evidence handling procedures for their effectiveness in maintaining chain of custody and supporting legal processes.
	CO4	Create incident response strategies and evidence-centric processes that integrate digital forensics within broader frameworks of Information Assurance and forensic readiness.

Module No.	Unit No.	Topics	Ref.	Hrs.
1		**Introduction to Digital Forensics:** Cyberspace and criminal behaviour, cyber crime investigation, forensics science, forensics methodologies, cyber incident handling and response, Controlling Contamination-the Chain of Custody, Legal perspectives and security compliance of digital forensics. ISO standards relating to Digital Evidence, SWDGE Model SOP for Computer Forensics,NIST Digital Evidence.	1,2	6
2		**Network Forensics:** Applying Forensic Science to Networks Preparation and Authorization,Identification Documentation, Collection,and Preservation Filtering and Data Reduction,Class/Individual Characteristics and Evaluation of Source,Evidence Recovery, Investigative Reconstruction,Reporting Results Network Forensics Analysis Toolkit (NFAT) **Practical Exercises:**	3,4, 5,6	6

		Collection of network traffic data (Live acquisition and dump) Analyzing the traffic and persevering original. Implementing network sensors. Correlate with host sensors. Develop an algorithm to filter and extract anomalous traffic. Use open source and commercial domain tools for NFA. Event reconstruction. OSINT Tools-tshark,wireshark,tcpdump & xplico **Windows Forensics:** Digital Evidence on Windows Systems, File Systems, Data Recovery,Log Files, Registry, Internet Traces, Program Analysis. Tools and methods. **UNIX/Linux OS Forensics:** UNIX Evidence Acquisition Boot Disk, File Systems,Overview of Digital Evidence Processing Tools,Data Recovery,Log Files, File System Traces Internet Traces Tools and methods. **Practical Exercises:** To perform live analysis, capture volatile data, make images of media, analyze filesystems, analyze network traffic, analyze files, perform memory analysis, and analyze malware for a Windows subject on a Linux system with readily available free and open source tools. Windows Registry analysis. To build an in-house forensic capability via a variety of free, open-source, and commercial tools provided within the SANS Windows SIFT Workstation, NST etc		
3		**Malware Forensics, Memory Forensics and Disk Forensics:** File Identification and Profiling in Windows and Linux Analysis of a Suspect Program,Discovering and Extracting Malware and Associated Artifacts from Linux Systems and Windows systems, Memory Forensics: Analyzing Physical and Process,Memory Dumps for Malware Artifacts,Malware Incident Response: Volatile Data Collection and Examination on a Live Windows Systems and Linux systems. Tools and techniques of conducting runtime behavioral malware analysis and static code analysis. **Practical Exercises:** Collecting data from system memory (system registers, cache, RAM) in raw form and then carving the data from Raw dump. Extracting data from storage media by searching active, modified, or deleted files. OSINT tools, TCT, Autopsy, Sleuth-kit, volatility tools. **Email Forensics:** Broad steps in email forensics, Investigate Email Headers, Sender IP address, Verify Emails, Fake Email Investigation, Hacked, Email Scams. Tools and methods used in Email Forensics. **Practical Exercises:** Identification of malicious code, to study their payload, viruses,	6,7, 8	**6**

		worms, etc. Recovery and analysis of emails, including deleted emails, calendars, and contacts		
4		**Mobile Forensics and Cloud Forensic**: Mobile Forensic: Inside Mobile device: Cell Phone Crime, SIM Card, SIM Security, Mobile forensic & its challenges,, Mobile phone evidence extraction process, Practical mobile forensic approaches, Mobile operating systems overview, Mobile forensic tool leveling system, Data acquisition methods. Cloud Forensics: Cloud Computing Basics-Cloud Architecture-SAAS,IAAS & PAAS Cloud types-Public, Private, Community and Hybrid.Usage of cloud forensics, Challenges and Issues with Cloud Forensics,The three dimensions of cloud forensics, Cloud forensic organizational structure, Chain of Dependencies,Multi-Jurisdiction and multi-tenancy,Digital forensics evidence acquisition in cloud storage service: examining and evaluating tools and techniques,		
5		**Self-Study:** **Anti Forensics** Anti-forensics techniques, detection and countermeasures **Incident Handling and Incident Response** Development of Incident Handling and Response methodology (PDCAERF) Incident Response is an integral part of information Assurance (IA) Forensic readiness. Evidence-centric procedures and processes. Proper evidence handling and management,Determining the scope Containment strategies. Event reconstruction,Review the incident. Follow up reporting. **Practical Exercises:** To develop digital forensics is part of the incident response (IR) capability & Automated report mechanism.	7,8, 9,10	**4**
			Total	**28**

Sr. No.	Title of the Laboratory Exercises
1	Setting up an electronic evidence forensics laboratory
2	Network Forensics using Open Source Tools-NFAT
3	Computer Forensics/Memory Forensics/Disk Forensics
4	To develop Python scripting to perform a variety of forensic collection and analysis tasks.
5	Linux Forensics/Windows Forensics
6	Application Forensic/Email Forensics
7	Malware Forensics
8	Application Development for Mobile Forensics and Investigation
9	Cloud Forensic : FROST , UFED Cloud Analyzer, diffy, MD-CLOUD
10	To develop digital forensics as part of the incident response (IR) capability & Automated report mechanism.

References:

[1] Crime Scene Forensics A Scientific Method Approach by Robert C. Shaler, CRC Press Taylor & Francis Group

[2] Computer Forensics and Cyber Crime:An Introduction, Third Edition by Marjie T. Britz, Ph.D. Pearson Publications

[3] Computer Forensics Evidence Collection & Preservation, CHFI, EC-Council Press, Cengage Learning

[4] Digital Evidence and Computer Crime Forensic Science, Computers and the Internet Third Edition by Eoghan Casey, Elsevier Publications.

[5] Computer Forensics: Computer Crime Scene Investigation Second Edition by John R. Vacca, Charles River Media

[6] Malware Forensics by James M. Aquilina, Syngress Publications

[7] Introduction to Network Forensics by The European Union Agency for Network and Information Security (ENISA) accessed in January 2019.

[8] Windows Forensics: The Field Guide for Corporate Computer Investigations by Chad Steel, John Wiley & Sons Publications

[9] Windows Forensics Analysis Training SANS FOR500

[10] Advanced Network Forensics Course Threat Hunting & Incident Response SANS FOR572

[11] https://www.cerias.purdue.edu/site/research/forensics/

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-I	Natural Language Processing	2	0	2	5	10	2	0	1	3
CS321		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	Probability & Statistics, Basic knowledge of Python

Course Objective: This course will provide students with the theoretical knowledge and practical skills required to process, analyze and generate human language using computational techniques. The course emphasis on the necessary linguistic intuitions as well as a broad coverage of statistical, sequential processing and deep learning models that enable practical applications in text analysis and AI-driven communication.

Course Outcomes (CO): ***At the End of the course students will be able to***

1	Use Natural language representations and perform language processing tasks.
2	Evaluate solutions for a range of natural language components using existing algorithms, techniques and frameworks, including part-of-speech tagging, language modeling, parsing and semantic role labeling.
3	Apply algorithms for text mining and single and multi-class classification problems.
4	Apply Deep Learning techniques for language generation tasks.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1	3	-	2	-	-	-	-	-	-	-	-	-
2	-	-	2	-	-	-	-	-	-	-	-	-
3	3	-	2	-	-	-	-	-	-	-	-	-
4	3	-	-	-	2	-	-	-	2	-	-	3

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
1	2	-	-	-	-	-	-
2	2	2	-	-	-	-	-
3	2	-	-	-	-	-	-
4	2	2	2	-	2	2	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	Introduction to NLP		8
	1.1	A brief history of natural language processing, language challenges, Ambiguity, Applications, classical vs statistical vs deep learning-based AI Ethics: What It Is, Why It Matters Bias, Fairness, Language Technology implications for NLP Basic concepts in linguistic data Structure: Morphology, syntax, semantics, pragmatics, Stemming, Tagging Parts of speech-identify parts of speech, Constituent structure, NLP pipeline	1,2	
	1.2	Regular Expression, Words, Corpora, Text Normalization, Minimum Edit Distance, Words and Vectors, Cosine Similarity, TF-IDF, Bag of words, CBOW, Word embeddings (Word2Vec, GloVe, FastText), Word Sense Disambiguation	1,2	
2	Title	**Computational Tools for Text Analysis**		6
	2.1	Corpora and other data resources, Uses of corpora: Training and evaluation, Basic corpus analysis: Frequency distribution building and analyzing a corpus. Natural Language Toolkit (NLTK), spaCy for classical NLP Scikit-learn for ML-based NLP,TensorFlow, PyTorch, Hugging Face for deep learning NLP	1,2	
	2.2	Stemmimg vs Lemmatization in NLP Stemming: Comparing stemmers, Tagging: RE tagging, Trained taggers and backoff, Transformation-based tagging.	1,2	
3	Title	**Statistically based techniques for text analysis**		8
	3.1	Fundamentals of machine learning: Naive Bayes classifiers, Hidden Markov models; Viterbi decoding, Information and entropy; Decision trees and maximum entropy classifiers, Unigram, Bigram, Trigram, N-gram language models, Advance smoothing for language model	1,2	
	3.2	Machine learning in action: document classification, Information extraction: Types of information extraction, Regular expressions for personal names, Information extraction as sequential classification: chunking and Named Entity Recognition (NER)	1,2	
4	Title	**Deep Learning for NLP**		6
	4.1	Recurrent Neural Networks, LSTM, GRU Transformer Models(BERT, GPT) Attention and self-attention mechanism	1,2	
	4.2	Application of NLP: Topic modeling, Machine translation, Chatbots & conversational AI, Summarization, Text generation	1,2	
6	Self Study	LLMs and their application, LLM Evaluation methods		5*
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Install NLTK and perform basic Corpus analysis using NLTK such as (i) frequency distribution (ii) learn about morphological features of a word by analysing it.
2	Implement stemming and lemmatization operations for a corpus. (Porter Stemmer, Snowball Stemmer, Lancaster Stemmer, Regexp Stemmer)
3	Use N-gram model to predict the next word.
4	Use HMM model for POS tagging of sentence.
5	Perform POS tagging using Viterbi decoding.
6	Apply Naive Bayes Classifier for document classification.
7	Perform information extraction using Named Entity Recognition.
8	Use CFG to generate parse trees for a given sentence.
9	Build a model for NLP task like Spam detection /Machine translation /Sentiment Analysis/ Text Summarization.
10	Create transformer model for NLP task. Evaluate and optimize the model (Translation/Text generation).

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Natural Language Processing with Python	First Edition	Steven Bird, Ewan Klein & Edward Loper	O'Reilly Media, Inc. ISBN: 9780596516 499	2009
2	Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition	Second Edition	Daniel Jurafsky & James H Martin	Pearson Education	2013

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Deep Learning for NLP and Speech Recognition	First Edition	Uday Kamath, John Liu, James Whitaker	Springer	2019
2	Linguistic Fundamentals for Natural Language Processing	First Edition	Emily M. Bender	Morgan and Claypool	2013
3	Hands-On Large Language Models: Language Understanding and Generation	First Edition	Jay Alammar Maarten Grootendorst	O'Reilly Media	2024
4	Neural Network Methods for Natural Language Processing	First Edition	Yoav Goldberg	Morgan and Claypool	2017

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus,
Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-II	Business Analytics with Python	2	0	2	4	8	2	0	1	3
CS332		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	

Course Objective:

1. Introduce students to the fundamentals of the Python programming language and its data manipulation capabilities
2. Familiarize students with the various data analysis and visualization libraries in Python, such as Pandas, Numpy, Matplotlib, Seaborn, and Plotly.
3. Develop students' statistical thinking and enable them to apply statistical techniques to business data, including hypothesis testing, regression analysis, and time series analysis.
4. Teach students to develop predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees, and random forests

Course Outcomes (CO): ***At the end of the course students will be able to***

1. Understand the basics of Python programming language, including variables, data types, operators, and control structures.
2. Manipulate and analyze data using Python's data manipulation libraries such as Pandas and Numpy.
3. Apply statistical techniques and methods to business data, including probability distributions, hypothesis testing, and regression analysis.
4. Visualize and present data using Python visualization libraries like Matplotlib, Seaborn, and Plotly

CS331.1	Understand the basics of Business analytics and its applications in various industries
CS331.2	Gain proficiency in using Python for data manipulation, analysis and visualization
CS331.3	Apply statistical techniques and machine learning algorithms to solve business Problems
CS331.4	Apply analytics tools to gain insight from data for decision making.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS331.1	2											
CS331.2		2										
CS331.3			2	2		2						
CS331.4	2		2	2	2	2		2				

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS331.1							
CS331.2							
CS331.3							
CS331.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply √	Analyze	Evaluate	Create

Theory Component:

Module No.	Unit No.	Topics	Ref.	Hrs.
	Title	**Introduction to Business Analytics**		
1	**1.1**	Overview and importance of business analytics , What is Business Analytics, Modern Business Analytics, Visual perspective of Business Analytics, Data for Business Analytics, Models in Business Analytics, Introduction to Python and its data science ecosystem, Setting up the development environment	**1**	**2**
	1.2	Python : Classes and Objects New Style Classes, Inheritance and Mixins, Properties and Slots Static and Class Methods ,Abstract Base Classes, Method Overriding Attributes and Functors, Decorators and Factories Descriptors and MetaClasses	**2,3**	**4**
	1.3	Difference between Business Analytics and Data Analytics: Scope and applications, Key objectives, Techniques used, Tools and technologies, Role in decision-making, Comparison of methodologies, Industry use cases. Case Studies in Business Analytics – Applications of Business Analytics in sectors like Banking (credit risk analysis), Telecom (churn prediction), and Sports Analytics (performance optimization).	**1,2**	**2**

	Title	**Data Preprocessing and Exploratory Data Analysis**		
2	**2.1**	Data cleaning and preprocessing techniques Handling missing values and outliers, Data Transformation Data Integration and Reshaping, Data Wrangling, Handling Text and Categorical data,	**2,3**	**3**
	2.2	Descriptive statistics and data visualization, Data Distribution and Outliers, Data exploration techniques, Data Cleaning and Preprocessing, Correlation and Relationships, Feature engineering and transformation, Dealing with Time Series Data,Dealing with Time Series Data, Time series Analysis.	**2,3**	**3**
	2.3	Case Studies in Data Preprocessing and Exploratory Data Analysis – Handling missing data in healthcare records, Feature engineering in retail customer segmentation, Time series analysis for stock market trends.	**3,4**	**1**
	Title	**Statistical Analysis using Python**		
3	**3.1**	Statistical Analysis process, types – descriptive, inferential, predictive, prescriptive, explorative, causal analysis, methods – mean, regression, sample size determination, standard deviation, hypothesis testing Prescriptive Analytics - Linear optimization, Integer optimization, Non-linear programming and Simulation	**5,6**	**3**
	3.2	Hypothesis testing- errors in hypothesis testing, Alpha and critical values, independent and dependent t-test, Chi-square test, Goodness of Fit test, Test of independence, Anova, One-way Anova, Two-way Anova. Introduction to Jamovi for statistical analysis, its interface, and application in hypothesis testing.	**5,6**	**3**
	3.3	Case Studies in Statistical Analysis – Hypothesis testing in clinical trials, Regression analysis in real estate pricing, ANOVA for product performance comparison.	**5,6**	**1**
	Title	**Tools for Business Analytics**		
4	**4.1**	Comparative study of NoSQL databases – Document database, Graph database, Key-value stores, Wide column database, Big Data Analysis using Hive Text Analytics- Sentiment Analysis, Naïve Bayes model for Sentiment Classification, Challenges of text analytics- using n-grams Forecasting- Forecasting using Moving Average, Calculating Forecast error- Mean Absolute Error, Exponential smoothing, Auto Regressive Integrated Moving Average Models	**1-4**	**4**
	4.2	Case studies in Business Analytics – Transport and logistics, Healthcare, Retail, Finance, E-commerce, Manufacturing, Marketing and Customer Analytics, Supply Chain Optimization, Fraud Detection, Human Resource Analytics.	**1-4**	**2**
5	**Self-Study**	Use of softwares like R, Power BI, SPSS for the model development and interpreting the outputs Visualization using Tableau		**4***
		Total		**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Exp. No.	Experiment Details
1	Data Importing and Exporting: ● Read a CSV file into a pandas Data Frame ● Export a Data Frame to an Excel file. ● Load JSON data into DataFrame.
2	Data Cleaning and Preprocessing: ● Handle missing values by imputing them with mean, median, or mode ● Removing outliers based on a specific threshold. ● Transform variables using log transformation or standardization. ● Remove duplicate records from a data frame. ● Standardize date formats across a dataset
3	Data Integration and Reshaping: ● Merge two or more data frames based on a common key. ● Concatenate multiple Data Frames vertically or horizontally. ● Pivot a Data Frame from long to wide format or vice versa ● Stack and unstack columns or levels in a Data Frame ● Data Wrangling
4	Dealing with Time Series Data ● Resample a time series to a different time frequency(eg. Daily, monthly) ● Compute moving averages or rolling sums over a time series ● Shift a time series forward or backward in time.
5	Data Quality Assurance ● Check for missing values and assess data completeness. ● Identify and handle inconsistent or erroneous data ● Calculate data quality metrics, such as data completeness or accuracy
6	Exploratory data analysis and visualization using Tableau/Power BI
7	Big data analysis using Hive/ Use of graph database
8	Forecasting, use of ARIMA model
9	Sentiment Analysis of YouTube, Facebook, Twitter Data
10	Regression Analysis on real time data e.g Stock Price

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Business Analytics Methods, Models and Decisions	III	James R. Evans	Pearson Publication	2021
2	Business Analytics The Science of Data- Driven Decision Making	II	Dinesh Kumar	Wiley Publications	2021
3	Python for Data Analysis: Data Wrangling with pandas, NumPy and Jupyter	I	Wes McKinney	O'REILLY	2022
4	Fundamentals of Business Analytics	II	N Prasad, Seema Acharya	Wiley publication	2016

Reference Book

Sr. No.	Title	Edition	Authors	Publisher	Year
1	An Introduction to Business Analytics ISBN : 9789082017939	I	Ger Koole	MG books Amsterdam	2019
2	Hands-On Data Analysis with Pandas: A Python data science handbook for data collection, wrangling, analysis and visualization	II	Stefanie Molin and Ken Jee	Packt Publishing	2021

Bharatiya Vidya Bhavan's

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-I	Fundamentals of Signal & Image Processing	2	0	1	4	7	2	0	1	3
CS341		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	
Course Objective: • Introduce computational techniques for processing digital signals and images. • Equip students with software required for real-world applications. • Explore various applications in signal and image processing.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CS341.1	Understand basic concepts of signals, systems, and their properties in both time and frequency domains.
CS341.2	Apply Fourier and other transforms to analyze signals and images for feature extraction and enhancement.
CS341.3	Implement filtering and enhancement techniques in spatial and frequency domains for signals and images.
CS341.4	Analyze and interpret the results of various signal and image processing techniques applied to real-life problems.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS341.1	3	2	-	-	-	-	-	-	-	-	-	1
CS341.2	3	3	-	-	2	-	-	-	-	-	-	1
CS341.3	3	3	2	-	3	-	-	-	-	-	-	1
CS341.4	3	3	3	3	3	-	-	-	-	-	-	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS341.1	3	2			-	-	-
CS341.2	3	3			-	-	-
CS341.3	3	3			2	-	-
CS341.4	3	3			3	-	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember ✔	Understand ✔	Apply✔	Analyze✔	Evaluate✔	Create✔

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Digital Signals**		8
	1.1	DT Signals, Signal Operations, Sampling and Reconstruction	T1, T2	
	1.2	Frequency Domain Processing: DFT, FFT, Real Time Filtering using FFT	T1, R1	
2	Title	**Image Fundamentals**		6
	2.1	Sampling and Quantization, Digital Image Representation, Image Coordinate System	T3, R2	
	2.2	Image Transforms: DFT, DCT, Walsh Transform, Wavelet Transform	T3, R2	
3	Title	**Image Enhancement - Spatial Domain**		6
	3.1	Zero memory Point Operations, Histogram Processing	T3, R1	
	3.2	Spatial Filtering, Homomorphic Filtering	T3, R1	
4	Title	**Image Segmentation**		4
	4.1	Detection of discontinuities, Edge linking and Boundary Detection, Thresholding, Region Oriented Segmentation	T3, R2	
	4.2	Binary Image Processing – Morphology Operators and Boundary Detection, Region Filling	T3, R2	
5	Title	**Image Compression**		4
	5.1	RLE, Huffman Coding, Arithmetic Coding	T3, R1	
	5.2	Predictive Coding and JPEG	T3, R1	
6	Self Study	Object Representation, Object Detection, Object Recognition, Applications of Image Processing	T3, R1, R2	

Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Basic Signal Operations
2	DFT
3	FFT and Spectrogram Analysis
4	Real time filtering using Overlap Add Method/Overlap Save Method
5	Image Enhancement: Histogram Equalization, Smoothing & Sharpening
6	Image Filtering in Spatial and Frequency Domains
7	Edge Detection (Sobel, Prewitt, Canny)
8	Segmentation using Thresholding and Watershed
9	Morphological Operations
10	Image Compression

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Digital Signal Processing: Principles, Algorithms, and Applications	4th	Proakis, Manolakis	Pearson	2007
2	Discrete-Time Signal Processing	3rd	Alan V. Oppenheim and Ronald W. Schafer	Pearson	2014
3	Digital Image Processing	4th	Gonzalez & Woods	Pearson	2017

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Handbook of Image and Video Processing	2nd	Al Bovik	Academic Press	2005
2	Feature Extraction & Image Processing for Computer Vision	4th	Nixon, Aguado	Academic Press	2019

Video Lectures

1. NPTEL - Signals and Systems (Prof. V. M. Gadre): https://nptel.ac.in/courses/108101174
2. NPTEL - Digital Image Processing (Prof. P. K. Biswas): https://nptel.ac.in/courses/117105079

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM	Cloud Computing	-	2	2	5	10	-	2	1	3
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
MDCE13		Theory	100		100		100		300	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	Machine Learning

Course Objective: This course introduces Concepts of cloud computing. This course focuses on the architecture of cloud computing along with the application of the same.

Course Outcomes (CO): ***At the End of the course students will be able to***

MC312.1	Demonstrate an understanding of cloud service models, deployment models, and mobile cloud computing through practical applications
MC312.2	Differentiate between various virtualization technologies and assess their suitability for different cloud environments.
MC312.3	Select and justify appropriate cloud computing services for a given scenario based on performance, cost, and security considerations
MC312.4	Design and implement cloud-based solutions using OpenStack, AWS, Google Cloud Platform and Microsoft Azure components for real-world applications.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
MC512.1	2	-	-	-	-	-	-	-	-	-	-	**1**
MC512.2	2	2	-	-	-	-	-	-	-	-	-	**1**
MC512.3	3	3	3	-	-	-	-	-	-	-	-	**2**
MC512.4	2	2	3	3	2	2	2	2	2	2	2	**2**

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
MC512.1					
MC512.2					
MC512.3					
MC512.4					

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✓

Tutorial-Cum-Lab-Based Approach to Cloud Computing

This course adopts a tutorial-cum-lab model to blend conceptual clarity with hands-on skills in cloud computing. It is structured to help students develop as solution architects with a focus on both technology and systems thinking.

Pre-Session: Students engage in pre-readings from curated books, technical papers, and white papers to build foundational knowledge aligned with each topic.

In-Session: Sessions emphasize solving concept-based review questions and real-world use case analysis. Focus areas include cloud service models, virtualization, mobile cloud computing, and platform-specific solutions (CO1–CO3).

Post-Session: Students summarize their learning, reinforcing understanding and connecting concepts to practical implementations (CO4).

This approach supports students in acquiring critical thinking, design, and decision-making skills essential for modern cloud solution development.

Tutorials

Sr. No.	Syllabus	Hrs.
1	Introduction to Distributed Computing and Cloud Fundamentals: ● Historical perspective of computing systems ● Distributed computing principles ● Introduction to cloud computing ● Key characteristics of cloud computing ● Cloud computing benefits and challenges ● Business drivers for cloud adoption	2
2	Virtualization Concepts and Levels ● Virtualization fundamentals ● Hypervisors and virtual machines ● Levels of virtualization ● Hosted virtualization ● Type-1 and Type-2 hypervisors ● Hardware-assisted virtualization	2

3	Advanced Virtualization Techniques ● Para-virtualization ● Full-virtualization ● CPU virtualization ● Memory virtualization ● Disk virtualization ● I/O mapping in virtual environments ● Network virtualization	2
4	Containerization and Modern Virtualization ● Introduction to containerization ● Container architecture ● Docker fundamentals ● Container vs. VM comparison ● Container lifecycle management ● Container orchestration concepts ● Container networking and storage	2
5	Cloud Computing Service Models ● Infrastructure as a Service (IaaS) ● Platform as a Service (PaaS) ● Software as a Service (SaaS) ● Anything as a Service (XaaS) ● Service model comparison ● Service model selection criteria ● Responsibilities in each service model	2
6	Cloud Deployment Models and Networking ● Public cloud ● Private cloud ● Hybrid cloud ● Community cloud ● Multi-cloud environments ● Cloud networking concepts ● Virtual networks and subnets ● Network security in the cloud	2
7	Data Center Technology and Cloud Architecture ● Data center technologies ● Cloud infrastructure components ● Monolithic architecture ● Microservices architecture ● Multi-tier application design ● Scalability patterns ● High availability in cloud applications	2

8	Cloud Programming and Optimization ● Cloud programming models ● Cloud-native application design principles ● Serverless computing ● Performance optimization in the cloud ● Cost optimization strategies ● Cloud migration approaches ● Refactoring applications for the cloud	2
9	DevOps for Cloud-Based Software Development ● DevOps fundamentals ● Continuous Integration (CI) ● Continuous Delivery/Deployment (CD) ● Infrastructure as Code (IaC) ● Configuration management ● DevOps tools and platforms ● DevOps best practices	2
10	Cloud Security and DevSecOps ● Cloud security challenges ● Shared responsibility model ● DevSecOps principles ● Security as Code ● Static Application Security Testing (SAST) ● Dynamic Application Security Testing (DAST) ● Vulnerability Assessment and Penetration Testing (VAPT) ● Secure Software Development Lifecycle (SecSDLC) ● Security tools and frameworks	2
11	AWS Cloud Services Fundamentals ● AWS Global Infrastructure ● AWS IAM (Identity and Access Management) ● AWS EC2 (Elastic Compute Cloud) ● AWS S3 (Simple Storage Service) ● AWS EBS (Elastic Block Store) ● AWS EFS (Elastic File System) ● AWS VPC (Virtual Private Cloud) ● AWS Security Groups ● AWS Cost Explorer and Calculator	2
12	AWS Advanced Services and Development ● AWS Lambda for serverless computing ● AWS RDS (Relational Database Service) ● AWS CodeCommit, CodeBuild, and CodeDeploy ● AWS CI/CD pipeline ● AWS ECS (Elastic Container Service) ● AWS EKS (Elastic Kubernetes Service) ● AWS SageMaker for ML/AI workloads ● AWS MLOps practices	2

13	Google Cloud Platform and Microsoft Azure ● Google Cloud Platform (GCP) overview ● Google App Engine (GAE) ● Google Compute Engine (GCE) ● Microsoft Azure fundamentals ● Azure core services ● Cross-cloud comparison ● Choosing the right cloud provider ● Transitioning between cloud providers	2
14	Multi-Cloud Strategy and Future Trends ● Multi-cloud architecture design ● Multi-cloud management tools ● Multi-cloud security considerations ● Cloud service brokerage ● XaaS (Anything as a Service) ● Edge computing and cloud ● Quantum computing in the cloud ● AI/ML and cloud convergence ● Future of cloud computing	2
15	**Mini-Project**	-
Total		**28**

Laboratory Component

Sr. No.	Title of the Experiment
1	**Creating and Running Virtual Machines on Hosted Hypervisors (Virtual Box, KVM, QEMU)** **Objective**: To create and run virtual machines on hosted hypervisors such as KVM, VMware Workstation, and Oracle VirtualBox.
2	**Docker and Docker Compose** **Objective:** To provide students with a thorough understanding of Docker and Docker Compose, enabling them to containerize applications, manage multi-container setups, and apply best practices for containerization in real-world scenarios. Participants will learn to build, deploy, and manage containerized applications efficiently.
3	**LXD and Multipass** **Objective:** To provide students with a thorough understanding of **Multipass** and **LXD**, enabling them to create lightweight virtual machines and manage Linux containers efficiently. Participants will learn to deploy and orchestrate virtualized and containerized workloads, applying best practices for managing scalable and secure environments.

4	**Automation using Ansible** **Objective:** Automation in Software Install/Configuration/operations. Description: Set up network applications, apache2, ftp,monitoring and network security devices various security devices (firewalls, IDS/IPS). To use Ansible playbooks for automating routine tasks, such as rule updates and device configuration.
5	**Kubernetes using Minikube** **Objective:** 1. Learn basic Kubernetes commands for resource inspection. 2. Understand the process of making deployed applications accessible both internally and externally through service exposure. Learn to deploy servers like NGINX on Kubernetes pods using YAML for effective resource management.
6	**Kubernetes using Minikube-Testing** Configure Horizontal Pod Autoscaling (HPA): 1. Test Case 1: Validate Load Balancing Across Pods 2. Test Case 2: High Load Performance 3. Test Case 3: Pod Failure and Recovery 4. Test Case 4: Service Degradation Under Node Failure 5. Test Case 5: Load Balancing with StatefulSets 6. Test Case 6: Network Latency Simulation
7	**OpenStack Implementation** **Objectives:** 1. Establish a Stable and Scalable Cloud Infrastructure. 2. Deploy OpenStack on Ubuntu servers 3. Creating a robust cloud infrastructure capable of handling diverse workloads 4. Optimize Resource Utilization and Efficiency
8	**Implementing and Managing IAM Roles, EC2 instance and S3 in AWS** Objectives: 1. Understand the importance and functionality of Identity and Access Management (IAM) roles in AWS. 2. Learn how to create and configure IAM roles to assign granular permissions for different AWS services and resources. 3. Ensure secure and appropriate access control for AWS resources using IAM roles.
9	**Building and Deploying Web Applications using AWS Elastic Beanstalk** **Objective:** 1. To design a lab on AWS Elastic Beanstalk. 2. The lab will focus on deploying, managing, and scaling web applications using AWS Elastic Beanstalk.
10	Implementing the AWS VPC and hosting multi tier application (public subnet-front-end and Private subnet- back-end)
11	**DevOps and DevSecOps in AWS** **Scenario:** A SaaS company wants to implement a CI/CD pipeline with security checks. Problem Statement: Build a secure CI/CD pipeline using AWS CodePipeline, CodeBuild, and security scanning tools. Expected Outcomes: ● Automated build, test, and deployment. ● Security vulnerabilities detected early. **Guided Solution:** 1. Set up AWS CodePipeline. 2. Integrate AWS CodeBuild and AWS Security Hub. 3. Deploy application to AWS Lambda or ECS.

12	**Deploying Containerized Applications with Kubernetes & Redshift** **Scenario:** An e-commerce company wants to containerize its application and use AWS Redshift for analytics. **Problem Statement:** Deploy a containerized app on EKS and set up a Redshift data warehouse. Expected Outcomes: ● Microservices running on AWS EKS. ● Redshift configured for analytics. **Guided Solution:** 1. Create a Kubernetes cluster on AWS EKS. 2. Deploy a sample microservices application. 3. Set up AWS Redshift and load sample data.
13	**Scenario-Based Lab** **1. Connecting On-Premises Databases to AWS Cloud** **Scenario:** A financial company wants to securely connect its on-premises PostgreSQL database to AWS RDS with minimal downtime. **Problem Statement:** Set up a secure connection between an on-premises database and AWS RDS using AWS Direct Connect and AWS VPN. **Expected Outcomes:** Secure connection established between on-premises and AWS RDS Data transfer with minimal latency **Guided Solution:** 1. Configure AWS Site-to-Site VPN. 2. Set up AWS Direct Connect for low-latency connection. 3. Use AWS DMS for database replication. **2. Data Migration from On-Premises to AWS Scenario:** A retail company needs to migrate its large MySQL database to AWS with minimal downtime. **Problem Statement:** Use AWS DMS and Snowball to migrate data efficiently. **Expected Outcomes:** ● Data successfully migrated with integrity. ● Reduced downtime during migration. **Guided Solution:** 1. Use AWS Schema Conversion Tool for compatibility. 2. Set up AWS DMS for real-time replication. 3. Use AWS Snowball for bulk data transfer. **3. Cloud Automation Using Terraform and Ansible** **Scenario:** A startup needs an automated AWS infrastructure setup. **Problem Statement:** Use Terraform to deploy AWS infrastructure and Ansible for configuration management. Expected Outcomes: ● Infrastructure as Code (IaC) deployed successfully. ● Configuration automated via Ansible. **Guided Solution:** 1. Write a Terraform script to provision an EC2 instance. 2. Use Ansible playbook to install required software. 3. Test deployment.

14	**Building Data Pipelines from Ingestion to Analytics** **Scenario:** A telecom company needs real-time analytics on customer call data. **Problem Statement:** Build a data pipeline using AWS Kinesis, Glue, and Redshift. **Expected Outcomes:** ● Real-time data ingestion and processing. ● Insights generated from Redshift queries. **Guided Solution:** 1. Stream data using Amazon Kinesis. 2. Process data with AWS Glue. 3. Store and analyze in Amazon Redshift. --- **Multi-Cloud Strategies & Hands-on** **Scenario:** A multinational corporation needs a hybrid cloud setup. **Problem Statement:** Deploy a multi-cloud environment using AWS and Azure with centralized logging. **Expected Outcomes:** ● Workloads running across AWS and Azure. ● Centralized logging enabled. **Guided Solution:** 1. Deploy Kubernetes clusters on AWS EKS and Azure AKS. 2. Configure a common logging and monitoring solution.
15	**Building a Video Streaming Platform for Startup** **Objectives:** 1. Design scalable video streaming architecture. 2. Utilize AWS services for storage, transcoding, and delivery. 3. Ensure high availability, scalability, and low latency. **Lab Outcomes:** 1. Deploy an MVP video streaming platform. 2. Configure transcoding and adaptive bitrate streaming. 3. Secure video uploads and delivery using AWS best practices.
16	**Unified Platform for Big Data Processing** **Objective:** Setting up and utilizing Apache Hadoop and Apache Spark to process large datasets efficiently. This lab focuses on understanding the integration of Hadoop's distributed storage with Spark's fast computation framework, enabling a unified platform for big data processing.
17	**Implementing the USE CASE using Google API & Google App Engine** **Objectives :** 1. Authenticate and integrate Google APIs. 2. Deploy applications on Google App Engine. 3. Implement a real-world use case using Google Cloud.

Textbooks:

[1] Cloud Computing: Concepts, Technology & Architecture by Thomas Erl
[2] Distributed and Cloud Computing by Kai Hwang
[3] Cloud Computing: A Hands-on Approach by Arshdeep Bahga and Vijay Maddisetti
[4] AWS Certified Solutions Architect - Associate Study Guide by Ben Piper & David Clinton
[5] AWS Certified Solutions Architect - Professional Study Guide by Brett McLaughlin & Jorge Acetozi
[6] Google Cloud Platform for Architects by Vitthal Srinivasan
[7] Microsoft Azure: Planning, Deploying, and Managing the Cloud by Marshall Copeland
[8] The Definitive Guide to Container Security by Gareth Baruch, Dave Dukinfield, and Mohit Bhasin, Wiley Publishing.

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-II	Deep Learning	2	-	2	2	5	-	-	1	3
CS322		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	Machine Learning
Course Objective: This course introduces Concepts of cloud computing. This course focuses on the architecture of cloud computing along with the application of the same.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
MDCS23.1	Understand the fundamentals of deep learning.
MDCS23.2	Understanding the working of Convolutional Neural Networks and RNN in decision making.
MDCS23.3	Implement, optimize and tune state-of-the-art deep neural network architectures
MDCS23.4	Implement major deep learning algorithms, the problem settings, and their applications to solve real world problems

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
MDCS23.1	2	-	2	2	-	-	-	-	-	-	-	1
MDCS23.2	2	2	-	-	2	-	-	-	-	-	-	1
MDCS23.3	3	3	3	2	2	-	-	-	-	-	-	2
MDCS23.4	2	2	3	3	2	2	2	2	2	2	2	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
MC512.1	1	-	-	2	-
MC512.2	1	2	-	2	-
MC512.3	1	-	-	2	-
MC512.4	1	2	-	2	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✓

Theory Component

Module No.	Topics	Ref.	Hrs.
1	Introduction	1,2	5
	Basic supervised classification task, optimizing logistic classifier using gradient descent, stochastic gradient descent, momentum, and adaptive sub-gradient method. Neural Networks: Feedforward neural networks, deep networks, regularizing a deep network, model exploration, and hyper parameter tuning.		
2	Introduction to Convolution Neural Networks CNN	1,2	6
	Architecture and Operations, convolutional layer, Pooling layer, Variants of the Convolution Model, Forward and Backward propagation, Training a deep neural network, hyper-parameter tuning, Generalization Gap – Under-fitting Vs Over-fitting – Optimization, Normalization.		
3	Optimization algorithms	1,2	6
	Mini-batch gradient descent, exponentially weighted averages, RMS prop, Learning rate decay, the problem of local optima, Batch norm – Parameter tuning process.		
4	Sequence Modeling: Recurrent Nets	1.2	7
	Unfolding computational graphs, recurrent neural networks (RNNs), bidirectional RNNs, encoder-decoder sequence to sequence architectures, deep recurrent networks, LSTM networks.		
5	Interactive Applications of Deep Learning	1.2	4
	Generative Adversarial Networks, Deep Reinforcement Learning.		
6	Self Study		
	Deep Learning Research: Autoencoders, Deep Generative Models: Boltzmann Machines Restricted Boltzmann Machines, Deep Belief Networks		
		Total	28

Laboratory Component

Sr. No.	Title of the Experiment
1	Study and Implement different classifier for decision tree algorithm eg. J48
2	Working and Implementation of Bayesian Hyperparameter Optimization
3	Implement Single-Layer Feedforward Neural Network
4	Implement Multi-Layer Feedforward Neural Network
5	Learning TensorFlow API and Keras API
6	Implement Optimization technique
7	Build and train RNNs, work with NLP and word embeddings
8	Use LSTMs and GRUs for implementing a real world problem
9	Implement a GAN for a given problem
10	Implement DCGAN for a given problem

Note: Course coordinators can use their own problem statements to solve listed lab sessions.

Textbooks

1. Deep Learning- Ian Goodfellow, YoshuaBengio and Aaron Courvile, MIT Press, 2016
2. Deep Learning with Python - Francois Chollet, Released December 2017, Publisher(s): Manning Publications, ISBN: 9781617294433

References

1. Bunduma, N. (2017). Fundamentals of Deep Learning
2. Heaton, J.(2015). Deep Learning and Neural Networks, Heaton Research Inc.

Additional References

1. Goodfellow, I. (2016). Deep Learning. MIT Press.
2. Deng, L., & Yu, D. (2009). Deep Learning: Methods and Applications (Foundations and Trends in Signal Processing). Publishers Inc.
3. Hall, M.L, (2011). Deep Learning. VDM Verlag

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-I	Big Data Analytics	2	0	1	4	7	2	0	1	3
		Examination Scheme								
CS331		Component	ISE		MSE		ESE		Total	
		Theory	75		75		150		300	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	CS204, AS201
Course Objective:	
Course Outcomes (CO): *At the End of the course students will be able to*	
CO1	Understand the building blocks of Big Data Analytics.
CO2	Choose appropriate storage structures to make sense out of big data.
CO3	Apply scalable algorithms based on Hadoop and Map Reduce to perform Big Data Analytics.
CO4	Achieve adequate perspectives of big data analytics in various applications like, recommender systems, social media applications etc.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1			3	3	3							
CO2			3	3	3							
CO3			3	3	3							
CO4			3	3	3							

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CO1							
CO2							
CO3							
CO4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✔️	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Introduction to Big data**		
	1.1	Big Data characteristics, types of Big Data, Traditional vs. Big Data, Big data challenges	1	**1**
2	**Title**	**Memory-efficient data structures**		
	2.1	Introduction to data streams, problems related to handling data stream, Mining Data Streams using DGIM algorithm, Need of dimensionality reduction - PCA, SVD - Singular Value Decomposition	3	
	2.2	Hash functions, universal / perfect hash families, Bloom filters, Sketches for distinct count, Flajolet Martin Sketch, Majority Algorithm, Misra-Gries sketch, Count-Min Sketch, Count Sketch, LSH, MinHash	2	**6**
3	**Title**	**Scaling with Big Data using Hadoop**		
	3.1	HDFS - Data in Hadoop, MapReduce: The Map Tasks, Grouping by Key, The Reduce Tasks, Combiners, YARN, Details of MapReduce Execution, Coping With Node Failures	1	
	3.2	Modern Big Data Architecture Data Lake House Architecture: Combining data lakes and data warehouses (Databricks Delta Lake, Apache Iceberg) Modern Data Stack: Introduction to ELT vs ETL pipelines Introduction to Serverless Big Data Computing: Overview of Cloud-native approaches - AWS Lambda, Azure, Google Cloud	1	**7**
4	**Title**	**Frequent Itemsets And Clustering**		
	4.1	Handling Larger Data sets in Main Memory Algorithm - Park, Chen and Yu Algorithm, The Multi stage Algorithm, The Multihash Algorithm. The SON Algorithm, BFR clustering algorithm, CURE algorithm	1	**4**
5	**Title**	**Big Data Analytics Applications**		
	5.1	Clustering of Social-Network Graphs - Clique Percolation Method, counting triangles, PageRank, Efficient Computation of PageRank, Topic-Sensitive PageRank, Link Spam, Hubs and Authorities	1	
	5.2	Scaling with Big Data using Apache SPARK SPARK Ecosystem, SPARK streaming - Flink, and Kafka Streams, Distributed Batch Processing with Spark		**5**

6	**Title**	**Big Data Machine Learning**		
	6.1	Decision Trees for Big Data Analytics - fraud detection, Big Data Predictive Analytics, K means using map reduce, Gradient Descent for Large-Scale Learning	1	5
7	**Self Study**	Recommendation Systems: Introduction, A Model for Recommendation Systems, Content based Recommendation System, Collaborative-Filtering System: Nearest Neighbour Technique	4	**4***
		Total		28

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Installation of Hadoop and execution of HDFS commands
2	Study of any latest research paper on memory efficient data structure for big data.
3	Demonstrate use of modern tools like Pandas for Exploratory Data Analysis.
4	Implement algorithms in Map-Reduce on Strings and integers
5	Implement algorithms in Map-reduce on Relational Algebra
6	Download a real world dataset and find insights using map reduce. For e.g for a movie dataset list all the movies and the number of ratings, list all the Movie IDs which have been rated (Movie Id with at least one user rating it), list all the Users who have rated the movies (Users who have rated at least one movie), list of all the User with the max, min, average ratings they have given against any movie
7	To use Apache Hbase and implement CRUD operations on database inside HDFS.
8	Extract facts in real world dataset using Hive
9	Extract sessions in real world dataset using Pig
10	Implement wordcount using Apache Spark.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mining of Massive Datasets	3rd	Anand Raja Raman and Jeff Ullman	Cambridge University	2019
2	Algorithms and models of computation	1st	Jeff Erickson	University of Illinois	2015
3	Introducing Data Science	3rd	Davy Cielen, Meysman Mohamed Ali	Dreamtech Press	2014
4	Learning Apache Spark 2	2nd	Muhammad Asif Abbasi	Packt Publishing	2017

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Data streams :Algorithms and applications.	2nd	Muthukrishnan. S.	now publishers Inc	2005
2	Introducing Data Science	3rd	Davy Cielen, Meysman, Mohamed Ali	Dreamtech Press	2015
3	Hadoop, the Definitive Guide	3rd	Tom White	O'Reilly	2013

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-II	Augmented Reality and Virtual Reality	2	0	2	4	8	2	0	1	3
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
CS332		Theory	50		50		100		200	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	Problem Solving using Object Oriented Programing
Course Objective:	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CS332.1	Identify and explain the differences among AR, VR, MR, and XR
CS332.2	Compare various AR/VR display and tracking technologies
CS332.3	Design intuitive interaction methods for immersive experiences
CS332.4	Evaluate the use of AR/VR in real-world applications such as healthcare, education, gaming, and simulation
CS332.5	Develop immersive scenes using Unity or Unreal

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO 8	PO9	PO10	PO11	PO12
CS332.1	2	-	-	-	-	-	-	-	2	-	-	2
CS332.2	2	2	-	-	-	-	-	-	2	-	-	2
CS332.3	2	2	2	2	-	-	-	-	2	-	-	2
CS332.4	2	2	-	2	-	-	-	-	2	-	-	2
CS332.5	3	2	2	2	3	-	-	-	2	-	2	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS332.1	**1**	**1**	**1**	**-**	-	**-**	**-**
CS332.2	**1**	**2**	**1**	**-**	-	**2**	**-**
CS332.3	**1**	**2**	**1**	**-**	-	**-**	**-**
CS332.4	**1**	**2**	**1**	**-**	-	**2**	**-**
CS332.5		**3**				**3**	

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create
	✔	✔	✔	✔	✔

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.	CO PO Mapped
1	Title	**Introduction to AR**		6	CO1
	1.1	Introduction to Augmented Reality, History of Augmented Reality, Taxonomy, Technology and Features of Augmented Reality, Difference Between AR and VR, Challenges With AR, Types of AR: Marker based, Marker Less etc.	T 1,2,3 R 2		
	1.2	**AR Concepts:** The Two-Step Process of Augmented Reality Applications, Three core components of implementing an AR Application (Sensor, Processor, Display), Computer Graphics, Dimensionality. AR Systems and Functionality, Augmented Reality Methods, Visualization Techniques For Augmented Reality, Mobile Projection Interfaces.	T 1		
2	Title	**AR Technologies and architecture**	T 1,2,3	6	CO2
	2.1	**Overview of devices** :HMDs, mobile AR, smart glasses. Audio Displays, Haptic Displays, Visual Displays, Visual Perception, Spatial Display Model. Tracking, Sensors Tracking, Calibration, and Registration, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors, Optical Tracking, Sensor Fusion.			
	2.2	**Introduction to AR SDKs** (ARCore, ARKit, Vuforia) Unity and Unreal Engine introduction.			
3	Title	**Perception, Immersion & Interaction Technologies**		5	CO3
	3.1	**Human Perception and AR/VR-** Visual perception and depth cues, Audio perception in virtual environments. Tracking, Haptics, Motion Controllers, Spatial audio, **AR Techniques** :Introduction to Marker Based Approach, Marker-Based Tracking, Types of Markers, Marker Camera Pose and Identification, Visual Tracking, Marker Types, Template Markers, 2D Barcode Markers, Imperceptible Markers. Marker Less Approach, Localization	T 1,3,4		

		Based Augmentation, Real World Examples, Tracking Methods Visual Tracking, Feature Based Tracking, Hybrid Tracking, Initialization and Recovery			
	3.2	Real-Time Rendering, Ray Tracing vs Rasterization, 3D Scene Composition, Stereoscopy, Shaders, Spatial Audio, Lighting Models	T 1		CO2
4	**Title**	**Introduction to Virtual Reality**	T 1,3,4	**6**	CO1, CO4
	4.1	Perception in virtual environments, Interaction techniques, Usability and ergonomics, Presence, immersion, motion sickness The three I's of virtual reality, Commercial VR technology, five classic components of a VR system. Input Devices, Trackers, Navigation, Gesture Interfaces, Three dimensional position trackers, Manipulation Interfaces, Output Devices, Graphics displays, Sound displays, Haptic feedback.	T 1,3,4 R 1		
	4.2	Mixed Reality (MR) and Extended Reality (XR)			CO4
5	**Title**	**Real-world Applications & Future Trends**		**5**	CO4,CO5
	5.1	Testbed Evaluation of Universal VR Tasks, VR Health and Safety Issues, Direct Effects of VR Simulations on User, VR in social aspects. VR applications in industry, Medical applications, Military applications, Robotics applications. Case studies in education, healthcare, entertainment, simulation,	T 1,2,3		
6	**Self Study**	Ethical considerations and safety, Future of AR/VR (metaverse, AI in AR/VR) Case studies: Applications across industries: healthcare, education, manufacturing, gaming, Spatial Computing and Apple Vision Pro, WebAR and WebXR for Browser-Based Immersive Experience, Generative AI in AR/VR, Digital Twins with AR/VR in Smart Manufacturing.	T 1,2,3	***5**	
			Total	**28**	

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment	CO Mapped
1	Explore Unity Interface and Create 3D Scene	CO2
	(First Certificate course of 'Unity Essential' can be started for all students as self study during lab hours. Successful completers (may be advanced learners) of the first course can be suggested to go ahead for the second course of 'Junior Programmer'. [Both courses are free].)	CO2

2	Develop a Marker-based AR Application using unity and view it using Vuforia view app (free)	CO2
3	Implement Markerless AR using ARCore or ARKit	CO2
4	Build a Simple VR Environment in Unity	CO2
5	Integrate User Interaction in VR (Teleportation/Grab Object)	CO2
6	Simulate Spatial Audio in VR Scene	CO2
7	Design a Virtual Showroom or Gallery (AR/VR)	CO2
8	Evaluate User Experience and Usability in AR/VR App (suggested article : Virtual Reality-Based Experience of AICTE Idea Development, Evaluation)	CO4, CO5
9	Study of Tracking Technologies (Sensor-based demo or simulation)	CO4,CO5
10	Mini Project: AR/VR App for a Real-World Use Case	CO4, CO5

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Understanding Augmented Reality: Concepts and Applications	Second	Alan B. Craig	Morgan Kaufmann	2013
2	Virtual Reality Technology	Third	Grigore C. Burdea and Philippe Coiffet	Wiley	2017
3	Augmented Reality: Principles and Practice	First	Dieter Schmalstieg and Tobias Hollerer	Addison-Wesley	2016
4	The VR Book: Human-Centered Design for Virtual Reality	First	Jason Jerald	ACM Books (Morgan & Claypool)	2015

Reference books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Augmented Human: How Technology Is Shaping the New Reality	First	Helen Papagiannis	O'Reilly Media	2017
2	Practical Augmented Reality: A Guide to the Technologies, Applications...	First	Steve Aukstakalnis	Addison-Wesley	2016
3	Augmented Reality: Where We Will All Live	First	Jon Peddie	Springer	2017

4	Learning Virtual Reality: Developing Immersive Experiences and...	First	Tony Parisi	O'Reilly Media	2015
5	3D User Interfaces: Theory and Practice	First	Doug Bowman, Ernst Kruijff, Joseph J. LaViola Jr., Ivan Poupyrev	Addison-Wesley	2004
6	Being Really Virtual: Immersive Natives and the Future of Virtual Reality	First	Frank Steinicke	Springer	2016
7	Extended Reality in Practice	First	Bernard Marr	Wiley	2021
8	Unity Virtual Reality Projects	Third	Jonathan Linowes	Packt Publishing	2020
9	Virtual and Augmented Reality,	First	Dr. Rajiv Chopra,	Khanna Book Publishing	2021
10	Virtual Reality-Based Experience of AICTE Idea Development, Evaluation	First	Kushal Birla, Snehal Kamalapur, Shirish Sane	Chapman and Hall/CRC Taylor & Francis	2024*

- **for case study mapping with theory course unit 4 & 5 and lab assignment 8**

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
CS308	DevOps Lab	0	1	2	2	5	0	1	1	2
		Examination Scheme								
		Component		ISE		MSE		ESE		Total
Skill Enhancement Course (SEC)		Tutorial		100		-		-		100
		Laboratory		80		-		20		100

Pre-requisite Course Codes, if any.	Fundamentals of Computer and Coding, CCN, OS, PSIPL, PSOOP
Course Objective: To provide hands-on experience with virtualization and containerization while leveraging a range of DevOps technologies for efficient and secure software development, deployment, and automation.	
Learning Outcomes (LO): At the End of the course students will be able to	
LO1	Implement virtualization and containerization techniques to optimize development and deployment workflows.
LO2	Examine the integration of CI/CD pipelines, infrastructure as code, and monitoring tools to enhance automation and efficiency.
LO3	Assess the performance, security, and scalability of various DevOps tools and methodologies in real-world scenarios.
LO4	Design and develop a fully automated DevOps workflow incorporating container orchestration, version control, and deployment strategies.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
LO1			1		3				2		1	
LO2			1		3				2		1	
LO3			1		3				2		1	
LO4			1		3				2		1	

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

LO	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
LO1							
LO2							
LO3							
LO4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Virtualization and Containerization**	1,2,3,4	4
	1.1	Concept of Virtualization, Types of Virtualization, VirtualBox, VMWare, KVM, QEMU, XEN		
	1.2	Container Technologies: Docker and podman, Minikube and Kubernetes		
2	Title	**DevOps Basics: Git, Jenkins, and SonarQube**	1,4	4
	2.1	Concept of DevOps, CI/CD Pipeline Introduction to Git: Distributed version control system. Importance in tracking changes, collaboration, and branch management.		
	2.2	Jenkins for Continuous Integration Introduction to Jenkins: Overview of Continuous Integration/Continuous Deployment (CI/CD), Key features and plugins		
	2.3	SonarQube for Code Quality Analysis:SAST and DAST Introduction to SonarQube, Importance of code quality and security Key metrics and issues tracked by SonarQube Implementing DAST in DevOps Pipelines Understanding how DAST works in real-time application security testing Common DAST tools: OWASP ZAP, Burp Suite, Nikto, Arachni		

		Automating DAST in CI/CD pipelines Detecting runtime vulnerabilities (SQL Injection, XSS, etc.) Generating and analyzing DAST security reports		
3	Title	**Advanced DevOps: Continuous Deployment and Infrastructure as Code**	1,4	6
	3.1	Ansible for Configuration Management: Introduction to Ansible Agentless configuration management tool. Overview of key features and architecture, Ansible Basics, Understanding Playbooks and Roles YAML syntax		
	3.2	Writing basic playbooks, Inventory Management Static and dynamic inventories, Advanced Ansible usage, Ansible Modules and Plugins		
	3.3	Terraform for Infrastructure as Code: Introduction to Terraform Terraform Basics, Understanding Terraform Configuration Files HCL syntax, Writing basic .tf files, Advanced Terraform Usage, Modules and State Management, Creating and using modules Managing state files and remote state,		
	3.4	Terraform in Action, Provisioning Infrastructure, Using providers (AWS, Azure, GCP), Applying and destroying infrastructure		
4	Self Study	Chef Configuration tool, Puppet Configuration tool, Grafana Monitoring tool, Prometheus Monitoring tool, NagiOS Configuration tool, OpenVAS, TravisCI tool, CircleCI Tool		*2 per week
Total				**14**

*** Excluding Self Study**

Laboratory Component (Minimum 10 Laboratory experiments are expected)- 2 hrs/week

Sr. No.	Title of the Experiment
1	Installation and configuration of Hypervisors (VirtualBox/VMWare/KVM): Loading the Virtual Machines (VMMs), Network configuration and management
2	Installation and Configuration of Docker (Container) and Docker-compose: Docker usage, docker images, docker containers, docker networking, docker monitoring, Dockerfile, Docker-compose, managing the containers

3	Develop and Manage Multipass and LXD Instances: 1. Deploy and manage lightweight Ubuntu VMs using Multipass, leveraging cloud-init for automation. 2. Create and manage LXD containers, utilizing profiles, storage pools, and networking configurations. 3. Understand how to migrate applications between LXD containers and traditional VMs. Implement Advanced Multipass and LXD Features: 1. Utilize Multipass cloud-init configurations to automate VM provisioning. 2. Leverage LXD features like profiles, snapshots, clustering, and remote instance management. 3. Configure LXD networking to allow communication between containers and external services. Troubleshoot and Optimize Multipass and LXD Environments: 1. Diagnose and resolve common issues in Multipass and LXD environments using logs and debugging tools. 2. Optimize performance by fine-tuning resource allocations, networking, and storage options. 3. Ensure security best practices by implementing access controls, container isolation, and secure configurations.
4	Kubernetes using Minikube: To set up a Kubernetes environment using Minikube with Docker as the driver. Confirm the cluster is up and running with `kubectl cluster-info` and `kubectl get nodes`. Deploy a sample application by creating a deployment with `kubectl create deployment hello-minikube --image=k8s.gcr.io/echoserver:1.4`. Expose this deployment to create a service accessible from outside the cluster with `kubectl expose deployment hello-minikube --type=NodePort --port=8080`. Obtain the service URL using `minikube service hello-minikube --url` and access the application through the provided URL. To clean up, delete the deployment and service with `kubectl delete deployment hello-minikube` and `kubectl delete service hello-minikube`, and stop the Minikube cluster using `minikube stop`.
5	Installation and configuration of Git: Basic Git Commands, Basic Operations, Working with Remote Repositories-Connecting to Remote Repos Setting up SSH keys Adding remote repositories: git remote add origin <url> Fetching and Pulling from Remotes Understanding git fetch vs. git pull ,Collaboration and Workflow,Branching Strategies,Feature branches GitFlow,Handling Merge Conflicts, Rebasing vs. Merging
6	Installing and Configuration of Jenkins: Installation Steps, Using Docker Configuring Jenkins ,Initial Setup,Jenkins Dashboard,Creating a Jenkins user, Setting Up Jobs and Pipelines,Freestyle projects,Pipeline projects,Jenkinsfile basics,Declarative vs. Scripted Pipelines Creating a Simple Pipeline,Stages, steps, and post actions,Integrating Jenkins with Git Setting Up Git Repositories in Jenkins, Polling SCM Webhooks,Building and Testing Code Automated builds,Running test suites, Advanced Jenkins Usage Using Plugins Commonly used plugins (Git plugin, Pipeline plugin, etc.) Distributed Builds, Master-slave configurations

7	Installation and configuration of SonarQube for Code Quality Analysis: Installing SonarQube, Installation Steps, Configuring SonarQube Initial Setup,SonarQube DashboardCreating a SonarQube user and projects Connecting to Databases,Configuring database settings (PostgreSQL, MySQL, etc.) Running SonarQube Analysis, Setting Up SonarQube Scanners SonarQube Scanner CLI, Integrating with Maven, Gradle, and other build tools Analyzing a Project Configuring sonar-project.properties Running the scanner and interpreting the results Integrating SonarQube with Jenkins Installing SonarQube Plugin in Jenkins Setting Up a Jenkins Job for SonarQube Analysis Configuring analysis parameters Running SonarQube analysis as part of Jenkins pipeline Advanced SonarQube Usage Custom Quality Profiles and Gates Creating and managing quality profiles Setting up quality gates
8	Installation and configuration of Ansible for Configuration Management: Agentless configuration management tool. Overview of key features and architecture, Ansible Basics, Understanding Playbooks and Roles,YAML syntax Writing basic playbooks, Inventory Management, Static and dynamic inventories, Advanced Ansible usage, Ansible Modules and Plugins, Continuous Monitoring- Observability Prometheus and Grafana, Elasticsearch, Logstash, Kibana (ELK) and LOKI
9	Terraform for Infrastructure as Code:Installation Steps,Terraform Configuration Files,HCL syntax Writing basic .tf files, Advanced Terraform Usage,Modules and State Management Creating and using modules,Managing state files and remote state Applying and destroying infrastructure,
10	Mini Project

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Learning DevOps - Second Edition: A comprehensive guide to accelerating DevOps culture adoption with Terraform, Azure DevOps, Kubernetes, and Jenkins	Second	M. Krief	Packt	2022
2	Practical Linux System Administration: A Guide to Installation, Configuration, and Management	First	Kenneth Hess	O'Reilly	2023

3	Linux For Beginners	Twenty Fourth	Papercut Limited	Papercut Limited	2024
4.	Building Cloud and Virtualization Infrastructure: A Hands-on Approach to Virtualization and Implementation of a Private Cloud Using Real-time Use-cases	First	Mrs.Lavanya S Dr. Venkatachalam K Dr. Saravanakumar N M Dr. Balamurugan S	BPB Publications	2021

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	The DevOps Handbook	First	Gene Kim, Jez Humble, Patrick Debois, John Willis, Nicole Forsgren	IT Revolution Press	2021

SEMESTER VII

SEM VII									
Course Category	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
Multidisciplinary Minor	MDM	MDEC4X	MDM-IV	To be defined others					4
Program Elective Courses	PEC	CS4X3	PE-III	2	0	1	4	7	3
Program Elective Courses	PEC	CS4X4	PE-IV	2	0	1	4	7	3
Open Elective	OE	OE1	OE-I	2	0	1	4	7	3
Experiential Learning	ELC	PR3-II	Main Project Stage II	0	0	6	4	10	3
			Total	**6**	**0**	**11**	**16**	**33**	**16**

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-III	Blockchain Technology	2	0	2	5	10	2	0	1	3
CS413		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	50		50		100		200	
		Laboratory	50		-		50		100	

Pre-requisite Course Codes, if any.	Cryptography and Network Security, CCN

Course Overview: This course introduces students to blockchain technology's theoretical foundations and practical applications. Students will gain a comprehensive understanding of blockchain architecture, consensus mechanisms, cryptographic foundations, and the development of decentralized applications. Through a combination of theoretical concepts and practical laboratory exercises, students will explore this transformative technology's technical, economic, and social implications.

Course Objectives: Develop a deep understanding of blockchain architecture and its underlying technologies. Gain practical experience in developing and deploying blockchain applications
Understand consensus mechanisms and their role in distributed systems. Explore the cryptographic foundations that secure blockchain networks. Analyze real-world blockchain use cases across various industries. Evaluate the ethical, legal, and regulatory considerations of blockchain implementation

Course Outcomes (CO): *At the End of the course students will be able to*	
CS413.1	Apply blockchain fundamentals to design a basic cryptocurrency or token system with appropriate features and security considerations.
CS413.2	Analyze various consensus mechanisms to determine their suitability for specific blockchain applications based on performance, security, and scalability requirements. Apply and implement a smart contract on the Ethereum test network
CS413.3	Evaluate the technical and business implications of implementing blockchain solutions for different industry use cases, critically assessing benefits and limitations.
CS413.4	Create a functioning decentralized application (DApp) that incorporates smart contracts and interacts with a blockchain network.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS474.1	2											
CS474.2			2									
CS474.3					2							
CS474.4		2	2		3	2						

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS474.1	2	-	-	-	-	2	-
CS474.2	2	-	-	-	-	2	-
CS474.3	2	-	-	-	-	2	-
CS474.4	2	-	-	-	-	2	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1		**Introduction to Blockchain Fundamentals**		10
	1.1	Introduction – basic ideas behind blockchain, and how it is changing the landscape of digitalization.	1,2,3,4	
	1.2	Introduction to distributed systems and decentralization. History and evolution of blockchain technology. Blockchain architecture and components Cryptographic foundations (hash functions, digital signatures, public/private keys) Types of blockchain networks (public, private, permissioned, consortium) Blockchain data structures and transaction flow.		
	1.3	Consensus Mechanisms and Security: Proof of Work (PoW) and its implementations Alternative consensus mechanisms (PoS, DPoS, PBFT, etc.) Security considerations in blockchain networks Attack vectors and mitigation strategies Scalability challenges and solutions Blockchain governance models Choosing an algorithm- Finality, Scalability, Speed, Performance		
2		**Cryptocurrency-Bitcoin and Ethereum Ecosystem**		06
	2.1	History of Cryptocurrency, Cryptography in blockchain, SHA hash Function, Merkle Tree, Bitcoin, Monetary Policy, The Halving, Block Frequency, Bitcoin Ecosystem, Bitcoin Network, Bitcoin Mining, Mining Pool, Mining Systems,GPU, FPGA and ASIC, Nonce range, Timestamp, Wallet, UTXOs, SegWit, Wallet Address, Bitcoin Network Payment and API, Bitcoin-core, bitcoind, bitcoin-cli, bitcoin-qt, Bitcoin Improvement Proposal (BIP) introduction, Types of BIP, BIP Lifecycle	1,2,3,4	

	2.2	**Ethereum Ecosystem:** Introduction to Ethereum, Ethereum Technology Stack, Ethereum Virtual Machine (EVM) - Advantages and Drawbacks of Ethereum. **Smart Contract**, ether, and solidity Wallets for Ethereum Solidity - Smart Contracts - some attacks on smart contracts. Testing of smart contracts, Blockchain: Enterprise use cases **NFTs**- Developing smart contracts for tokenization of assets Reviewing the Ethereum improvement proposal process and standards Understanding fungible and non-fungible tokens. Exploring ERC standard tokens ERC20 and ERC721 for fungible and non-fungible assets Designing and developing ERC721-compliant real estate token **DeFi-** What is DeFi, What you can do with DeFi, How does DeFi work, Ethereum and DeFi, **Metaverse**- Metaverse and Digital Evolution, Blockchain, and Metaverse, enabling technologies of the metaverse, the motivation behind the integration of blockchain in the metaverse, the use cases of application of blockchain in the metaverse-Financial System, Smart Contract deployment, NFTs, Healthcare, The metaverse projects Decentraland (https://decentraland.org/), SandBox, Axie Infinity, Illuvium		
3		**Hyperledger Fabric**		**06**
	3.1	Transaction Flow. Hyperledger Fabric Details, Fabric - Membership and Identity Management, Hyperledger Fabric Network Setup, Fabric Demo on IBM Blockchain Cloud. Hyperledger Composer – Application Development. Hyperledger Composer - Network Administration	1,2,3,4	
	3.2	Blockchain: Enterprise use cases Alternative coins – Ethereum and Smart contracts Alternative coins – Ethereum continued		
4		**Applications Blockchain Technology**		**06**
		Uses of Blockchain in E-Governance, Land Registration, Medical Information Systems, and smart cities, smart industries, anomaly detections, FinTech, Shaping the Financial World, IoT	1,2,3,4	
	Self Study	Emerging trends in blockchain technology Layer 2 scaling solutions Zero-knowledge proofs and privacy technologies Tokenomics and cryptoeconomics Quantum computing and its implications for blockchain Regulatory landscape and legal considerations Scaling the blockchain: payment channels and state channels Scaling the blockchain using optimism and using zk-SNARK Privacy in public blockchain: de-anonymizing the blockchain and mixing		**04***
				28

Books

[1] Mastering Blockchain 4/e by Imran Bashir, Packt Publications
[2] Blockchain in Action by Bina Ramamurthy, Manning Publications
[3] Mastering Bitcoin: Unlocking Digital Cryptocurrencies, by Andreas Antonopoulos Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder.
Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press, 2016. (Free download available)
[4] Blockchain by Melanie Swa, O'Reilly
[5] Hyperledger Fabric - https://www.hyperledger.org/projects/fabric

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Lab1: Track Ethereum transactions using API
Lab2: Ethereum Private Blockchain Part-I
Lab3: Ethereum Private Blockchain Part-II
Lab4: Development of Smart Contract using Solidity and Remix IDE
Lab5: Development of Smart Contract using Solidity and Truffle-Suite
Lab6: Development of Ethereum dApp
Lab7: Tokenization of Asset: NFTs
Lab8: Hyperledger Fabric Setup
Lab9: Blockchain and Cybersecurity
Lab10: Testing Smart Contracts

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-III	Generative AI	2	0	2	4	8	2	0	1	3
CS423		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	–
Course Objective: This course provides an introduction to the principles, techniques, and applications of Artificial Intelligence (AI) in healthcare. Understand how Artificial Intelligence (AI) and Machine learning (ML) can be used for implementing applications in healthcare and to analyze the healthcare data and process it using data analysis and statistical tools.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
1	Demonstrate a solid understanding of generative models and their mathematical foundations.
2	Develop and fine-tune AI models for text, image, and multimedia content generation.
3	Evaluate generative AI outputs for performance, quality, and ethical considerations.
4	Deploy and optimize generative AI applications using modern frameworks like TensorFlow and PyTorch.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1												
2												
3												
4												

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
1							
2							
3							
4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✔	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Mo	**Fundamentals of Generative AI**	1,2	6
	1.1	Introduction to Generative AI: Definition, Evolution, and Applications.Key Architectures: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Transformers. Probability Distributions in Generative Models (Gaussian, Multimodal, etc.) Latent Space Representation & Sampling Techniques.		
2	Title	**Deep Generative Models**	3,4	8
	2.1	**Variational Autoencoders (VAEs)-** Concept of Latent Variables, KL Divergence, and ELBO.Training VAEs with PyTorch/TensorFlow		
	2.2	**Generative Adversarial Networks (GANs)-**Architecture of GANs (Generator & Discriminator).Training Challenges: Mode Collapse, Vanishing Gradients, Wasserstein Loss. Variants: DCGAN, StyleGAN, CycleGAN		
3	Title	**Generative Transformers & Large Models**	5,6	8
	3.1	**Transformers for Generative AI-** Self-Attention Mechanism & Multi-Head Attention. Pretrained Models: GPT, BERT, T5, LLaMA.Prompt Engineering & Fine-Tuning LLMs		
	3.2	**Diffusion Models & Image Generation**-Denoising Diffusion Probabilistic Models (DDPMs). Stable Diffusion & MidJourney		
4	Title	**Applications & Ethical Considerations**	7	6
	4.1	AI in Text Generation (ChatGPT, Bard, etc.).Image & Video Synthesis (DeepFakes, AI Art).Generative AI for Code		

		Generation (Codex, Copilot). Ethical Concerns: AI Bias, Deepfake Detection, Responsible AI.		
5	**Self Study**	Auto-Regressive & Auto-Encoding Models, AI-Generated Music & Creativity, Bias & Fairness in Generative AI, Future Careers & Research Directions in Generative AI		**4**
			Total	**28**

Laboratory Component

Sr. No.	Title of the Experiment
1	Implementing Variational Autoencoders (VAEs) for Image Generation
2	Training a Basic Generative Adversarial Network (GAN) on MNIST Dataset
3	Fine-Tuning GPT-2 for Custom Text Generation
4	Generating AI Art using Stable Diffusion Models
5	Building a DeepFake Model for Face Swapping using Autoencoders
6	Implementing Text-to-Image Generation using DALL·E Mini
7	Training a CycleGAN for Style Transfer between Images
8	Generating AI Music using OpenAI Jukebox
9	Developing an AI-Powered Chatbot using Transformer-based Language Models
10	Deploying a Generative AI Model as a Web API using FastAPI

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Deep Learning	1st	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	2016
2	Probabilistic Machine Learning: An Introduction	2nd	Kevin P. Murphy	MIT Press	2022
3	Generative Deep Learning	2nd	David Foster	O'Reilly	2023
4	Deep Learning with Python	2nd	François Chollet	Manning	2021
5	Natural Language Processing with Transformers	2nd	Lewis Tunstall, Leandro von Werra, Thomas Wolf	O'Reilly	2022
6	Transformers for Natural Language Processing	2nd	– Denis Rothman	Packt	2022
7	The Alignment Problem: Machine Learning and Human Values	2nd	Brian Christian, W.W. Norton		2020

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-III	Data Warehouse and Mining	2	0	1	4	7	2	0	1	3
		Examination Scheme								
		Component		ISE		MSE		ESE		Total
CS433		Theory		50		50		100		200
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	CS307 Machine Learning
Course Objective: To extract knowledge from data repository for data analysis using frequent pattern, classification, clustering and prediction.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CS433.1	To utilize strategic data in an enterprise and build a data warehouse.
CS433.2	Make use of data lakes for scalable, secure, and efficient data storage for analytical processing.
CS433.3	To organize the data needed for data mining using preprocessing techniques.
CS433.4	To identify the appropriate data mining methods like classification or frequent pattern mining on large data sets.
CS433.5	To apply clustering for pattern discovery and optimization algorithms to improve model performance

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS433.1					3	3						3
CS433.2					3	3						3
CS433.3	2	2	2	2	3	3					2	3
CS433.4	2	2	2	2	3	3					2	3
CS433.5	2	2	2	2	3	3					2	3

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2
CS433.1					3	
CS433.2					3	
CS433.3					3	
CS433.4					3	
CS433.5					3	

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze✔	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction of Data Warehousing:**	1,2	6
	1.1	Data Warehouse architecture, data warehouse environment and its components , ETL Process, Top-down and bottom-up approaches, data mart, business impact of data warehouse, evolution of data warehouse from a business perspective, federated data warehouse.		
	1.2	Difference between OLAP and OLTP. Dimensional analysis- Define cubes, Drill-down, rollup, slice, dice, rotation, OLAP models- ROLAP and MOLAP.		
2	Title	**Data Lakes:**	1,2	8
	2.1	Introduction to Data Lakes, Data Lake vs. Data Warehouse, Structured, Semi-structured, and Unstructured Data Benefits & Challenges of Data Lakes. Data Ingestion: Batch vs. Streaming (Kafka, Flume, Kinesis), Storage Layers: Raw, Processed, Curated Data, Schema-on-Read vs. Schema-on-Write Metadata Management & Data Governance		
	2.2	Storage: AWS S3, Azure Data Lake, Google Cloud Storage, Processing: Apache Spark, Databricks Delta Lake, Presto Query Engines: Apache Hive, Athena. Security & Access Control: Role-based Access, Data Masking. How Data Warehouses and Data Lakes Work Together, Introduction to Lakehouse Architecture (Delta Lake, Apache Iceberg)		
3	Title	**Data Mining:**	1,2	5
	3.1	Steps in Data mining process, Data Mining Functionalities, Architecture of a Typical Data Mining Systems, Classification of Data Mining Systems, Knowledge Discovery in Databases (KDD), KDD Process, Data Preprocessing, Data Cleaning, Data Transformation,		
	3.2	Data Compression, Dimension Reduction with Principal Component Analysis, TSNE, Binning Methods.		
4	Title	**Data Mining Techniques:**	1,2	5
	4.1	Association Rule Mining: Association rules, market basket analysis, Apriori algorithm, FP-Growth algorithm.		
	4.2	Classification by Decision Tree, Bayesian networks, Lazy Learners. Bagging using random forest, and Boosting using Adaboost, Gradient Boosting, XGBoost.		
5	Title	**Cluster Analysis:**	1,2	4
	5.1	Basic Concepts; Partitioning Methods: K-Means, K-Medoids; Hierarchical Methods: Agglomerative.		
	5.2	Optimization formulations Gradient and search-based optimization, Linear, quadratic, nonlinear, and mixed integer program, multi-objective and multi-criteria decision-making evolutionary tools		

6	**Self Study**	Anomaly detection, Local Outlier Factor, Isolation Forest, Reinforcement Learning, Hidden Markov Model, Gaussian Mixture Models		**7**
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	To implement a Data Warehouse and perform OLAP queries.
2	To perform Drill-down, rollup, slice and dice operations.
3	Implement ROLAP/MOLAP using Pentaho BI or Power BI.
4	To build a Data Lake with AWS S3 and AWS Glue for Data Ingestion and Transformation
5	To build a Data Lake on Hadoop with Apache Spark for data processing and visualize using Tableau, Power BI, or Jupyter Notebooks (with PySpark)
6	To perform data preprocessing on a dataset and dimension reduction using Principal Component Analysis/TSNE on a large dimension dataset.
7	To find relationships between variables in datasets, with applications spanning diverse fields like retail, healthcare, cybersecurity, etc. enabling insights into customer behavior, diagnosis, fraud detection etc. using association rule mining.
8	To apply various classification algorithms to applications like image recognition, spam detection, medical diagnosis etc. and compare them using performance metrics.
9	To group similar data points, leading to insights and solutions in various fields like market segmentation, social network analysis, anomaly detection etc.
10	To assess the effectiveness of different optimization formulations (linear, quadratic, nonlinear, multi-objective).

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1.	Data Warehousing: Fundamentals for IT Professionals	2nd Edition	Paulraj Ponniah	Wiley India.	2012
2.	Data Mining Introductory and Advanced Topics	1st Edition	M.H. Dunham	Pearson Education.	2006

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1.	Data Mining Concepts and Techniques	4th Edition	Han, Kamber	Morgan Kaufmann Pub	2022
2.	Introduction to Data Mining	2nd Edition	Pang-Ning Tan, Michael Steinbach and Vipin Kumar	Pearson Education.	2021

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Course(Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
	Computer Vision									
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
To be filled		Theory								
		Laboratory								

Pre-requisite Course Codes, if any.	Exposure to basic mathematics: calculus, linear algebra and probability. Ability to program in C/C++/MATLAB and/or willingness to learn MATLAB in (Image Processing),
Course Objective: To develop mathematical foundation of system and design digital filters	
Course Outcomes (CO):*At the End of the course students will be able to*	
CO.1	**Apply** geometric computer vision techniques such as camera calibration, stereo imaging, and structure from motion to estimate depth
CO.2	**Analyze and compare** various feature detection and description method
CO.3	**Implement and evaluate** image processing, segmentation, and deconvolution techniques including classical and deep learning-based models
CO.4	**Design and deploy** advanced object detection, image inpainting, and synthesis models using deep learning architectures

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO.1												
CO.2												
CO.3												
CO.4												
CO.5												
CO.6												**2**

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CO.1							
CO.2							
CO.3							
CO.4							
CO.5							
CO.6							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	✔ Apply	✔ Analyze	✔ Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Geometric Techniques in Computer Vision**		04
	1.1	Color Photography overview, Camera Projections Camera Calibration Depth from Stereo Two View Structure from Motion Object Tracking		
	1.2	Visual Features and Representations: Edge, Blobs, Corner Detection; Scale Space and Scale Selection; SURF; HoG, LBP, etc. Visual Matching: Bag-of-words, VLAD; RANSAC, Hough transform; Pyramid Matching; Optical Flow		
2	Title	**Features**	1, 3	7
	2.1	Local Descriptors: Corner, SIFT, LBP, HOG Edge Detection and Linking: LoG, Canny Image Features: Steerable Filters, Shape & Texture, DL based Perceptual Features		
	2.2	Motion: Optical Flow, Block Matching, Parametric Motion, Global Motion, Flownet Depth: Depth from Structure, DL based Depth, Structure & Depth from Egomotion Full-reference Quality: SSIM, FSIM		
3	Title	**Processing and segmentation**		7
	3.1	Generic and video filters, Deconvolution Techniques: Unsharp Masking,LMMSE-based Deconvolution, Bayesian-based Deconvolution, CNN-based Deconvolution		
	3.2	Segmentation: Superpixels, Mean Shift and Mode Seeking Segmentation, Low-level segmentation, energy minimization and clustering based methods, semantic segmentation, CNN-based Semantic Image Segmentation, CNN-based Semantic Video Segmentation, clustering-based methods, semantic segmentation, Object Detection Semantic Segmentation		
4	Title	**Object Detection and Recognition**		7
	4.1	Contrast based Salient Object Detection (SOD), DL based SOD, Video SOD, You Only Look Once (YOLO), Region based CNN (R-CNN) Variants Retargeting and Inpainting: Seam Carving for		
	4.2	Image and Video Retargeting, Image Inpainting, DL based Inpainting Categorization and Captioning: Bag of words, Efficient Net on ImageNet, Image Captioning		
5	Title	**Synthesis**		3
	5.1	GAN overview, 3D shape synthesis, integrating viewpoint and texture, semantic image synthesis (4		
	Self Study	Concept of transfer learning, Using pre-trained models for computer vision tasks- VGG, ResNet and Inception .		
		Applications of CNNs in Computer Vision-Image classification, Object detection, Semantic segmentation.		
			Total	**28**

Laboratory Component

Sr. No.	Title of the Experiment
1	Camera Calibration and Epipolar Geometry: Calibrate camera using OpenCV and explore stereo vision to compute depth from stereo image pairs.
2	Structure from Motion: Use tools like COLMAP or OpenCV to perform 3D reconstruction from multiple images.
3	Object Tracking and Motion Estimation: Implement object tracking using Kalman Filter and OpenCV tracking algorithms; analyze optical flow (Lucas-Kanade / Farneback).
4	Feature Detection and Matching: Detect and match keypoints using SIFT, SURF, ORB with FLANN or Brute Force matching.
5	Homography Estimation using RANSAC: Use feature correspondences and RANSAC to estimate homography for image stitching or transformation.
6	Image Filtering and Deconvolution: Apply Gaussian, Median, Bilateral filters; explore Wiener and CNN-based image deconvolution techniques.
7	Image Segmentation Techniques: Perform segmentation using Watershed, Mean Shift, GrabCut, and evaluate semantic segmentation using U-Net or DeepLabV3+.
8	Object Detection with Deep Learning: mplement YOLOv5 or SSD for object detection in real-time image/video streams.
9	Saliency Detection and Inpainting: Perform saliency detection using CNNs or transformers; explore traditional and DL-based image inpainting methods.
10	Generative Adversarial Networks (GANs): Train a basic GAN on MNIST/CIFAR; extend to image-to-image translation with Pix2Pix or CycleGAN.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Computer Vision: Algorithms and Applications	2nd Edition	Richard Szeliski	Springer	2022
2	Multiple View Geometry in Computer Vision	2nd Edition	Richard Hartley, Andrew Zisserman	Cambridge University Press	2003
3	Deep Learning for Vision Systems	1st Edition	Mohamed Elgendy	Manning Publications	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Digital Image Processing	4th Edition	Rafael C. Gonzalez and Richard Woods	Pearson	2018
2	Deep Learning	1st Edition	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	2016

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PE)	IT Infrastructure Monitoring and Management	2	0	2	5	9	2	0	1	3
		Examination Scheme								
		Component	ISE		MSE		ESE			Total
CS414		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	Basic Networking, Operating Systems, Cloud Computing Fundamentals
Course Description: This course provides a comprehensive understanding of modern infrastructure monitoring and management techniques, tools, and best practices. Students will learn to design, implement, and maintain monitoring solutions for various IT infrastructures, analyze performance metrics, troubleshoot issues, and optimize system resources. The course combines theoretical concepts with hands-on practical exercises to develop proficiency in infrastructure monitoring and management. **Course Objectives:** Develop a thorough understanding of IT infrastructure monitoring principles, methodologies, and frameworks. Master the use of industry-standard monitoring tools and technologies for different types of infrastructure components	
Course Outcomes (CO): ***At the End of the course students will be able to***	
EL33T.1	Apply monitoring and management principles to design comprehensive infrastructure monitoring solutions for diverse IT environments
EL33T.2	Analyze performance metrics and system logs to identify and troubleshoot infrastructure issues and bottlenecks
EL33T.3	Evaluate the effectiveness of monitoring strategies and tools, making data-driven recommendations for infrastructure improvements
EL33T.4	Create customized monitoring dashboards, alerting mechanisms, and automation workflows to address specific infrastructure management requirements

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
EL33T.1												
EL33T.2												
EL33T.3												
EL33T.4												

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
EL33T.1							
EL33T.2							
EL33T.3							
EL33T.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	Fundamentals of IT Infrastructure Monitoring Introduction to Infrastructure Monitoring and Management Definition, importance, and evolution Types of IT infrastructure and monitoring requirements Challenges in modern infrastructure monitoring Monitoring Principles and Frameworks Key monitoring concepts: metrics, logs, traces Monitoring frameworks and methodologies Service Level Indicators (SLIs), Service Level Objectives (SLOs), and Service Level Agreements (SLAs) Monitoring Architecture and Components Monitoring system architecture Data collection mechanisms Data storage and retention strategies Visualization and reporting		6
	1.1	IT vs. OT Infrastructure (Key differences, convergence trends)		
	1.2	Layered Monitoring Approach (Business, Service, Network, Element, Physical)		
	1.3	Key Protocols & Standards (SNMP, NetFlow, IPFIX, Syslog, Modbus)		
	1.4	Monitoring vs. Observability (Logs, Metrics, Traces)		
2	Title	**Business & Service Management Layer**		6
	2.1	Business Metrics (QoS, QoE, CapEx, OpEx, ROI)		
	2.2	Service-Level Agreements (SLAs) (Uptime, Downtime, Availability)		
	2.3	Reliability Engineering (MTBF, MTTR, MTTF)		
	2.4	Cost Optimization (Cloud vs. On-Premise TCO Analysis)		
3	Title	**Network & Element Management Layer**		6
	3.1	Network Performance Monitoring (Latency, Jitter, Packet Loss)		
	3.2	Protocols & Tools (SNMP, sFlow, Wireshark, Nagios, SolarWinds)		
	3.3	Element Management Systems (EMS) (Device-level monitoring)		
	3.4	SDN & NFV Monitoring (Software-Defined Networking, Virtualization)		

4	**Title**	**Physical & Industrial Infrastructure Monitoring**		**6**
	4.1	Physical Layer Monitoring (Cabling, Power, Environmental Sensors)		
	4.2	Industrial (OT) Monitoring (SCADA, ICS, IIoT)		
	4.3	Real-time Monitoring in OT (Siemens MindSphere, OSIsoft PI)		
	4.4	Safety & Compliance (NIST, IEC 62443)		
5	**Title**	**Cloud, Hybrid & Mobile Infrastructure**		**4**
	5.1	Cloud Monitoring (AWS CloudWatch, Azure Monitor, GCP Operations)		
	5.2	Hybrid Infrastructure Challenges (Unified monitoring with Datadog, Splunk)		
	5.3	Mobile Device Management (MDM) (Microsoft Intune, Jamf)		
	5.4	Container & Serverless Monitoring (Prometheus, Grafana, OpenTelemetry)		
6	**Self Study**	Advanced Monitoring Frameworks & Automation Observability Stacks (ELK, Loki-Grafana-Tempo-Mimir (LGTM), OpenTelemetry), AIOps & Predictive Analytics (Anomaly detection, Auto-remediation), Security Monitoring (SIEM) (Splunk, IBM QRadar, Elastic Security), Case Studies & Emerging Trends (AI-driven monitoring, Edge computing)		***6**
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)
Suggested Laboratory Exercises
Lab 1: Setting up a basic monitoring environment with Prometheus and Grafana
Lab 2: Configuring server monitoring using node_exporter and system metrics collection
Lab 3: Implementing network monitoring using SNMP and Netflow
Lab 4: Building comprehensive dashboards for infrastructure visualization
Lab 5: Configuring alerting rules and notification channels
Lab 6: Log aggregation and analysis using ELK stack (Elasticsearch, Logstash, Kibana)
Lab 7: Implementing distributed tracing with Jaeger or Zipkin
Lab 8: Cloud resource monitoring using cloud-native tools
Lab 9: Container and Kubernetes monitoring with Prometheus Operator
Lab 10: Performance testing and benchmark analysis
Lab 11: Implementing anomaly detection for system metrics
Lab 12: Developing automated remediation scripts for common issues
Lab 13: Creating custom exporters for application-specific metrics
Lab 14: Designing and implementing a complete monitoring solution for a multi-tier application
Lab 15: Capacity planning and resource optimization based on historical data

Recommended Resources

Textbooks
[1] "Monitoring Distributed Systems" by Rob Ewaschuk
[2] Turnbull, J. (2018). The Art of Monitoring. Turnbull Press. ISBN: 978-0988820234
[3] "Site Reliability Engineering" by Google

Online Resources
Prometheus documentation
Grafana Labs learning resources
CNCF Observability technical reports

Tools
Prometheus + Grafana
Nagios
Zabbix
ELK Stack
SolarWinds

Reference Books

1. Ligus, S. (2012). *Effective Monitoring and Alerting: For Web Operations*. O'Reilly Media. ISBN: 978-1449333522
2. Arundel, J., & Domingus, J. (2019). *Cloud Native DevOps with Kubernetes: Building, Deploying, and Scaling Modern Applications in the Cloud*. O'Reilly Media. ISBN: 978-1492040767
3. Beyer, B., Jones, C., Petoff, J., & Murphy, N. R. (2016). *Site Reliability Engineering: How Google Runs Production Systems*. O'Reilly Media. ISBN: 978-1491929124
4. Sridharan, C. (2018). *Distributed Systems Observability*. O'Reilly Media. ISBN: 978-1492033646
5. Rabenstein, J., & Pahl, J. (2021). *Prometheus: Up & Running: Infrastructure and Application Performance Monitoring*. O'Reilly Media. ISBN: 978-1492034148
6. Subramanian, M. (2012). *Network Management: Principles and Practice*. Pearson Education. ISBN: 978-0132221368
7. Stallings, W. (1998). *SNMP, SNMPv2, SNMPv3, and RMON 1 and 2*. Addison-Wesley Professional. ISBN: 978-0201485349
8. Cisco Systems. (2019). *Network Management Fundamentals*. Cisco Press. ISBN: 978-1587201370

Online Resources

1. Prometheus Documentation: https://prometheus.io/docs/
2. Grafana Documentation: https://grafana.com/docs/
3. ELK Stack Documentation: https://www.elastic.co/guide/index.html
4. Kubernetes Monitoring Guide:
   https://kubernetes.io/docs/tasks/debug-application-cluster/resource-usage-monitoring/
5. Cloud Provider Monitoring Documentation:
   - AWS CloudWatch: https://docs.aws.amazon.com/cloudwatch/
   - Azure Monitor: https://docs.microsoft.com/en-us/azure/azure-monitor/
   - Google Cloud Monitoring: https://cloud.google.com/monitoring/docs

Academic Journals and Papers

1. Journal of Network and Systems Management (Springer)
   https://www.springer.com/journal/10922
2. IEEE Transactions on Network and Service Management
   https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=4275028
3. Campos, J., et al. (2022). "AIOps: The Evolution of Infrastructure Monitoring." IEEE Access, 10, 45280-45295.
4. Aceto, G., et al. (2021). "Cloud monitoring: A survey." Computer Networks, 19(2), 2093-2115.

Industry Resources

1. DevOps Institute - Observability Certification Materials
   https://www.devopsinstitute.com/certifications/
2. New Relic University - Observability Materials
   https://learn.newrelic.com/
3. Datadog Learning Center
   https://learn.datadoghq.com/
4. The Open Telemetry Project Documentation
   https://opentelemetry.io/docs/
5. Site Reliability Engineering Workbook by Google
   https://sre.google/workbook/table-of-contents/

Online Courses

1. Coursera: "Cloud Monitoring and Observability" by Google Cloud
   https://www.coursera.org/learn/cloud-monitoring
2. edX: "Infrastructure Monitoring Best Practices" by Linux Foundation
   https://www.edx.org/professional-certificate/linuxfoundationx-devops-and-sre
3. Udemy: "Complete Guide to Infrastructure Monitoring" by David Clinton
   https://www.udemy.com/course/infrastructure-monitoring/
4. LinkedIn Learning: "Monitoring Cloud Infrastructure" by David Linthicum
   https://www.linkedin.com/learning/monitoring-cloud-infrastructure

Industry Reports and White Papers

1. Gartner (2024). "Market Guide for Infrastructure Monitoring Tools"
2. Forrester (2023). "The Forrester Wave™: Infrastructure Monitoring Platforms"
3. IDC (2023). "Infrastructure Monitoring: Market Analysis and Forecast"
4. Flexera (2024). "State of IT Infrastructure Monitoring Report"

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PE-IV	Explainable AI	2	0	2	4	8	2	0	1	3
CS424		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		50		50		100		200
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	NNFL, ML
Course Objective: This course is designed to provide fundamental concepts of explainable AI.The students should be able to grasp the basic concepts of explainable AI, including its definition, importance, and the challenges involved in developing an AI system that is explainable. The students will be able to understand interpretability , explainability and attribution methods for AI models.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
CO1	Understand the concept and importance of XAI and its applications in various domains.
CO2	To explore the various XAI techniques and tools used for tabular, image and text dataset.
CO3	To gain practical experience with XAI frameworks..
CO4	To evaluate the trade-offs between explainability and accuracy.
CO5	To Understand ethical and legal aspects of XAI

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	–	2	–	2	–	–	–	–	–	2
CO2	3	3	2	2	3	–	–	–	–	–	–	2
CO3	–	2	3	3	3	–	–	–	2	–	2	3
CO4	3	3	2	2	2	–	–	2	–	–	–	3
CO5	–	–	–	–	–	2	2	3	–	–	–	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CO1							
CO2							
CO3							
CO4							
CO5							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	Introduction to Explainable Artificial Intelligence		4
	1.1	Why Explainable AI?, What Is Explainable AI?, Who Needs Explainability?, Challenges in Explainability, Types of explanation	T1-T3, R1,R2	
2	Title	Explainability for Tabular Data		8
	2.1	Permutation Feature Importance: Permutation Feature Importance from Scratch, Permutation Feature Importance in scikit-learn	T1-T3 R1-R5	
	2.2	Shapley Values: SHAP (Shapley Additive exPlanations), Visualizing Local Feature Attributions, Visualizing Global Feature Attributions, Interpreting Feature Attributions from Shapley Values, Managed Shapley Values		
	2.3	Explaining Tree-Based Models: From Decision Trees to Tree Ensembles, SHAP's TreeExplainer		
	2.4	Partial Dependence Plots and Related Plots: Partial Dependence Plots (PDPs), Individual Conditional Expectation Plots (ICEs)		
3	Title	Explainability for Image Data		8
	3.1	Integrated Gradients (IG), Choosing a Baseline, Accumulating Gradients, Improvements on Integrated Gradients, (Extremal regions with attributions importance) XRAI, How XRAI Works, Implementing XRAI, Grad-CAM, How Grad-CAM Works, Implementing Grad-CAM	T1-T3 R1,R2	
	3.2	LIME: How LIME Works, Implementing LIME, Guided Backpropagation and Guided Grad-CAM, Guided Backprop and DeConvNets, Guided Grad-CAM		
4	Title	Explainability for Text Data		8
	4.1	Overview of Building Models with Text: Tokenization, Word Embeddings and Pretrained Embeddings, LIME, How LIME Works with Text, Gradient x Input, Intuition from Linear Models From Linear to Nonlinear and Text Models	T1-T3, R1,R5	
	4.2	Layer Integrated Gradients: A Variation on Integrated Gradients, Layer-Wise Relevance Propagation (LRP), How LRP Works, Deriving Explanations from Attention, Which Method to Use?, Language Interpretability Tool		
5	Self Study	Building with Explainability in Mind, The ML Life Cycle AI Regulations and Explainability, What to Look Forward To in Explainable AI, Natural and Semantic Explanations, Interrogative Explanations, Targeted Explanations		4
			Total	28

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	To implement a simple decision tree model for predicting whether a patient has diabetes based on their medical information. Note : students could implement decision trees using a dataset, visualize the decision tree, and explain the results
2	Feature Importance: To implement SHAP algorithm to calculate the feature importance on a given data set. Note:students could analyze a dataset and calculate the feature importance using an algorithm such as permutation importance or SHAP (SHapley Additive exPlanations), and explain the results.
3	Non linear relationship between features and predicted model: Interpretation and explanation of ML models [marginal effect of a single feature on the predicted outcome while holding all other features constant] To implement PDP(partial dependance plot) to calculate the feature importance on a given data set. Note: students could perform sensitivity analysis on a trained model to understand how the output varies with changes in the input features.
4	Non linear relationship between features and predicted model: Interpretation and explanation of ML models To implement ICE (Individual Conditional Expectation Plots) to calculate the feature importance on a given data set.[conditional effect of a single feature on the predicted outcome for each individual sample in a dataset] Note: students could perform sensitivity analysis on a trained model to understand how the output varies with changes in the input features.
5	To implement an IG algorithm for attributions analysis for a given problem statement.
6	To implement the XRAI algorithm for attributions analysis for a given problem statement.
7	To implement a grad-CAM method to generate class activation maps for a given problem statement.
8	Local Interpretable Model-agnostic methods: To implement LIME to explain the predictions of a given problem statement. Note: students could use model-agnostic methods such as LIME (Local Interpretable Model-Agnostic Explanations) or Anchors to explain the predictions of a machine learning model
9	To implement LIME to analyze text data, such as social media posts or customer reviews, and classify them into different categories.
10	To study LRP and demonstrate LIT by Google for interactive visualization, interpretation and understanding NLP models.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1.	Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions	1st Edition	Michael Munn, David Pitman	OReilly Media, Inc	2022
2.	Explainable Artificial Intelligence: An Introduction to Interpretable Machine Learning	1st Edition	Uday Kamath, JohnLiu	Springer Link	2021
3.	Explainable AI: Interpreting, Explaining and Visualizing Deep Learning	1st Edition	Wojciech Samek, Grégoire Montavon, Andrea Vedaldi, Lars Kai Hansen, Klaus-Robert Müller	Springer Link	2019

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1.	Interpretable Machine Learning: A Guide for Making Black Box Models Explainable	1st Edition	Christoph Molnar, published	Springer	2019
2.	Explainable AI: Interpreting, Explaining and Visualizing Deep Learning	1st Edition	Konstantinos G. Margaritis and Evangelos N. Daskalakis, published	Wiley	2020
3.	Explainable AI in Healthcare: An Interdisciplinary Approach	1st Edition	Kristin H. Jarman and Julia A. Lane	Springer	2021
4.	XAI - Explainable Artificial Intelligence: Foundations, Methods, and Applications	1st Edition	Fatma Bouali and Sihem Mesnager	Springer	2020
5.	Explainable AI: Interpreting, Explaining and Visualizing Deep Learning	1st Edition	Kalyan Veeramachaneni, published by	O'Reilly Media	2021

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
IT24	Program Elective-IV: AI for Healthcare Analytics	2	0	2	4	8	2	0	1	3
PE-IV		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	–
Course Objective: This course provides an introduction to the principles, techniques, and applications of Artificial Intelligence (AI) in healthcare. Understand how Artificial Intelligence (AI) and Machine learning (ML) can be used for implementing applications in healthcare and to analyze the healthcare data and process it using data analysis and statistical tools.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
AI306.1	To study the fundamental tools within modern artificial intelligence (AI) with a focus on intelligent agents and systems for healthcare analytics
AI306.2	To analyze the healthcare data for prediction and future use
AI306.3	To explore the applications of AI and ML with respect to the healthcare domain

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
AI306.1												
AI306.2												
AI306.3												

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
AI306.1							
AI306.2							
AI306.3							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✔	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Mo	**AI for Healthcare**		2
	1.1	Introduction to Artificial Intelligence (AI) and Machine learning (ML) Importance and Applications of AI and ML in Healthcare	1,2	
2	Title	**Statistics for AI in Healthcare Analytics in Excel**		4
	2.1	Linear Algebra and Multivariable Calculus and statistics (Hypothesis testing, ANOVA,ChiSqaure etc.)	4	
3	Title	**Types of Machine Learning for Healthcare Analytics**		8
	3.1	Introduction to supervised, unsupervised, and reinforcement learning for Data Analytics in Healthcare	1,3,6	
	3.2	Decision Tree, Bayesian Classifier, K-NN and Regression for Data Analytics in Healthcare	4, 5	
4	Title	**Data Representation and Analytics**		5
	4.1	Introduction to data, data frames, Data standardization, Dealing with noise and missing values and Transforming and normalizing data	1, 7, 8	
	4.2	Overview of tools like R, Python and Statistical and Visualization tools like Power BI and Tableau		
5	Title	**Healthcare Data Analysis:**		9
	5.1	Sources of the healthcare data, Pre-processing of the healthcare data, Handling of the healthcare data and Creation of analysis-ready datasets	3, 5,8	
	5.2	Case studies - Future trends in AI Healthcare		
6	Self Study	Reinforcement Learning, conversational AI methods, Various Chatbot, statistical software tools for various healthcare domain like TB analysis from X-ray images, identifying tumors, root canal of tooth by dentist, cardiologist for heart risk analysis etc		4
			Total	**28**

Laboratory Component:

Sr. No.	Title of the Experiment
1	To implement the regression for healthcare datasets
2	To implement the decision Tree for healthcare datasets
3	To implement the K-NN for healthcare datasets
4	Hypothesis Testing in Excel for healthcare datasets - ANOVA and Chi Square
5	Data Transformation and preparation for analysis of healthcare data
6	Healthcare data analysis using Python/ R Programming
7	Healthcare data analysis using Power BI
8	Healthcare data analysis using Tableau
9	To implement KERAS or tensorflow model for automatic detection of abnormalities
10	To implement NLTK, spaCY (NLP libraries in Python) to extract important information like patient diagnosis,medications and lab results.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Artificial Intelligence: A Modern Approach	4th	Russell, S. and Norvig,	Pearson	2022
2	Healthcare Data Analytics	1st	Chanda Reddy, Charu Aggarwal	CRC Press	2020
3	Artificial Intelligence in Healthcare	1st	Parag Mahajan	Parag Mahajan	2022
4	The Elements of Statistical Learning	2nd	Hastie, T., Tibshirani, R. and Friedman, J.	Springer	2017
5	Machine Learning for Healthcare Analytics - Projects	1st	Eduonix Academic Solutions	PackT	2018
6	Artificial Intelligence Fundamentals and Applications	1st	Cherry Bhargava, Pradeep Kumar	CRC Press	2021
7	Hands on Healthcare Data	1st	Andrew Neugen	Shroff Publications	2015
8	Artificial Intelligence in Healthcare: Review and Prediction Case Studies	NA	Guoguang Rong, Arnaldo Mendez, Elie Bou Assi, Bo Zhao, Mohamad Sawan	www.sciencedirect.com	2020

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PE4)	Visual Intelligence	2	0	2	4	8	2	0	1	3
CS434		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		75		75		150		300
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	–	
Course Objective: This course explores the principles and technologies behind how machines and humans interpret visual data. Topics include computer vision, deep learning, object recognition, visual perception, cognitive models of vision, and practical applications in areas like robotics, AI, and augmented reality. Students will gain hands-on experience with techniques for analyzing and understanding visual data.		
Course Outcomes (CO): ***At the End of the course students will be able to***		
CS434.1	Apply the principles of computer vision and deep learning	
CS434.2	Analyze human visual perception and compare it to machine interpretation	
CS434.3	Design and implement solutions for real-world visual intelligence problem.	

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS434.1	-	2	-	2	3	-	-	-	-	-	-	-
CS434.2	-	2	2	-	3	-		3	3	3	3	3
CS434.3	-	2	2	-	3	-	2	-		2	2	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CS434.1	2						
CS434.2	2						2
CS434.3		2					2

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply✔	Analyze ✔	Evaluate✔	Create

Theory Component

Prerequisites:

- Basic understanding of programming (Python or similar).
- Familiarity with linear algebra and statistics.
- Basic knowledge of machine learning concepts

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Visual Intelligence**		3
	1.1	Overview of the visual perception system: Human and Machine.		
	1.2	Applications: Autonomous vehicles, medical imaging, robotics, augmented reality.		
	1.3	Introduction to computer vision and deep learning.		
	1.4	Image transformations: Scaling, rotation, and translation. Color spaces and histograms, Image filtering and enhancement techniques.		
2	Title	**Machine Learning for Vision**		4
	2.1	Introduction to machine learning for visual data.		
	2.2	Supervised vs unsupervised learning.		
	2.3	Feature extraction methods (SIFT, HOG, Gabor filters)		
	2.4	Dimensionality reduction (PCA, t-SNE).		
3	Title	**Convolutional Neural Networks (CNNs)**		7
	3.1	Deep learning overview. Introduction to CNNs: Architecture and components (convolution, pooling, and fully connected layers).		
	3.2	Applications of CNNs in image classification, segmentation, and object detection		
	3.3	Transfer learning in CNNs.		
4	Title	**Object Detection and Recognition**		7
	4.1	Object detection frameworks: YOLO, SSD, Faster R-CNN.		
	4.2	Applications: Face detection, pedestrian detection, and object tracking.		
	4.3	Image segmentation and instance segmentation.		

5	**Title**	**Visual Perception in Humans vs. Machines**		**7**
	5.1	Cognitive models of visual perception: How humans perceive the world.		
	5.2	Theories of visual attention and object recognition in cognitive psychology.		
	5.3	Comparing human and AI-based perception.		
	5.4	Visual illusions and machine interpretation.		
6	**Self Study**	**Generative Models in Vision** Generative Adversarial Networks (GANs) for image generation. Variational Autoencoders (VAEs). Style transfer and image-to-image translation. Ethical implications of generated images		
			Total	**28**

Laboratory Component:

Sr. No.	Title of the Experiment
1	Image Preprocessing and Enhancement
2	Object Detection with Pre-trained CNNs
3	Image Classification with CNNs
4	Face Recognition using Deep Learning
5	Semantic Image Segmentation
6	Object Tracking with Optical Flow
7	Depth Estimation and 3D Reconstruction(stereo vision techniques.)
8	Augmented Reality with Object Recognition
9	Robotic Navigation and Obstacle Detection Using LiDAR
10	Implement any one application of visual intelligence baed on Research papers and online resources

Reference/Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Deep Learning for Computer Vision		Rajalingappaa Shanmugamani.	Packt Publishing	2018
2	Computer Vision: Algorithms and Applications"		Richard Szeliski	Springer	2011
3	Pattern Recognition and Machine Learning"		Christopher Bishop	Springer NP Exclusive (CBS)	2009

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
PR3-II	Main Project Stage II	0	0	6	4	10	0	0	6	3
		Examination Scheme								
		Component		Phase 1		Phase 2		ESE		Total
ELC		Major Project		40		40		20		100

Pre-requisite Course Codes	All courses
Course Objectives	To provide an opportunity for students to apply theoretical knowledge to real-world or research-based problems. It emphasizes project planning, technical implementation, documentation, and evaluation with real-world relevance
Course Outcomes (CO): ***At the End of the course students will be able to***	
PR3-I.1	Identify and analyze real-world problems to define clear, feasible project objectives.
PR3-I.2	Apply knowledge of engineering fundamentals and modern tools to design and develop effective solutions.
PR3-I.3	Manage project activities including teamwork, communication, and time management.
PR3-I.4	Test, validate, and evaluate the developed system.
PR3-I.5	Present the project work through structured documentation and presentation.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PR3-I.1	3	3	2	2	2	2	1	1	2	2	1	2
PR3-I.2	3	3	3	3	3	2	1	1	2	2	2	3
PR3-I.3	2	2	2	2	2	1	1	2	3	3	3	2
PR3-I.4	3	3	3	3	3	2	1	2	2	2	2	2
PR3-I.5	2	2	2	2	2	2	2	3	3	3	2	3

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
PR3-I.1	3	2	2	2	2
PR3-I.2	3	3	2	3	3
PR3-I.3	2	3	3	2	2
PR3-I.4	3	3	2	3	3
PR3-I.5	2	2	3	2	3

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create✓

Phase III Evaluation Rubric

P1 (02)	P2 (20)	P3 (10)	P4 (05)	Total Marks (Max=40) (Passing= 16)
Completion of All Functionalities/objectives	Work done (Total 80 % Implementation/ Meets project objectives/result analysis/test cases/solution shows originality)	Presentation (Coverage of topic/use of graphics/Delivery &; Speaking skills/Presentation length) and Ethics (Attitude/Deadline/Distribution of Work)	Quality of Documentation / Report (in LaTeX)	

Phase IV Evaluation Rubric

P1 (07)	P2 (12)	P3 (06)	P4 (05)	P5 (10)	Total Marks (Max=40) (Passing= 16)
User Interface/ Usability	Work done (100 % Implementation/ Meets project objectives/result analysis/test cases/ solution shows originality)	Presentation (Coverage of topic/use of graphics/ Delivery & Speaking skills/ Presentation length) and Ethics (Attitude/ Deadline/Distribution of Work)	Report (in LaTeX)	Research paper (in IEEE format with Plagiarism report)	

ESE Evaluation Rubric

P1 (02)	P2 (05)	P3 (08)	P4 (02)	P4 (03)	Total Marks (Max=20) (Passing= 08)
Problem Definition (Idea/Originality/ Novelty/Feasibility/ Application in Society)	Presentation (Demo/Oral/ Written) and Ethics (Attitude/ Deadline/ Distribution of Work)	Work Done (Survey/Design/ Simulation/ Experimentation/ Tools Usage/ Budget Plan/ Execution Plan)	Research paper (in IEEE format with Plagiarism report)	Knowledge (Question-Answer)	

SEMESTER VIII

SEM VIII										
Sr. No.	**Course Category**	**Abbreviation**	**Course Code**	**Course Name**	**L**	**T**	**P**	**O**	**E**	**C**
1	Open Elective	OE	OE2	OE-II**	2	0	1	4	7	3
2	Experiential Learning	ELC	INTR/ INTI/PR4	Research/ Industry Internship/ Major Project***	0	0	24	12	36	11
				Total	**2**	**0**	**25**	**16**	**43**	**14**

- **Summer Research Phase II of minimum 2 month and A course on Optimization for the "Honors by Research" for 7(4+3) credits HR31**
- **For Enrollment to Honors by research, Minimum CGPA must be 8.25**

** To be completed from MOOCs

***Students neither taking research or industry internships nor willing to extend their project work can earn additional 11 credits from Swayam Platform or NPTEL or registering courses from any peer institution of higher learning., besides open elective program elective courses offered by the institute.

PROGRAM ELECTIVE COURSES

4 Electives are sufficient to specialize in a particular domain.

Track	**PE-I** **CS3X1**	**PE-II** **CS3X2**	**PE-III** **CS4X3**	**PE-IV** **CS4X4**
Emerging Networking Technologies	CS311: Digital Forensic	CS312: Cloud Computing	CS413: Block chain Technology	CS414: IT Infrastructure Monitoring and Management
Emerging AI	CS321: Natural Language Processing	CS322: Deep Learning	CS423: Generative AI	CS424: Explainable AI
Data Analytics	CS331: Big data Analytics	CS332: Business analytics with Python	CS433: Data Warehouse and Mining	CS434: AI for Healthcare Analytics
Fundamentals of Visualization	CS341: Fundamentals of Signal & Image Processing	CS332: Augmented Reality & Virtual Reality [AR-VR]	CS433: Computer Vision	CS434: Visual Intelligence

Indicative list of Multidisciplinary Minors

MDM Sequels for EXTC

- Computer Engineering
- AIML
- Data Science
- Interface and Experience Design
- IT Infrastructure

Course Category of Multidisciplinary Minor	MDM-I (Semester IV)	MDM-II (Semester V)	MDM-III (Semester VI)	MDM-IV (Semester VII)
Computer Engineering	**M051:** Database Management Systems	**M052:** Data Structures and Algorithms	**M053:** Cloud Computing	**M054:** Internet and Web Technology
Artificial Intelligence and Machine Learning	**M061:** Fundamentals of NNFL (NN, Fuzzy)	**M062:** Machine Learning	**M063:** Deep Learning	**M064:** Generative AI and application
Data Science	**M071:** Fundamentals of Data Science	**M072:** Data Analytics and Visualization	**M073:** Decision Making and Business Intelligence	**M074:** Social Media Analytics
Interface and Experience Design	**M081:** UI/UX Fundamentals	**M082:** Fundamentals of Experience Design	**M083:** Human Machine Interaction	**M0844:** Prototyping and Interaction Design [Capstone project]

MDM Sequels for CE/CSE

- Industrial IoT
- Digital Signal Processing
- Electronics Communication
- VLSI
- Mathematics and Statistics
- Finance
- Economics

Course Category of Multidisciplinary Minor	**MDM-I (Semester IV)**	**MDM-II (Semester V)**	**MDM-III (Semester VI)**	**MDM-IV (Semester VII)**
Industrial IoT	**M011:** Introduction to Internet of Things	**M012:** Embedded "C" and Micro Python for IoT	**M013:** Communication Protocols for IoT	**M014:** IoT Applications and Analytics
Signal Processing and Communication	**M021:** Digital Signal Processing	**M022**: Digital Communication	**M023:** Digital Image & Video Processing	**M024:** Wireless Communication
VLSI	**M031:** Hardware Description Language programming	**M032:** Digital CMOS VLSI Design	**M033:** VLSI Physical Design	**M044:** ASIC Verification

SYLLABUS
MDM Sequels for EXTC

Course Category of Multidisciplinary Minor	MDM-I (Semester IV)
Computer Engineering	**M051:** Database Management Systems
Artificial Intelligence and Machine Learning	**M061:** Fundamentals of NNFL (NN, Fuzzy)
Data Science	**M071:** Fundamentals of Data Science
Interface and Experience Design	**M081:** UI/UX Fundamentals

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-I	Database Management Systems	2	0	2	4	08	3	0	0	3
M051		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	
Course Objective:	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
M051:.1	Demonstrate understanding of given system to construct a database model
M051:.2	Apply various Relational and SQL commands on the populated database
M051.3	Examine the functional dependencies to make a normalized database system and transaction processing techniques on a database
M051.4	Illustrate query processing and optimization method on a database

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Database Concepts and ER Modeling**	1,2	05
	1.1	Introduction to basic concept of Database, File system V/s Database system, Users of Database system, Database Administrator, Data Independence		
	1.2	The Entity-Relationship (ER) Model, Extended Entity-Relationship (EER) Model , Database integrity		
2	Title	**Relational Algebra And SQL**	1,2	07
	2.1	Relational model, Relational query languages, Relational algebra, Tuple and domain calculus		
	2.2	Structured Query Language: Data Definition Commands, Data Manipulation commands, Data Control commands, Join expressions, views, Triggers		
3	Title	**Normalization And Transaction Processing**	1,2	05
	3.1	Design guidelines for relational schema, Functional dependencies		
	3.2	Normal Forms- 1NF, 2 NF, 3NF, BCNF		
4	Title	**Transaction Processing**	1,2	06
		Transactions, ACID properties, Concurrency Control, Recovery		

		Serializability, Recoverability, Lock-based, Timestamp-based, Validation-based protocols.		
5	**Title**	**Indexing And Query Optimization:**	1,2	**05**
	4.1	Row-wise and column database, database buffering. Indexing, B+-tree indices		
	4.2	Query Processing, Query Optimization		
6	**Self Study**	NO SQL-Data type, Database creation, Basic command for creation, updating and querying the database, Mongo dB, Applications of Hyper Graph DB, cloud database		
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Formulate a case study and create an E-R Diagram. Mapping of E-R model to Relational Model.
2	To create a database and populate using SQL commands (With constraints) • Data Definition Language- Create, Alter, Drop, Rename, Truncate • Data Manipulation Language- Insert, Update, Delete, Select Constraints-Not Null, Unique Key, Primary Key, Foreign Key, Check, Dropping a Constraint
3	To perform DCL, TCL commands • Data Control Language: Grant, Revoke, Roles • Transaction Control Language: Commit, Rollback, Save point
4	To perform Date, Time, Arithmetic and Set operation on database.
5	To perform Aggregate function and Group by- Having clause on database
6	To perform Join operations on database. • Equijoins, Non-Equijoins, Self Joins, Outer Join, cross Join
7	To retrieve data using a Subquery.
8	To Create a different view of the database.
9	To examine the integrity of the database using Triggers.
10	To study distributed databases/Normalization

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Database System Concepts	7th	Korth, Slberchatz, Sudarshan	McGraw – Hill	2019
2	Fundamentals of Database Systems	6th	Elmasri and Navathe	PEARSON Education	2011

©Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Database Management Systems	3rd	Raghu Ramkrishnan and Johannes Gehrke	TMH	2003
2	Database Management Systems	1st	G. K. Gupta	McGraw – Hill.	2018

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-1	Fundamentals of NNFL	2	0	2	4	8	2	0	1	3
M061		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Basic understanding of mathematics, probability, and programming.	
Course Objective: This course focuses on the fundamental of Neural networks and Fuzzy Logic along with its application. Students will be able to design an expert system using neural networks and Fuzzy logic system for implementation of real-world applications.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
M061.1	Identify the various characteristics of Neural Network techniques in building intelligent machines
M061.2	Apply the supervised and unsupervised Neural Network Learning algorithm to solve real world engineering problems.
M061.3	Design Fuzzy Logic Controller System

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO12	PSO1
M061.1	2	2	--	2	--	--	--
M061.2	2	2	--	2	3	2	1
M061.3	2	2	2	2	3	2	1

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	**Apply**	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Neural Networks**	T1,T2 T3,R1	**12**
	1.1	**Basics of Neural Networks:** Introduction to Neural Networks, Biological Neural Networks, Models of ANN with its terminologies, Activation functions and its types		
	1.2	McCulloch-Pitts Model, Linear separability, Hebb Network		
	1.3	**Supervised Learning algorithms:** Perceptron (Single-Layer perceptron: Learning Rule and Applications), Multi-Layer Perceptrons (MLPs), Backpropagation Algorithm: Training MLPs Activation Functions Vanishing Gradient Problem and Solutions		
	1.4	**Un-Supervised Learning algorithms:** Hebbian Learning, Winner takes all, Self-Organizing Maps KSOFMN, Learning Vector Quantization.		
2	**Title**	**Fuzzy Logic, Classical Set and Fuzzy Relations**	T2,T3, T4,R1 , R3	**10**
	2.1	Introduction to Fuzzy Logic, Classical and Fuzzy Sets, Membership Functions, Classical and Fuzzy set operations, and properties of classical and Fuzzy sets. Fuzzy Logic Operators: AND, OR, NOT		
	2.2	Classical and Fuzzy Relations**:** Cartesian product of relation, Fuzzy Max-Min and Max-Product Composition, Fuzzy extension principle		
3	**Title**	**Fuzzy control system design**	T2,T3, T4	**6**
	3.1	Fuzzy Inference System and its types, Fuzzification, Defuzzification, Designing Fuzzy logic control systems.		
	3.2	Mamdani and Sugeno Fuzzy Inference Systems		
4	**Self Study**	Associative Memory Network, Architecture of Neuro-Fuzzy Networks(ANFIS and CANFIS), Performance Metrics for Neural Networks and Fuzzy Logic Systems, Model Evaluation Techniques: Cross-Validation, ROC Analysis, Optimization Techniques: Genetic Algorithms, Particle Swarm Optimization	-	**5**
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No	Title of the Experiment
1	To implement Transfer/Activation Functions for a given problem statement.
2	To design ANN to implement logic gates.
3	To implement Hebb Network for a given problem statement.
4	To design and implement ANN (perceptron) for a given problem statement using Joone Editor
5	Write a program using Single Layer perceptron
6	To implement the Supervised Learning algorithm.
7	To implement the Unsupervised Learning algorithm
8	To implement Fuzzy Sets for a given problem statement
9	To implement Fuzzy Relations for a given problem statement
10	To design and implement Fuzzy Logic controller for a given problem statement

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
T1	Introduction to Artificial Neural Systems	1st	Jacek M. Zurada	Jaico Publisher	1994
T2	Principles of Soft Computing	3rd	Sivanandan and Deepa	Pearson Edition	2019
T3	Fuzzy logic with engineering applications	3rd	Ross, Timothy J	John Wiley & Sons	2011
T4	Neural Networks, Fuzzy Logic and Genetic Algorithms	Kindle	S.Rajasekaran and G.A.Vijayalakshmi Pai	PHI Learning	2013

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
R1	Neural Network Design	2nd	Hagan, Demuth, Beale	CENGAGE Learning	2014
R2	Neuro-Fuzzy and Soft Computing	1st	J.-S.R.Jang .	Pearson	1996
R3	Introduction to Soft Computing	1st	Sameer Roy	Pearson	2013

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-I	Fundamentals of Data Science	2	0	2	5	10	2	0	1	3
M071		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	–
Course Objective: To provide students with a comprehensive understanding of the fundamental concepts, tools, and techniques used in data science and data visualization. This course is designed to introduce students to the basic principles of data science and data visualization, including libraries used for Data Science, data exploration, data preprocessing, EDA, data visualization and basic model building.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
M071.1	Demonstrate the need of libraries used in Data Science.
M071.2	Make use of different tools and techniques for Data Visualization
M071.3	Analyze the data performance using the EDA process.
M071.4	Develop Linear Regression and Logistic Regression Models for a given case study.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M071.1	-	2	-	2	3	-	-	-	-	-	-	-
M071.2												
M071.3	2	2	2	2	3	-	-	-	-	-	-	-
M071.4	-	2	2	-	3	-	-	-	1	1	-	-

LO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
M071.1							
M071.2							
M071.3							
M071.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✔	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Python for Data Science**		06
	1.1	Introduction to Data Science and its roles in Modern Society, Descriptive Statistics		
	1.2	Working with Numpy Basics of Numpy, Creating NumPy Arrays, Mathematical Operations on NumPy Arrays	1,2,3	
	1.3	Working with Pandas Basics of Pandas, Data Frames, Indexing and Slicing, Operations on Data Frames, GroupBy and Aggregate Functions, Merging Data Frames, Pivot Tables	1,2,3	
2	Title	**Data Visualization**		06
	2.1	Data Visualization using Matplotlib Bar Graphs, Scatter Plots, Line Graphs, Histogram, Box Plots, Sub Plots	1,3,7	
	2.2	Data Visualization using Seaborn Distribution Plots, Pie Charts, Bar Charts, Scatter Plots, Box Plots, Pair Plots, Heat Maps, Line Charts	1,3,8	
4	Title	**Data Preprocessing and Exploratory Data Analysis**		06
	4.1	Data Sourcing Public Data and Private Data, Web Scraping	5,6	
	4.2	Data Cleaning Fixing the rows and columns, Overfitting and Underfitting, Impute/Remove missing values, Feature Engineering Techniques, Feature Reduction Techniques, Handling Outliers, Standardizing values	5,6	
	4.3	Univariate Analysis Categorical Ordered and Unordered Univariate Analysis	5,6	
	4.4	Bivariate and Multivariate Analysis Numerical-Numerical Analysis, Numerical-Categorical Analysis, Categorical-Categorical Analysis	5,6	
5	Title	**Regression**		10
	5.1	Linear Regression Simple Linear Regression, Multiple Linear Regression	5,6	
	5.2	Logistic Regression Univariate Logistic Regression, Multivariate Logistic Regression Model Building and Evaluation	5,6	
6	Self Study	Plotly, Different Visualization visualization techniques other than Python and Tableau.	1,2,5	
			Total	**28**

Laboratory Component

Sr. No.	Title of the Experiment
1	Working with Python Basics
2	Working with Python Data Structures
3	Working with Python Numpy Library
4	Working with Python Pandas Library
5	Working with Python Matplotlib Library
6	Working with Python Seaborn Library
7	Exploratory Data Analysis on the given Case Study
8	Develop a Linear Regression Model on the given Case Study
9	Develop a Logistic Regression Model on the given Case Study
10	Mini Project based on a case study.

Text Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Python for Data Science Handbook: Essentials Tools for Working with Data	Second	Jake VanderPlas	O'Reilly	2022
2	Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter	Third	Wes McKinney	O'Reilly	2022
3	Python Data Visualization Essentials Guide	First	Kalilur Rahman	BPB	2021
4	Data Visualization Through TABLEAU	First	George Peck	McGraw Hill	2020
5	Hands-On Exploratory Data Analysis with Python	First	Suresh Kumar Mukhiya, Usman Ahmed	Packt	2020
6	Python Data Science Essentials	Third	Alberto Boschetti, Luca Massaron	Packt	2018

Web References:

1) https://matplotlib.org/
2) https://seaborn.pydata.org/

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM I	UI/UX Fundamentals	2	0	2	4	8	2	0	1	3
M081		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	50		50		100			200
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	AI305
Course Objective: The aim of the UI/UX course is to provide students with the knowledge of user-centered design, user -centered methods in design, graphic design on screens, simulation and prototyping techniques, usability testing methods, interface technologies and user centered design in corporate perspective. The course is organized around a practical project with iterative design of a graphical user interface to organize information about users into useful summaries with affinity diagrams, to convey user research findings with personas and scenarios and to learn the skill of sketching as a process for user experience design. The students will be given exposure to wireframing and Prototyping software in the various UI/UX Design tools.	
Learning Outcomes (LO): ***At the End of the course students will be able to***	
M081.1	Understand iterative user-centered design of graphical user interfaces
M081.2	Apply the user Interfaces to different devices and requirements
M081.3	Design prototype for the given design problems.

LO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation) CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M081.1	2	-	--	2	--	--	--	--	--	--	--	--
M081.2	-	-	2	-	2	--	--	--	--	--	--	--
M081.3	-	-	-	-	2	--	--	--	-	-	--	2

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply ✓	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**Title**	**Introduction to UI and UX**	T1-T4	**06**
	1.1	What is User Interface Design (UI) -The Relationship Between UI and UX, Roles in UI/UX, Menus, Tabs, Bottom tab bar, Buttons (including "Call to action" or CTA), Accordion, Carousel, Breadcrumbs, Modals Forms, etc		
	1.2	A Brief Historical Overview of Interface Design, Interface Conventions, Designer vs. developer, Skills to be a top designer.		
2	**Title**	**User Interface Design Elements**	T2,T4	**06**
	2.1	Approaches to Screen Based UI, Template vs Content, Formal Elements of Interface Design, Active Elements of Interface Design, Composing the Elements of Interface Design, UI Design Process, Visual Communication design component in Interface Design, Spacing and the grid.		
3	**Title**	**Colors, Typography & Fonts**	T1-T4	**05**
	3.1	Display Text (Such as Headings) versus Body Text, Legibility , Type Trends, Typeface Selection & Pairing, Where to Get Web Fonts, Ideal Line Height, Column Width (Line Length), Hyphenation & Justification		
	3.2	Color Harmonies, Creating Contrast with Color, Guidelines for Proper Color Usage		
4	**Title**	**Design Guidelines and Process**	T1-T4	**06**
	4.1	UX Basics- Foundation of UX design, Good and poor design, Understanding Your Users, Designing the Experience, Elements of user Experience, Visual Design Principles, Functional Layout, Interaction design, Introduction to the Interface, Navigation Design, User Testing, Developing and Releasing Your Design		
5	**Title**	**Design Tools**	T2,T4	**03**
	5.1	Adobe Photoshop, Illustrator, Figma, AdobeXD, Pencil Project tool.		
	Self Study	Mobile Ecosystem: Platforms, Application frameworks: Types of Mobile Applications: Widgets, Applications.		**02**
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Case Study to understand and know about UI elements for Zomato, Swiggy app.
2	To create logo for restaurant based on given scenario using Adobe XD.
3	To create basic app for creating your own profile using Adobe XD.
4	To create different icons for giving ratings using Pencil tool. Use your own creativity.
5	To customize typography for logo for fitness app in UI design using Pencil tool. Use your own creativity.
6	To evaluate Good and Bad Design of UI elements for IRCTC website
7	To customize typography for logo for selling variations of plants, pots and fertilizers in UI design using Illustrator tool.
8	To create GUI for creating feedback form for taking feedback from students and teachers by using only icons which are familiar to users and choose colors based on the scenario and describe the reason for such selection for the same.
9	To create simple Web UI for newly developed Pencil company (like apsara, doms etc.) using different kinds of Menus in an application using Figma tool.
10	To create simple flower selling application in Mobile app layout, laptop screen layout and Desktop layout and compare using Figma tool.

Refernces

1. A Project Guide to UX Design: For user experience designers in the field or in the making (2nd. ed.). Russ Unger and Carolyn Chandler. New Riders Publishing, USA, 2012.
2. The Elements of User Experience: User-Centered Design for the Web and Beyond, Second Edition Jesse James Garrett, Pearson Education. 2011.
3. The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques, Third Edition Wilbert O. Galitz , Wiley Publishing, 2007.
4. The UX Book Process and Guidelines for Ensuring a Quality User Experience, Rex Hartson and Pardha S. Pyla, Elsevier, 2012.

Course Category of Multidisciplinary Minor	MDM-II (Semester V)
Computer Engineering	**M052:**Data Structures and Algorithms
Artificial Intelligence and Machine Learning	**M062:**Machine Learning
Data Science	**M072:**Data Analytics and Visualization
Interface and Experience Design	**M082:** Fundamentals of Experience Design

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(MDM-II)	Data Structures and Algorithms	3	0	2	5	10	3	0	1	4
		Examination Scheme								
M052		Component	ISE		MSE		ESE		Total	
		Theory	50		50		100		200	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	Problem-solving using imperative programming
Course Objective: To introduce the fundamentals and abstract concepts of Data Structures for Problem Solving.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
M052.1	Apply various operations of linear and non-linear data structures to given problems.
M052.2	Apply the concepts of Trees and Graphs to a given problem.
M052.3	Analyze the algorithm for given Problem statement
M052.4	Apply the Divide and Conquer, Greedy method, Dynamic Programming strategy to solve given problems.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M052.1	1	2	2					2				2
M052.2	1	2	2					2				2
M052.3	1	2	2					2				2
M052.4	1	2	2					2				2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
	EXTC	ETRX	EXTC	ETRX	EXTC	ETRX	EXTC
M052.1	1	1			1	1	
M052.2	1	1			1	1	
M052.3	1	1			1	1	
M052.4	1	1			1	1	

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✓	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Data Structures**		7
	1.1	Concept of Linear and Non-linear Data Structures, **Stack:** Stack as ADT, operations on the stack, **Queue:** Queue as ADT, Operations on Queue,	1,2	4
	1.2	**Linked List:** Linked List as ADT, Operations on Singly Linked List. Types of the linked list- Linear and circular linked lists, Doubly Linked List	1,2	3
2	Title	**Trees (**use of visualization tools (e.g., Graphviz, VisuAlgo) for better comprehension of tree and graph representations is encouraged)		3
	2.1	Trees as ADT, General tree v/s Binary Tree Terminology, Traversal of Binary Tree, Operations on Binary tree, Binary Search Tree and its operations.	1,2	3
3	Title	**Graphs**		3
	3.1	Graph as ADT, Introduction To Graph, Representation of Graph- Adjacency Matrix, Adjacency List, Graph Traversal Technique	1,2	3
4	Title	**Hashing**		4
	4.1	Introduction to Hash Table, Hash functions, Collision Resolution Technique.	1,2	4
5		**Introduction to Analysis of algorithm**		8
	5.1	Role of Algorithms in Computing, Performance analysis-space and time complexity, Growth of Functions: Asymptotic Notation, Analysis of sorting algorithms such as Selection sort and insertion sort, Master Method	1,2	4
	5.2	**Divide and Conquer Approach –** General Method, Merge Sort and its Analysis, Quick sort and its Analysis, Binary search algorithm and its analysis	1,2	4
6		**Greedy and Dynamic Programming Approach**		9
	6.1	**Greedy Approach:** Basic strategy, Knapsack problem, single-source shortest path-Dijkstra's algorithm. Minimum cost spanning trees-Kruskal algorithm	1,2	4
	6.2	**Dynamic Programming:** 0/1 knapsack problem, Assembly-line scheduling, Longest common subsequence	1,2	5
7		**Backtracking and Branch-and-bound**		8
	7.1	Backtracking – General method, 8-queen problem (N-queen problem), Sum of subsets, Graph coloring	1,2	4
	7.2	Branch-and-bound – General method, Travelling Salesman problem, 15 puzzle problem	1,2	4
8	Self Study	AVL Tree, Optimal Binary Search Tree, Approximation and NP-Completeness		5
			Total	**42**

Laboratory Component, if any (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Implement a given problem statement using Stack.
2	Implement a given problem statement using Queue
3	Implement a given problem statement using Linked List.
4	Implement a given problem statement using Binary Trees.
5	Apply Graph Traversal Technique on a given problem statement to solve the problem
6	Implement and analyze insertion sort selection sort
7	Implement and analyze problem based on Divide and Conquer strategy - Merge and Quick sort
8	Implement a given problem statement using Greedy Strategy.
9	Implement a given problem statement using Dynamic Programming.
10	Implement a given problem statement using Backtracking /Branch & bound

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Algorithms	Third Edition	Thomas H. Cormen, Charles E. Leiserson, Ronald L Rivest, Clifford Stein	MIT Press	2009
2	Fundamentals of Computer Algorithms	Second Edition	Horowitz E, Sahni S and S.Rajasekaran	Galgotia Publications	2010

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Classic Data Structures	Second	Samanta Debasis	PHI	2009
2	Data Structures With C	First	Seymour Lipschutz	Schaum's Outline Series	2010

Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai) Bhavan's Campus,
Munshi Nagar, Andheri (West), Mumbai-400058

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-II	Machine Learning	3	-	2	4	9	3	-	1	4
M062		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		100		100		100		300
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	
Course Objective: To learn fundamentals of Data Structures.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
M062.1	Understanding Machine Learning algorithms with their associated mathematical foundations for appreciating these algorithms.
M062.2	Develop and apply various supervised and unsupervised algorithms using the concept of Dimension reduction techniques
M062.3	Develop and apply reinforcement learning algorithms for learning to control complex systems
M062.4	Enable formulating real world problems to implement basic algorithms using machine learning libraries.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M062.1	2	-	-	-	-	-	-	-	-	-	-	-
M062.2	2	2	**2**	**2**	**2**	-	-	-	1	-	1	-
M062.3	2	2	**2**	**2**	**2**	-	-	-	1	-	1	-
M062.4	-	2	**2**	-	**2**	-	-	-	1	-	1	**1**

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
M062.1	**1**	-	-	2	-
M062.2	**1**	2	-	2	-
M062.3	**1**	-	-	2	-
M062.4	**1**	2	-	2	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✓

Theory Component

Module No.	Topics	Ref.	Hrs.
1	**Introduction To Machine Learning** Need of machine learning, machine learning vs AI, machine learning vs Deep learning ,Learning types : Supervised Learning, Unsupervised learning, Reinforcement learning, What makes Machine Learning tick purpose or objectives, variety of algorithms- learning style, similarity style, Applications of machine learning, General Steps or Process of Machine Learning- SourceX -> Feature Extraction -> Feature Correlation -> Feature TransformX-> Train Model-> Ensemble-> Evaluate Capacity, Overfitting and Underfitting, Hyperparameters and Validation Sets. Estimators, Bias and Variance, likelihood, Stochastic Gradient Descent.	1,2	9
2	**Supervised Learning** Hypothesis testing, Training versus Testing, Gradient Descent, Over fitting & Regularization ,Regression: Regression fundamentals, Linear Regression, Polynomial regression, Regularization technique (LASSO), Classification: Classification fundamentals, Logistic Regression, Decision trees,-CART,-Random Forest, Naïve Bayes , Support Vector Machine, Time Series, Neural Networks, Types of errors (TYPE I and TYPE II), Case Study(Classification)	1,2	11
3	**Unsupervised Learning** Clustering basics: K-means clustering, K-Nearest Neighbor ,Association Rule Learning , Hierarchical	1,2	9
4	**Dimensionality Reduction** Feature Engineering, Feature Selection methods, - Filters; Wrappers, Embedded, PCA, SVD, -tSNE, Case Study (Clustering/Anomaly/Fraud Detection)	1.2	7
5	**Reinforcement Learning** Markov Decision, Monte Carlo Prediction, Case Study (next best offer, dynamic pricing)	1.2	7
6	**Self-Study Topics** Machine Learning Applications across Industries (Healthcare, Retail, Financial Services, Hospitality), Watson, Explore tools used in ML, TensorFlow, Keras, Scikit learn		
		Total	**42**

Laboratory Component

Sr. No.	Title of the Experiment
1	Introduction to Python/R Introduction Python/R, Python/R data types and objects, reading and writing data, Python/R Packages
2	Python/R flow control Control structures, scoping rules, dates and times, data manipulation in Python/R
3	Functions and Modules Loop functions, debugging tools, Mathematical Functions, Data Processing and handling
4	Apply Linear regression
5	Apply Logistic regression
6	Apply decision tree for given problems
7	Apply Random Forest for given problems
8	Apply Naïve Bayes for given problems
9	Apply K means clustering for given problem
10	Apply PCA for given problem

Note: Course coordinators can use their own problem statements to solve listed lab sessions.

Text Books

[1] Shai Shalev-Shwartz and Shai Ben-David," Understanding Machine Learning: From Theory to Algorithms", Cambridge University Press, 1st Edition,2014

[2] Mehryar Mohri Afshin , Rostamizadeh ,Ameet Talwalkar," Foundation of Machine Learning", The MIT Press, 2nd Edition,2018

Reference Books

[1] Gareth James, Daniela Witten, Trevor Hastie Robert Tibshirani," An Introduction to Statistical Learning", Springer, 7th Edition,2007

[2] Andrew Ng, Machine Learning Yearning, Deeplearning.ai, Draft v0.5, 2018

[3] Dr. Dinesh Kumar," Machine-learning-using-python", WileyIndia,1st Edition, 2019

Web References

[1] https://www.altexsoft.com/blog/datascience/comparing-machine-learning-as-a-service-amazonmicrosoft-azure-google-cloud-ai-ibm-watson/

[2] https://cloud.google.com/products/ai.

Bharatiya Vidya Bhavan's

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-II	Data Analytics and Visualization	3	0	2	4	9	3	0	1	4
M072		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		75		75		150		300
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	MDCS31
Course Objective: This course aims to provide foundational knowledge and practical skills in data analytics and visualization. Students will learn to analyze, preprocess, and visualize data using tools like Python, SQL, and Power BI. Emphasis is placed on real-world applications, effective storytelling with data, and ethical considerations.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
M072.1	Illustrate the concepts of data analytics, types of data, and preprocessing techniques used in real-world applications.
M072.2	Apply data visualization principles and use libraries/tools like Matplotlib, Seaborn, Plotly, and Power BI to create effective visualizations.
M072.3	Analyze time series and geographic data using appropriate visualization and storytelling techniques.
M072.4	Design and implement a complete data analytics project, integrating machine learning, visualization, and ethical considerations.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M072.1												
M072.2												
M072.3												
M072.4												

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
M072.1							
M072.2							
M072.3							
M072.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✓

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Data Analytics**	1,2,4,5,6	8
	1.1	**Overview of Data Analytics** Definition and importance of data analytics Real-world applications (business, healthcare, finance, etc.) Data types and sources		
	1.2	**Data Collection and Storage** Structured vs. unstructured data Data collection techniques Databases and data warehouses		
	1.3	**Data Cleaning and Preprocessing** Handling missing values and duplicates Data transformation and normalization Exploratory Data Analysis (EDA) basics		
	1.4	**Tools and Technologies in Data Analytics** Introduction to Excel, SQL, Python, and R Overview of popular libraries (Pandas, NumPy, Matplotlib, Seaborn) Cloud computing and big data platforms		
2	Title	**Data Visualization Principles**	2,7	10
	2.1	**Importance of Data Visualization** Why visualization matters in data analysis Types of visualizations and their use cases		
	2.2	**Chart Types and Best Practices** Bar charts, line graphs, scatter plots, histograms Heatmaps, box plots, and pie charts Best practices for effective visualization		
	2.3	**Data Visualization with Matplotlib and Seaborn** Creating plots with Matplotlib Enhancing visualizations with Seaborn Customizing charts (colors, labels, annotations)		
	2.4	**Interactive Visualizations with Plotly and Power BI** Introduction to Plotly for interactive charts Basics of Power BI dashboarding DAX functions		
3	Title	**Advanced Data Visualization Techniques**	3,7	12
	3.1	**Storytelling with Data** How to present data insights effectively Common mistakes to avoid		
	3.2	**Time Series Analysis and Visualization**		

		Time series components (trend, seasonality) Visualizing time series data		
	3.3	**Geographic Data Visualization** Using maps for data visualization Tools like Folium and Power BI for geospatial analysis		
	3.4	**Dashboard Design with Power BI/Tableau** Creating interactive dashboards Connecting data sources and filters		
	3.5	**Real-World Case Study: Building a Business Dashboard** Hands-on project with Power BI or Tableau		
4	**Title**	**Applied Data Analytics & Project**	1,4,5,8	**12**
	4.1	**Data Analytics in Different Industries** Business, finance, healthcare, e-commerce use cases		
	4.2	**Machine Learning for Data Analytics** Introduction to classification and clustering Basic models (Logistic Regression, K-Means)		
	4.3	**Ethics and Data Privacy** Data security, GDPR, and ethical concerns		
	4.4	**Final Project** Working on a real dataset Presentation of insights and visualization		
5	**Self Study**	Data Ethics and Bias in Machine Learning, Data Pipeline concepts, Big Data Ecosystem, Google Data Studio		
			Total	**42**

Laboratory Component

Sr. No.	Title of the Experiment
1	Data Cleaning & Preprocessing
2	Data Exploration & Summary Statistics
3	SQL Data Analysis
4	Data Visualization with Matplotlib & Seaborn
5	Power BI/Tableau Dashboard Creation
6	Time Series Analysis & Forecasting
7	Sentiment Analysis on Text Data
8	Clustering with K-Means
9	A/B Testing for Business Decisions
10	End-to-End Data Analytics Project

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Data Science from Scratch: First Principles with Python	Second	Joel Grus	O'Reilly Media	2019
2	Python for Data Analysis	Third	Wes McKinney	O'Reilly Media	2022
3	Storytelling with Data: A Data Visualization	First	Cole Nussbaumer Knaflic	Wiley	2015

	Guide for Business Professionals				

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Data Analytics Made Accessible	Forth	Anil Maheshwari	Amazon/KDP	2021
2	Practical Statistics for Data Scientists	Second	Peter Bruce, Andrew Bruce, Peter Gedeck	O'Reilly Media	2020
3	Learning SQL	Third	Alan Beaulieu	O'Reilly Media	2020
4	Hands-On Data Visualization	First	Jack Dougherty and Ilya Ilyankou	O'Reilly Media	2021
5	Interactive Data Visualization for the Web	Second	Scott Murray	O'Reilly Media	2017

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-II	Fundamentals of Experience Design	3	0	2	4	9	3	0	1	4
M082		Examination Scheme								
		Component	ISE		MSE		ESE			Total
		Theory	75		75		150			300
		Laboratory	50		--		50			100

Pre-requisite Course Codes		Software Engineering
At the End of the course students will be able to		
Course Outcomes	M082.1	Analyze UX design life cycle and its process for users.
	M082.2	Apply UX design process for the given scenario
	M082.3	Create real life application with end-to-end understanding of User experience practices.
	M082.4	Evaluate UX design process for best experience.

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1		**UX Design and Life Cycle**	1,2	6
	1.1	What is UX (User Experience), Ubiquitous interaction, A UX process lifecycle template, The system complexity space, Meet the user interface team		
2		**The UX Design Process – Understand Users**	1	12
	2.1	**Contextual Inquiry:** Introduction, the system concept statement, User work activity gathering, Abridged contextual inquiry process		
	2.2	**Contextual analysis:** Introduction, Creating and managing work activity notes, Constructing your WAAD (Work Activity Affinity Diagram)		
	2.3	**Extracting Design Interaction requirements:** Formal Requirements Extraction, Abridged method for requirement extraction		
	2.4	**Design Informing Model:** User Model (Social Model), Usage Model (Flow Model, Task Interaction Model), Work Environment Model.		

3		**The UX Design Process-Design Thinking, Ideation and Sketching**	1,3	12
	3.1	Design Paradigm, Design thinking, Design perspective, User personas, Ideation, Sketching		
	3.2	Mental Models and Conceptual Design		
	3.3	Storyboards, Wireframes		
4		**The UX Design Process- Prototyping, Evaluation and Agile development**	1,3, 5	12
	4.1	Fidelity of Prototype, To make effective paper prototype,		
	4.2	UX Evaluation and Improve UX Goals, Metrics and Targets		
	4.3	UX Evaluation Techniques- Formative vs Summative		
	4.4	UX Method for Agile Development: Introduction, Basics of agile SE methods, drawbacks of agile SE methods from UX perceptive.		
5		**Self-Learning**		
	5.1	Affordance, Integration of UX into agile SE methods		
			Total	42

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	To Study of open source UX tools (Justinmind Prototype, Pidoco, Marvel Prototype) and create UX design for a given problem definition.
2	Design Mobile/Web UI for your own Travelling agent considering adding map and localization feature along with descriptions if required. (e.g, Make my Trip,Tripadvisor,thrillophilia etc.)
3	Design a Map based UI(Web User) for Mumbai Dabbawalas with localization feature.
4	Pick a website/app that you use on a daily basis (eg. facebook, gmail, whatsapp, zomato, etc). Evaluate the product based on user experience principles and give suggestions for improvement. Explain usability testing process for the same.
5	Analyze and redesign the (IRCTC/IndianRail) website for better User Experience and create the heuristic report using Nielsen's Heuristic
6	Design UI for student to teach mathematics in rural areas/ to sell the products of farmers directly to consumers.
7	Design UI for student to sell the products of farmers directly to consumers.
8	Design UI/UX Mobile App along with making logo for the same for your own newly opened restaurant. (Consider all the UX Parameters while designing).
9	Design UI for any differently abled users.
10	Design UI for analysis of number of children suffering from juvenile diabetic children in India. The design aims at providing solutions for improving quality treatment and making the treatment affordable.

Note: All the experiments need to be tested for usability.

References:

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	The UX Book	1st	Rex Hartson and Pardha Pyla	MK Publication	2012
2	A Project Guide to UX Design: For user experience designers in the field or in the making.	2nd	Russ Unger and Carolyn Chandler	O'reilly, Series Editor	2012
3	UI Design:Key to captivate user understanding	1st	Jain, Kalbande,	SybGen Publications	2021

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Smashing UX Design	1st	Jesmond Allen and James Chudley	John Wiley & Sons	2012
2	Agile Experience Design	1st	Brian Fling	, O'Reilly Media Inc.,	2009.

Course Category of Multidisciplinary Minor	MDM-III (Semester VI)
Computer Engineering	**M053:**Cloud Computing
Artificial Intelligence and Machine Learning	**M063:**Deep Learning
Data Science	**M073:**Decision Making and Business Intelligence
Interface and Experience Design	**M083:**Human Machine Interaction

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-III	Cloud Computing	-	1	2	2	5	-	-	1	4
		Examination Scheme								
		Component	ISE		MSE		ESE			Total
M053		Theory	100		100		100			300
		Laboratory	50		--		50			100

Pre-requisite Course Codes, if any.	Machine Learning
Course Objective: This course introduces Concepts of cloud computing. This course focuses on the architecture of cloud computing along with the application of the same.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
M053.1	Understand the different Cloud Computing environment
M053.2	Analyze virtualization technology and install virtualization software
M053.3	Apply security in cloud applications
M053.4	Use advance techniques in Cloud Computing

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M053.1	2	-	-	-	-	-	-	-	-	-	-	1
M053.2	2	2	-	-	-	-	-	-	-	-	-	1
M053.3	3	3	3	-	-	-	-	-	-	-	-	2
M053.4	2	2	3	3	2	2	2	2	2	2	2	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
M053.1					
M053.2					
M053.3					
M053.4					

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✓

Theory Component

Module No.	Topics	Ref.	Hrs.
1	Cloud Computing Architecture and Management Cloud architecture, Layer, Anatomy of the Cloud, Network Connectivity in Cloud Computing, Applications, on the Cloud, Managing the Cloud, Managingthe Cloud Infrastructure Managing the Cloud application, Migrating Application to Cloud, Phases of Cloud Migration Approaches for Cloud Migration.	1,2	4
2	Cloud Service Models Infrastructure as a Service, Characteristics of IaaS. Suitability of IaaS, Prosand Cons of IaaS, Summary of IaaS Providers, Platform as a Service, Characteristics of PaaS, Suitability of PaaS, Pros and Cons of PaaS, Summary of PaaS Providers, Software as a Service, Characteristics of SaaS, Suitability of SaaS, Pros and Cons of SaaS, Summary of SaaS Providers,Other Cloud Service Models.	1,2	11
3	Virtualization in Cloud Computing Definition of Virtualization, Adopting Virtualization, Types of Virtualization, Virtualization Architecture and Software, Virtual Clustering, Virtualization Application, Pitfalls of Virtualization. Grid, Cloud and Virtualization: Virtualization in Grid, Virtualization in Cloud, Virtualization and Cloud Security. Virtualization and Cloud Computing: Anatomy of Cloud Infrastructure, Virtual infrastructures, CPU Virtualization, Network and Storage Virtualization.	1,2	11
4	Security in Cloud Computing Risks in Cloud Computing: Risk Management, Enterprise-Wide Risk Management, Types of Risks in Cloud Computing. Data Security in Cloud: Security Issues, Challenges, advantages, Disadvantages, Cloud Digital persona and Data security, Content Level Security. Cloud Security Services: Confidentiality, Integrity and Availability, Security Authorization Challenges in the Cloud, Secure Cloud Software Requirements, Secure Cloud Software Testing.	1.2	8
5	Advanced Techniques in Cloud Computing Future Trends in cloud Computing, Mobile Cloud, Automatic Cloud Computing: Comet Cloud. Multimedia Cloud: IPTV, Energy Aware Cloud Computing, Jungle Computing, Distributed Cloud Computing Vs Edge Computing, Containers, Docker, and Kubernetes, Introduction to DevOps. IOT and Cloud Convergence: The Cloud and IoT in your Home, The IOT and cloud in your Automobile, PERSONAL: IoT in Healthcare.	1.2	8
6	Self Study Features of Cloud and Grid platforms, Programming support for Google App Engine, Programming on Amazon AWS, Programming on Microsoft Azure, Emerging Cloud software Environments, Understand the need of Cloud Computing, Existing Cloud Applications and opportunities for new Applications		4*
		Total	**42**

Laboratory Component

Sr. No.	Title of the Experiment
1	Working and Implementation of Infrastructure as a service
2	Working and Implementation of Software as a service.
3	Working and Implementation of Platform as a services
4	Practical Implementation of File sharing and Storage as a Service
5	Working and Implementation of identity management.
6	Practical Implementation of cloud security.
7	Installing and Developing Application Using Google App Engine.
8	Managing and working of cloud xen server
9	Using OpenNebula to manage heterogeneous distributed data centre Infrastructure.
10	Working with Aneka and demonstrate how to Managing cloud computing Resources.

Textbooks

1. A. Srinivasan, J. Suresh, “Cloud Computing: A Practical Approach for Learning and Implementation”, Pearson, ISBN: 978-81-317-7651-3
2. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, “Mastering Cloud Computing”, McGraw Hill Education, ISBN-13:978-1-25-902995-0

Reference Books

1. James Bond ,“The Enterprise Cloud”, O'Reilly Media, Inc. ISBN: 9781491907627
2. Dr. Kris Jamsa, “Cloud Computing: SaaS, PaaS, IaaS, Virtualization and more”, Wiley Publications, ISBN: 978-0-470-97389-9
3. Anthony T. Velte Toby J. Velte, Robert Elsenpeter, “Cloud Computing: A Practical Approach”, 2010, The McGraw-Hill.
4. Gautam Shrof, “ENTERPRISE CLOUD COMPUTING Technology Architecture, Applications”, Cambridge University Press, ISBN: 9780511778476
5. Tim Mather, Subra K, Shahid L.,”Cloud Security and Privacy”, Oreilly, ISBN-13 978-81- 8404-815-5
   e-Books:
   https://sjceodisha.in/wp-content/uploads/2019/09/CLOUD-COMPUTING-Principles-andParadigms.pdf
   https://studytm.files.wordpress.com/2014/03/hand-book-of-cloud-computing.pdf
   https://arpitapatel.files.wordpress.com/2014/10/cloud-computing-bible1.pdf
   https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.500-291r2.pdf
   MOOCs Courses link: Cloud Computing https://onlinecourses.nptel.ac.in/noc21_cs14/preview?
   Cloud Computing and Distributed System: https://onlinecourses.nptel.ac.in/noc21_cs15/preview?
   https://www.digimat.in/nptel/courses/video/106105167/L01.html
   https://www.digimat.in/nptel/courses/video/106105167/L03.html
   https://www.digimat.in/nptel/courses/video/106105167/L20.html

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		L	T	P	O	E	L	T	P	Total
MDM-III	Deep Learning	2	-	2	2	5	-	-	1	3
		Examination Scheme								
		Component		ISE		MSE		ESE		Total
M063		Theory		50		50		100		200
		Laboratory		50		--		50		100

Pre-requisite Course Codes, if any.	Machine Learning
Course Objective: This course introduces Concepts of cloud computing. This course focuses on the architecture of cloud computing along with the application of the same.	
Course Outcomes (CO): ***At the End of the course students will be able to***	
M063.1	Understand the fundamentals of deep learning.
M063.2	Understanding the working of Convolutional Neural Networks and RNN in decision making.
M063.3	Implement, optimize and tune state-of-the-art deep neural network architectures
M063.4	Implement major deep learning algorithms, the problem settings, and their applications to solve real world problems

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M063.1	2	-	2	2	-	-	-	-	-	-	-	1
M063.2	2	2	-	-	2	-	-	-	-	-	-	1
M063.3	3	3	3	2	2	-	-	-	-	-	-	2
M063.4	2	2	3	3	2	2	2	2	2	2	2	2

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
M063.1	1	-	-	2	-
M063.2	1	2	-	2	-
M063.3	1	-	-	2	-
M063.4	1	2	-	2	-

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✓

Theory Component

Module No.	Topics	Ref.	Hrs.
1	Introduction	1,2	5
	Basic supervised classification task, optimizing logistic classifier using gradient descent, stochastic gradient descent, momentum, and adaptive sub-gradient method. Neural Networks: Feedforward neural networks, deep networks, regularizing a deep network, model exploration, and hyper parameter tuning.		
2	Introduction to Convolution Neural Networks CNN	1,2	6
	Architecture and Operations, convolutional layer, Pooling layer, Variants of the Convolution Model, Forward and Backward propagation, Training a deep neural network, hyper-parameter tuning, Generalization Gap – Under-fitting Vs Over-fitting – Optimization, Normalization.		
3	Optimization algorithms	1,2	6
	Mini-batch gradient descent, exponentially weighted averages, RMS prop, Learning rate decay, the problem of local optima, Batch norm – Parameter tuning process.		
4	Sequence Modeling: Recurrent Nets	1.2	7
	Unfolding computational graphs, recurrent neural networks (RNNs), bidirectional RNNs, encoder-decoder sequence to sequence architectures, deep recurrent networks, LSTM networks.		
5	Interactive Applications of Deep Learning	1.2	4
	Generative Adversarial Networks, Deep Reinforcement Learning.		
6	Self Study		
	Deep Learning Research: Autoencoders, Deep Generative Models: Boltzmann Machines Restricted Boltzmann Machines, Deep Belief Networks		
		Total	**28**

Laboratory Component

Sr. No.	Title of the Experiment
1	Study and Implement different classifier for decision tree algorithm eg. J48
2	Working and Implementation of Bayesian Hyperparameter Optimization
3	Implement Single-Layer Feedforward Neural Network
4	Implement Multi-Layer Feedforward Neural Network
5	Learning TensorFlow API and Keras API
6	Implement Optimization technique
7	Build and train RNNs, work with NLP and word embeddings
8	Use LSTMs and GRUs for implementing a real world problem
9	Implement a GAN for a given problem
10	Implement DCGAN for a given problem

Note: Course coordinators can use their own problem statements to solve listed lab sessions.

Textbooks

1. Deep Learning- Ian Goodfellow, YoshuaBengio and Aaron Courvile, MIT Press, 2016
2. Deep Learning with Python - Francois Chollet, Released December 2017, Publisher(s): Manning Publications, ISBN: 9781617294433

References

1. Bunduma, N. (2017). Fundamentals of Deep Learning
2. Heaton, J.(2015). Deep Learning and Neural Networks, Heaton Research Inc.

Additional References

1. Goodfellow, I. (2016). Deep Learning. MIT Press.
2. Deng, L., & Yu, D. (2009). Deep Learning: Methods and Applications (Foundations and Trends in Signal Processing). Publishers Inc.
3. Hall, M.L, (2011). Deep Learning. VDM Verlag

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-III	Decision Making and Business Intelligence	2	0	2	4	8	2	0	1	3
M073		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
		Theory	50		50		100		200	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	Applied Mathematics, Programming

Course Objective:
1.Introduce students to the principles and concepts of decision making and business intelligence.
2.Familiarize students with the various types of data and information used in decision making and business intelligence, including structured, semi-structured, and unstructured data.
3.Teach students how to collect, process, and analyze data to support decision making and business intelligence, using tools such as data warehouses, ETL processes, and data modeling.

Course Outcomes (CO): ***At the End of the course students will be able to***
1.Understand the principles and concepts of decision making and business intelligence.
2.Collect, process, and analyze data to support decision making and business intelligence, using various business intelligence techniques

M073.1	Identify various decision making methods in industries.
M073.2	Summarize the characteristics of complex business problems.
M073.3	Apply prediction and optimization techniques for decision making.
M073.4	Design expert system for decision making using expert techniques.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M073.1	2										2	
M073.2	2			2	2	2					2	
M073.3	2			2	2	2					2	
M073.4	2			2	2	2					2	

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

CO	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
M073.1							
M073.2							
M073.3							
M073.4							

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze ✓	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction to Decision Making**		4
	1.1	Introduction: Introduction to decision making methods, AHP	1-4	
	1.2	SAW,WPM methods and examples.		
2	Title	**Characteristics of Complex Business Problems**		5
	2.1	Number of Possible Solutions, Time-Changing Environment , Problem-Specific Constraints	1-4	
	2.2	Multi-objective Problems , Modeling the Problem, A Real-World Example		
3	Title	**Prediction Methods and Models**		6
	3.1	Data Preparation, Different Prediction Methods, Mathematical Methods, Distance Methods	1-4	
	3.2	Logic Methods, Modern Heuristic Methods, Additional Considerations, Evaluation of Models		
4	Title	**Modern Optimization ,Computational Intelligence and Expert Techniques**		
	4.1	Local Optimization Techniques, Stochastic Hill Climber, Simulated Annealing, Tabu Search, Constraint Handling	1-4	6
	4.2	Design of an expert system for decision making using Neural network, genetic algorithm and Fuzzy Logic		7
6	Self Study	Applying Business Intelligence : Marketing Campaigns , Manufacturing, Investment Strategies, Emergency Response Services, Credit Card Fraud		4
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	To implement SAW method for a given problem.
2	To implement WPM method for a given problem
3	To implement AHP method for a given problem.
4	To implement Business Intelligence using prediction model for a given problem.
5	To implement optimization technique for a given problem.
6	To implement an expert system for decision making using Neural Network.
7	To implement an expert system for decision making using Fuzzy Logic.
8	To implement an expert system for decision making using genetic algorithm.
9	Mini Project on decision making and BI.
10	Mini Project on decision making and BI.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Adaptive Business Intelligence	1st	Zbigniew Michalewicz, Martin Schmidt, Matthew Michalewicz, Constantin Chiriac	Springer Publication	2006
2	Decision Making in the Manufacturing Environment: Using Graph Theory and Fuzzy Multiple Attribute Decision Making Methods	2nd	Venkata Rao	Springer publications	2014
3	Computational Intelligence in Complex Decision Systems	1st	Da Ruan	Atlantis Press, Amsterdam Press, World Scientific.	2010
4	Fuzzy sets, decision making and expert system	1st	Hans- Jurgen Zimmermann ,Kluwer	Academic Publishers, Boston	2011

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Business Intelligence: Data Mining and Optimization for Decision Making	1st	Carlo Vercellis	Wiley Publications	2009

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-III	Human Machine Interaction (HMI)	2	0	2	4	8	2	0	1	3
M083		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		50		50		100		200
		Laboratory		50		--		50		100

Pre-requisite Course Codes	Software Engineering
Course Objectives	This course provides an opportunity to learn and apply the design principles of Human-Machine Interaction. Learners will learn the basic human psychology of everyday actions and will be able to design a UI prototype of an application. This course covers the discussion of various interaction design concepts. The laboratory experiments are designed to practice the concepts and to adopt the systematic approach for interface design using various HMI-tools.
Course Outcomes (CO): ***At the End of the course students will be able to***	
M083**.1**	Identify the various design principles used for interacting between humans and machines.
M083**.2**	Apply human psychology of everyday actions and UI design process for real-world applications.
M083**.3**	Implement mobile, windows, and web-based application
M083**.4**	Evaluate and justify UI design
M083**.5**	Create applications for social and technical tasks.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO5
M083**.1**	2	-	2	-
M083**.2**	2	-	2	-
M083**.3**	2	3	-	**2**
M083**.4**	2	-	2	-
M083**.5**	2	3	-	-

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PSO1	PSO2
CS306.1					
CS306.2					
CS306.3					
CS306.4					
CS306.5					

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply ✓	Analyze	Evaluate	Create

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Introduction**	T1-T4	06
	1.1	Introduction to Human Machine Interface, Hardware, software and operating environment to use HMI in various fields.		
	1.2	**The psychopathology of everyday things** – complexity of modern devices; human-centered design; fundamental principles of interaction		
	1.3	**The psychopathology of everyday things** – complexity of modern devices; human-centered design; fundamental principles of interaction;		
2	Title	**Graphical User Interface and Web Interface**	T1,T2, T4	06
	2.1	The Graphical User Interface: Popularity of graphics, the concept of direct manipulation, characteristics of GUI,		
	2.2	Web user Interface: Interface popularity, characteristics. Principles of user interface design.		
3	Title	**Understanding Goal-Directed Design:**	T1,T2, T4	05
	3.1	Goal-directed design; Implementation models and mental models; Beginners, experts, and intermediates – designing for different experience levels;		
	3.2	Understanding users; Modeling users – personas and goals.		
4	Title	**Design Guidelines**	T1,T2, T3,T4	06
	4.1	perception, Gestalt principles, visual structure		
	4.2	color, vision, memory, six behavioral patterns, recognition and recall.		
5	Title	**Interaction Styles and Communication**	T2,T4	05
	5.1	**Interaction Styles 1:** Menus, Windows,		
	5.2	**Interaction Styles 2:**Device-based and Screen-based Controls.		
	5.3	**Communication:** Text messages, Feedback, and Guidance.		
6	Self Study	UX tools: Figma, Just In Mind, and any open-source tool for prototype designing, Mobile Ecosystem: Platforms, Application frameworks: Types of Mobile Applications: Widgets, Applications.		
			Total	**28**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	To Study of open-source UX tools (Justinmind Prototype, Pidoco, Marvel ,Figma Prototype) and create a simple design for a given problem definition.
2	**Know your client.** a. Design an app that can teach mathematics to children of 4-5 years age in schools in Rural Sector. b. Design an app that can teach mathematics to children of 4-5 years age in schools in Urban Sector. c. Design a site that can help people to sell their handmade products in metro cities. d. Design a site that can connect housewives and keep them engaged. **Note : Students should be able to do the following for any given problem statement** i. Analysis of user's/client's behavior eg their preferences, interests etc ii. What kind of interfaces will they like and why? iii. Existing apps - analyze and rate them. iv. What will be your choice of screen elements? v. How will your app/web design be better than the existing one?
3	**Goal-oriented design** - Design an experience for passengers whose flight /train is delayed.
4	**Design Principles** - Understand principles of good UI design by heuristic evaluation. Design UI for a given problem statement.
5	**Menus & Navigation** – Redesign of a user interface (Suggest and implement changes in Existing User Interface) for a given problem statement.
6	**Windows & Screen controls** – Design UI for a given problem statement. **a.** Design a navigator for a student new in your Institute. **b.** Design a navigator for a person new in tourist city/ village. **c.** Motor paralysis for differently able people. **d.** Vaccination App design with localization
7	**Icons** - Design appropriate icons pertaining to a given domain. (Eg. Greeting cards, Travelling, restaurants, Education, Medical, security at Airport, Malls etc)
8	**Colors** – Design a personal website for any socio-technical problem. Use color guidelines with statistical graphics for better visualization.
9	Design a Map-based UI(Web User) for the given problem statement. Example: Mumbai Dabbawallas with localization feature. Pet Care New Visitors to Hospital
10	To calculate the screen complexity of the existing Graphical User Interface and redesign the interface to minimize the screen complexity.

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Human Computer Interaction	3rd	Alan Dix, J. E. Finlay, G. D. Abowd, R. Beale	Peason,Prentice Hall	2003
2	The Essential Guide to User Interface Design	3rd	Wilbert O. Galitz,	Wiley publication	2007
3	Design of everyday things	2nd	Donald A. Normann	Basic Books; Reprint edition	2013
4	Galitz's Human Machine Interaction	1st	Kalbande,Kanade,Iyer	Wiley Publications	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Interaction Design: Beyond Human Computer Interaction	5th	Rogers Sharp Preece	Wiley publications	2019
2	Mobile Design and Development	1st	Brian Fling	, O'Reilly Media Inc.,	2009.

Course Category of Multidisciplinary Minor	MDM-IV (Semester VII)
Computer Engineering	**M054:**Internet and Web Technology
Artificial Intelligence and Machine Learning	**M064:**Generative AI and application
Data Science	**M074:**Social Media Analytics
Interface and Experience Design	**M0844:**Prototyping and Interaction Design [Capstone project]

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-IV	Internet and Web Technology	3	0	2	5	8	3	0	1	4
M054		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		--		--		--		--
		Laboratory		100		--		100		200

Pre-requisite Course Codes, if any.	CS208
Course Objective:	
Course Outcomes (CO): ***At the End of the course, students will be able to***	
M054.1	Develop a sophisticated web UX
M054.2	Create, integrate, and test REST-based web services
M054.3	Design a secure web application/ web services
M054.4	Demonstrate the behavior of web crawlers and testing of web applications

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M054.1					3							
M054.2					3							
M054.3					3							
M054.4					3							

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
M054.1	3	3	3		2	2	
M054.2	3	3	3		2	2	
M054.3	3	3	3		2	2	
M054.4	3	3	3		2	2	

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✔

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1		**Designing UI**		**6**
	1.1	Fundamentals of UX Design, Defining UX Solutions, Design Communication, and Visualizing Ideas	1	
2		**Web content management system**		**4**
	2.1	Introduction to Web CMS, different types of Web CMS	2	
3		**Web services**		**5**
	3.1	Introduction to web service, REST architecture,	3	
4		**Web mashups**		**4**
	4.1	Introduction to web mashups, server-side mashups, client side mashups,	2	
5		**Secured Web application**		**6**
	5.1	Introduction to Web Tokens, Auth2.0, OAuth, Access token	2	
6		**Integration of web services**		**6**
	6.1	Introduction to Mule ESB, Introduction to Anypoint Studio, Integrating Web Services using Anypoint studio	4	
7		**Web crawlers**		**6**
	7.1	Introduction to web crawler, role of crawler in the internet, concept of page ranking	3	
8		**Testing web applications**		**5**
	8.1	Introduction to different types of testing, manual testing, and automated testing testing, performance testing, and functional testing, open source tools used for testing, web-specific test	2	
Total				**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Design web pages using HTML, CSS and javascript
2	Create a website using web CMS (Node Js/Angular Js/React Js/Flask/Django/Wordpress/Joomla etc.)
3	Design UX for a given problem definition by using open source UX tools
4	Create a Restful service to demonstrate different HTTP methods
5	Testing of restful web service using Postman/ARC
6	Create a web mashup of web services using open source framework
7	**Implement a Secure Authentication System** – Develop a web application using **OAuth 2.0, JWT (JSON Web Tokens), and Multi-Factor Authentication (MFA)** for secure user login.
8	Integration of web services using open-source integration tools like Mulesoft
9	Demonstrate the behavior of Web Crawlers/ spiders (use XPATH,CSS PATH),extract information and store it in the database.
10	End to end testing of the web application using open-source testing tools like Selenium, Test runner and Junit

Sr. No.	Title of the Experiment
1	Design web pages using HTML, CSS and javascript
12	**Containerizing a Web Application Using Docker** **Statement:** Develop and deploy a web application using **Docker** by creating a **Dockerfile**, building a container image, running the container, and exposing the application on a specific port. _ignore the comment as there is no cloud technology prerequisite covered in the course

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Sketching the User experiences	2nd edition	Bill Buxton	Diane Cerra	2010
2	Rich Internet Application AJAX and Beyond	3rd edition	Dana Moore, Raymond Budd, Edward Benson	WROX Publisher	2017
3	Web Technology	2nd Edition	Srinivasan	Pearson	2014
4	API Recipes with Mulesoft(r) Anypoint Platform	1st Edition	WHISHWORKS Editorial Board	White falcon	2017

Reference Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	https://nodejs.org				
2	https://angularjs.org/				
3	https://reactjs.org/				
4	Internet Technology And Web Design	1st edition	R. K. JAIN	Khanna Book Publishing Company	2015
5	Understanding the Internet: A Clear Guide to Internet Technologies	1st edition	Keith Sutherland	A Butterworth-Heinemann Title	2016
6	RESTful Web APIs: Services for a Changing World	3rd edition	Leonard Richardson, Mike Amundsen, Sam Ruby	O'REILLY	2013

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-IV	Social Media Analytics	2	0	2	4	8	2	0	2	3
		Examination Scheme								
		Component	ISE		MSE		ESE		Total	
M074		Theory	50		50		100		200	
		Laboratory	50		--		50		100	

Pre-requisite Course Codes, if any.	
Course Objective:	
Course Outcomes (CO): *At the End of the course students will be able to*	
M074.1	Explain the concept of social media Analytics and its significance.
M074.2	Use different Social media analytics tools effectively and efficiently
M074.3	Use different effective Visualization techniques to represent social media analytics.
M074.4	Develop content and structure based social media analytics models.
M074.5	Design social media analytics applications for business

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
M074.1	3	3										
M074.2		3	3		2							
M074.3		3	3		2							
M074.4		3	3	3	2							
M074.5		3	3	3	2							

CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2
M074.1	**3**		**3**		**2**	**2**
M074.2	**3**		**3**		**2**	**2**
M074.3	**3**		**3**		**2**	**2**
M074.4	**3**		**3**		**2**	**2**
M074.5	**3**		**3**		**2**	**2**

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create ✔

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	**Social Media Analytics: An Overview**		5
		Core Characteristics of Social Media, Types of Social Media, Social media landscape, Need for Social Media Analytics (SMA), SMA in small & large organizations. Purpose of Social Media Analytics, Social Media vs. Traditional Business Analytics, Seven Layers of Social Media Analytics, Types of Social Media Analytics, Social Media Analytics Cycle, Challenges to Social Media Analytics, Social Media Analytics Tools		
2	Title	**Social Network Structure, Measures & Visualization**		5
		Basics of Social Network Structure - Nodes, Edges & Tie Describing the Networks Measures - Degree Distribution, Density, Connectivity, Centralization, Tie Strength & Trust Network Visualization - Graph Layout, Visualizing Network features, Scale Issues. Social Media Network Analytics - Common Network Terms, Common Social Media Network Types, Types of Networks, Common Network Terminologies, Network Analytics Tools.		
3	Title	**Social Media Text, Action & Hyperlink Analytics**		8
		Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text Analysis Tools Social Media Action Analytics - What Is Actions Analytics? Common Social Media Actions, Actions Analytics Tools Social Media Hyperlink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools		
4	Title	**Social Media Location & Search Engine Analytics**		5
		Location Analytics - Sources of Location Data, Categories of Location Analytics, Location Analytics and Privacy Concerns, Location Analytics Tools Search Engine Analytics - Types of Search Engines, Search Engine Analytics, Search Engine Analytics Tools		
5	Title	**Social Information Filtering**		5
		Social Information Filtering - Social Sharing and filtering , Automated Recommendation systems, Traditional Vs social Recommendation Systems Understanding Social Media and Business Alignment, Social Media KPI, Formulating a Social Media Strategy, Managing Social Media Risks		
6	Self Study	Social Media Analytics Applications and Privacy- Social media in public sector - Analyzing public sector social media, analyzing individual users, case study. Business use of Social Media - Measuring success, Interaction and monitoring, case study. Privacy - Privacy policies, data ownership and maintaining privacy online.		5
			Total	**33**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Study various – i) Social Media platforms (Facebook, twitter, YouTubeetc) ii) Social Media analytics tools (Facebook insights, google analytics net lytic etc) iii) Social Media Analytics techniques and engagement metrics (page level, post level, member level) iv) Applications of Social media analytics for business. e.g. Google Analytics https://marketingplatform.google.com/about/analytics/ https://netlytic.org
2	Data Collection-Select the social media platforms of your choice (Twitter, Facebook, LinkedIn, YouTube, Web blogs etc), connect to and capture social media data for business (scraping, crawling, parsing).
3	Data Cleaning and Storage- Preprocess, filter and store social media data for business (Using Python, MongoDB, R, etc
4	Exploratory Data Analysis and visualization of Social Media Data for business.
5	Develop Content (text, emoticons, image, audio, video) based social media analytics model for business. (e.g. Content Based Analysis :Topic, Issue, Trend, sentiment/opinion analysis, audio, video, image analytics)
6	Develop Structure based social media analytics model for any business. (e.g. Structure Based Models -community detection, influence analysis)
7	Develop a dashboard and reporting tool based on real time social media data.
8	Design the creative content for promotion of your business on social media platform
9	Analyze competitor activities using social media data.
10	Develop social media text analytics models for improving existing product/ service by analyzing customer‘s reviews/comments

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Seven Layers of Social Media Analytics_ Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data, Gohar F. Khan,(ISBN-10: 1507823207).		Gohar F. Khan		2015
2	Analyzing the Social Web	1st	Jennifer Golbeck	Elseveir	2013
3	Mining the Social Web_	3rd	Matthew A Russell,	O‘Reilly	2019
4	Social Network Data Analytics		Charu Aggarwal	Springer	2011

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Social Media Analytics [2015], Techniques and Insights for Extracting Busi		Matthew Ganis, Avinash Kohirkar	IBM press	2015
2	Social Media Analytics Strategy_ Using Data to Optimize Business Performance		Alex Gonçalves	Apress	2017
3	Social Media Data Mining and Analytics		Szabo G., G. Polatkan, O. Boykin & A. Chalkiopoulus	Wiley	2019

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Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
MDM-IV	Prototyping_Interaction Design [Capstone Project]	3	0	2	4	9	3	0	1	4
M084		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		75		75		150		300
		Laboratory		50		--		50		100

Pre-requisite Course Codes		Software Engineering
At the End of the course students will be able to		
Course Outcomes	M084.1	Apply design research and usability principles to real-world problems
	M084.2	Develop low to high-fidelity prototypes using appropriate tools
	M084.3	Work in interdisciplinary teams and deliver a functional design project
	M084.4	Communicate design decisions through documentation and presentations.

Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1		**Project Framing and Research**	1,2	6
	1	Understand the design challenge, user needs, and context.Capstone project proposal (problem statement, context, significance)Stakeholder mapping Secondary research (literature, trends, existing solutions) Initial user interviews / contextual inquiry		
2		**Ideation and Concept Development**	1	12
	2	Generate, explore, and select ideas. Design goals & refined "How Might We…" questions" Sketching ideas and interaction flows Storyboarding use scenarios Low-fidelity prototypes,Storyboards, Wireframes		
3		**Low-Fidelity Prototyping and User Feedback**	1,3	12
	3.1	Create testable low-fidelity interactive prototypes. Wireframes (using tools like Balsamiq, Figma)Click-through interactive prototype,Usability testing plan,Conduct first round of user testing		
	3.2	Explore and plan Gamification and microinteractions		

4		**High-Fidelity Prototyping and Design Iteration**	1,3,5	12
	4.1	Visual design system (colors, typography, icons, spacing) High-fidelity interactive prototype (Figma, Adobe XD, etc.) Interactive states and transitions,Iteration based on feedback		
	4.2	UX Evaluation techniques and Improve UX Goals, Metrics and Targets		
5		**Final Presentation and Reflection**		
	5.1	Final prototype with interactive demo Video walkthrough (optional)Reflection journal (what worked, what didn't, what they'd do next),Design documentation & style guide,Accessibility considerations,Pitch deck for evaluation		
			Total	**42**

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

Sr. No.	Title of the Experiment
1	Sample Capstone Project : Redesign of a healthcare appointment app.Financial literacy app for teens. Students can be given the freedom to explore and work on the project.
2	Capstone project continue
3	Capstone project continue
4	Capstone project continue
5	Capstone project continue
6	Capstone project continue
7	Develop the capstone project in any software language with full functionality
8	Continue
9	continue
10	Presentation and design documentation with complete testing guide

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	The Essentials of Interaction Design (4th Ed.)	4th	Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel	Wiley	2014
2	A Project Guide to UX Design: For user experience designers in the field or in the making.	2nd	Russ Unger and Carolyn Chandler	O'reilly, Series Editor	2012
3	Communicating Design: Developing Web Site Documentation for Design and Planning	2nd	Dan M. Brown	New Riders (Pearson)	2010

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	A Project Guide to UX Design: For User Experience Designers in the Field or in the Making	2nd	Russ Unger, Carolyn Chandler	Pearson	2012
2	The UX Book: Agile UX Design for a Quality User Experience	2nd	Rex Hartson, Pardha Pyla	Morgan Kaufmann.	2019